

**Search for high-mass leptoquark with
tau-lepton and missing transverse
momentum using 140 fb of pp collision
data at $\sqrt{s} = 13$ TeV in the LHC-ATLAS
experiment**

(LHC-ATLAS実験における重心系エネルギー13 TeVの140 fbの陽子陽子衝突データを用いたタウレプトンおよび消失横運動量を使った高質量レプトクォーク探索)

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Abstract

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Chapter 1

Introduction

1.1 Introduction

Modern high-energy physics seeks to understand the elementary constituents of matter and the fundamental interactions that govern their behaviour. At present, the most successful theoretical description of particle physics is the Standard Model (SM), a quantum field theory based on the gauge symmetry

$$SU(3)_C \times SU(2)_L \times U(1)_Y. \quad (1.1)$$

Within this framework, matter is described by three generations of quarks and leptons, while the electromagnetic, weak, and strong interactions are mediated by gauge bosons. The electroweak symmetry is spontaneously broken through the Higgs mechanism, giving masses to the weak gauge bosons and to the fermions through their interactions with the Higgs field. The discovery of the Higgs boson by the ATLAS and CMS experiments in 2012 completed the particle content predicted by the SM and marked a major milestone in modern particle physics.

The SM has been tested with remarkable precision over a wide range of energy scales and has successfully described a vast number of experimental observations. Nevertheless, it is widely regarded as an incomplete theory. It does not provide a particle candidate for dark matter, does not explain the observed matter–antimatter asymmetry of the universe, and does not account for the origin of non-zero neutrino masses. In addition, the pattern of fermion masses and flavour mixing remains unexplained, and the large separation between the electroweak and Planck scales suggests the possible existence of new physics beyond the SM. These open questions motivate direct searches for new particles and interactions at the highest experimentally accessible energies.

The Large Hadron Collider (LHC) at CERN provides a unique environment for probing such new phenomena. By colliding protons at tera-electronvolt centre-of-mass energies, the LHC allows the production of particles that may be associated with physics beyond the SM. During Run 2 of the LHC from year 2015 to 2018, proton–proton (pp) collisions were delivered at a centre-of-mass energy of $\sqrt{s} = 13$ TeV.

ATLAS is a detector located at one of the interaction points of the LHC. It is designed to observe and study a wide range of physics processes, mainly from the pp collisions, from precision measurements of SM interactions to searches for new particles. The detector consists of several sub-detectors surrounding the collision interaction point, including an inner tracking detector, electromagnetic and hadronic calorimeters, and a muon spectrometer. Together with a powerful magnet system, these components allow the reconstruction and identification of electrons, photons, muons, hadronically decaying tau leptons (τ_{had}), jets, and missing transverse momentum, denoted by $E_{\text{T}}^{\text{miss}}$.

Among the many possible extensions of the SM, leptoquarks (LQs) provide a well-motivated class of new particles connecting the quark and lepton sectors. They are hypothetical bosons that couple directly to a quark and a lepton, and they appear naturally in theories such as grand unified theories and models of quark–lepton unification. Leptoquarks are also of particular phenomenological interest because they can modify semileptonic processes and may therefore be connected to the observed tensions in flavour observables.

A search for the leptoquarks is presented in the thesis using 140 fb^{-1} of pp collision data recorded by the ATLAS detector at a centre-of-mass energy of $\sqrt{s} = 13$ TeV. This thesis focuses on the vector singlet leptoquark U_1 as the benchmark model. This model can contribute to processes involving b quarks, τ leptons, and neutrinos. Therefore, the analysis targets events with a τ_{had} , large missing transverse momentum, and jets. The observed data are compared with SM background predictions in signal regions designed to enhance sensitivity to possible leptoquark contributions. In the absence of a significant excess over the SM expectation, constraints are placed on the allowed parameter space of the U_1 model, expressed in terms of the leptoquark mass and its couplings.

This dissertation is organised as follows. Chapter 2 provides an overview of the SM’s theoretical framework and the physics beyond it. Chapter 3 describes the leptoquark models and the searches performed at LHC. Chapter 4 introduces the LHC and the ATLAS detector. Chapter 5 provides the dataset and simulated samples. Chapter 6 details the reconstruction and the identification of the physics objects that are used for event selection. Chapter 7 discusses the optimisation for the event selection used for signal signature extraction. Chapter 8 presents the background estimation strategy. Chapter 9 discusses the systematic

290 uncertainties for the background and signals. Chapter 10 presents the statistical interpretation
291 of the results. Chapter 11 summarises this analysis.

Chapter 2

Theoretical Background

This chapter introduces the theoretical background which motivates this search performed in the thesis. The discussion begins with the Standard Model (SM), which provides the current theoretical framework for describing elementary particles and their interactions. The limitations of the SM are then summarised, followed by a discussion of several possible extensions beyond the SM. Particular emphasis is placed on leptoquark models, which provide a direct connection between the quark and lepton sectors and motivate the experimental search presented in this thesis.

2.1 The Standard Model

The Standard Model (SM) is a relativistic quantum field theory describing the known elementary particles and their strong, weak, and electromagnetic interactions. Its structure is determined by local gauge invariance under

$$SU(3)_C \times SU(2)_L \times U(1)_Y. \quad (2.1)$$

The group $SU(3)_C$ describes the colour symmetry of the strong interaction, while $SU(2)_L \times U(1)_Y$ describes the electroweak interaction [1–3]. In quantum field theory, particles are classified not only by their mass and spin, but also by their transformation properties under these gauge symmetries. Particles with half-integer spin are fermions and obey Fermi–Dirac statistics, while particles with integer spin are bosons and obey Bose–Einstein statistics [4]. The matter particles of the SM are spin-1/2 fermions, the interaction carriers are spin-1 gauge bosons, and the Higgs boson is a spin-0 scalar boson.

The minimal SM Lagrangian used here is constructed from the SM field content and is invariant under the local gauge group. Therefore, instead of introducing the Lagrangian as

an isolated expression, this section first summarises the field content and gauge quantum numbers of the SM. The principle of local gauge invariance is then introduced, followed by the gauge transformations of the strong sector, based on $SU(3)_C$, and of the electroweak sector, based on $SU(2)_L \times U(1)_Y$. The covariant derivative and field-strength tensors are then constructed from these symmetries, leading to the full SM Lagrangian. Finally, electroweak symmetry breaking and the generation of fermion masses are discussed.

2.1.1 Particle content and gauge group of the SM

The gauge quantum numbers of a field specify how it transforms under the SM gauge group. In this thesis, the electric charge is written as

$$Q = T^3 + Y, \quad (2.2)$$

where T^3 is the third component of weak isospin and Y is the weak hypercharge. This convention is used throughout the thesis.

For each generation, the SM fermions are arranged as

$$Q_L^i = \begin{pmatrix} u_L^i \\ d_L^i \end{pmatrix} \sim (3, 2, 1/6), \quad u_R^i \sim (3, 1, 2/3), \quad d_R^i \sim (3, 1, -1/3), \quad (2.3)$$

$$L_L^i = \begin{pmatrix} \nu_L^i \\ e_L^i \end{pmatrix} \sim (1, 2, -1/2), \quad e_R^i \sim (1, 1, -1), \quad (2.4)$$

where $i = 1, 2, 3$ is the generation index, and the notation denotes the representation under $SU(3)_C \times SU(2)_L \times U(1)_Y$, which are the QCD representation, the weak isospin representation and the weak hypercharge, respectively. In the minimal SM, no right-handed neutrino field is included. The scalar sector contains one complex Higgs doublet,

$$\Phi = \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix} \sim (1, 2, 1/2). \quad (2.5)$$

The gauge bosons are associated with the generators of the three gauge groups. The eight gluon fields G_μ^a , with $a = 1, \dots, 8$, are associated with $SU(3)_C$. The three weak gauge fields W_μ^I , with $I = 1, 2, 3$, are associated with $SU(2)_L$. The hypercharge gauge field B_μ is associated with $U(1)_Y$. Before electroweak symmetry breaking, W_μ^1 , W_μ^2 , W_μ^3 , and B_μ are gauge eigenstates. After electroweak symmetry breaking, they are reorganised into the physical electroweak bosons $W^\pm$, Z , and γ .

The main particle content of the SM is summarised in Table 2.1. For fermions and the Higgs boson, the table lists the corresponding electroweak quantum numbers using the convention in Eq. (2.2). For gauge bosons, the entries describe their gauge fields and the electric charges of the physical states where appropriate.

Table 2.1 Particle content of the SM and the corresponding spin and electroweak quantum numbers. For fermions and the Higgs doublet, T^3 and Y denote the third component of weak isospin and the weak hypercharge, respectively, with the electric charge defined as $Q = T^3 + Y$. For gauge bosons, only the electric charges of the physical states after electroweak symmetry breaking are shown.

Category	Type	Generation			Spin	Quantum numbers		
		1st	2nd	3rd		T^3	Y	Q
Fermion	Left-handed quark	u_L	c_L	t_L	1/2	+1/2	1/6	+2/3
		d_L	s_L	b_L		-1/2		-1/3
	Right-handed up-type quark	u_R	c_R	t_R	1/2	0	2/3	+2/3
	Right-handed down-type quark	d_R	s_R	b_R	1/2	0	-1/3	-1/3
	Left-handed lepton	ν_{eL}	$\nu_{\mu L}$	$\nu_{\tau L}$	1/2	+1/2	-1/2	0
		e_L	μ_L	τ_L		-1/2		-1
	Right-handed charged lepton	e_R	μ_R	τ_R	1/2	0	-1	-1
Gauge boson	Gluon	$g^a, a = 1, \dots, 8$			1	-	-	0
	Photon	γ			1	-	-	0
	Charged weak boson	$W^\pm$			1	-	-	± 1
	Neutral weak boson	Z			1	-	-	0
Scalar field	Higgs doublet	ϕ^+			0	+1/2	1/2	+1
		ϕ^0				-1/2		0

The quantum numbers listed above determine which interactions each field can have. They also determine which terms can be written in the SM Lagrangian. For example, explicit Dirac mass terms for charged fermions are not gauge invariant before electroweak symmetry breaking, because the left- and right-handed fermions transform differently under $SU(2)_L \times U(1)_Y$. Fermion masses instead arise from gauge-invariant Yukawa interactions with the Higgs field.

2.1.2 Local gauge invariance

The guiding principle in constructing the SM is local gauge invariance. A global transformation acts in the same way at every point in spacetime, while a local gauge transformation depends on the spacetime coordinate x . For a field $\psi(x)$ transforming under a gauge group with generators T^A , a generic local transformation is

$$\psi(x) \rightarrow \psi'(x) = U(x)\psi(x), \quad U(x) = \exp \left[i\alpha^A(x)T^A \right]. \quad (2.6)$$

Here $\alpha^A(x)$ are arbitrary real, spacetime-dependent gauge parameters. They specify how the field is transformed locally along each generator T^A of the gauge group, and the repeated index A is summed over all generators.

The ordinary derivative of the transformed field is

$$\partial_\mu \psi'(x) = U(x) \partial_\mu \psi(x) + (\partial_\mu U(x)) \psi(x). \quad (2.7)$$

The second term in Eq. (2.7) shows that $\partial_\mu \psi$ does not transform in the same way as ψ . Therefore, the kinetic term constructed using the ordinary derivative is not invariant under a local gauge transformation.

To restore local gauge invariance, the ordinary derivative is promoted to a covariant derivative D_μ satisfying

$$D_\mu \psi(x) \rightarrow D'_\mu \psi'(x) = U(x) D_\mu \psi(x). \quad (2.8)$$

This requirement introduces gauge fields. The gauge fields are therefore not added by hand as arbitrary new particles; they are required in order to compare fields at neighbouring spacetime points in a way that is independent of the local gauge convention. The following subsections describe how this principle is realised for the strong and electroweak gauge sectors of the SM.

2.1.3 The $SU(3)_C$ gauge sector

The strong interaction is described by quantum chromodynamics (QCD), a gauge theory based on the colour group $SU(3)_C$. Quark fields transform as colour triplets, while leptons and the Higgs field are colour singlets. For a colour triplet quark field $q(x)$, the local $SU(3)_C$ transformation is

$$q(x) \rightarrow q'(x) = U_C(x) q(x), \quad U_C(x) = \exp \left[i \alpha_C^a(x) \frac{\lambda^a}{2} \right], \quad (2.9)$$

where λ^a are the Gell-Mann matrices and $a = 1, \dots, 8$. The corresponding gauge fields are the eight gluon fields G_μ^a .

It is useful to introduce the matrix-valued gluon field

$$G_\mu(x) = G_\mu^a(x) \frac{\lambda^a}{2}. \quad (2.10)$$

With the convention

$$D_\mu = \partial_\mu - ig_s G_\mu^a \frac{\lambda^a}{2}, \quad (2.11)$$

where g_s is the strong coupling constant, the gluon field transforms as

$$G_\mu(x) \rightarrow G'_\mu(x) = U_C(x) G_\mu(x) U_C^\dagger(x) - \frac{i}{g_s} (\partial_\mu U_C(x)) U_C^\dagger(x). \quad (2.12)$$

This transformation law ensures that $D_\mu q$ transforms in the same way as q under local $SU(3)_C$ transformations.

The $SU(3)_C$ field-strength tensor is

$$G_{\mu\nu}^a = \partial_\mu G_\nu^a - \partial_\nu G_\mu^a + g_s f^{abc} G_\mu^b G_\nu^c, \quad (2.13)$$

where f^{abc} are the structure constants of $SU(3)$, defined by the commutation relation

$$\left[\frac{\lambda^a}{2}, \frac{\lambda^b}{2} \right] = i f^{abc} \frac{\lambda^c}{2}. \quad (2.14)$$

The last term in Eq. (2.13) arises from the non-Abelian nature of $SU(3)_C$. It gives rise to gluon self-interactions, which are a defining feature of QCD and are essential for asymptotic freedom at high energies and confinement at low energies.

2.1.4 The electroweak gauge sector

Before electroweak symmetry breaking, the weak and electromagnetic interactions are described together by the gauge group $SU(2)_L \times U(1)_Y$ [1–3]. The $SU(2)_L$ factor is the non-Abelian weak-isospin gauge symmetry, while the $U(1)_Y$ factor is the Abelian hypercharge gauge symmetry. They are introduced separately below, but the physical photon and Z boson arise only after electroweak symmetry breaking, when the neutral fields W_μ^3 and B_μ mix.

The $SU(2)_L$ gauge transformation

The $SU(2)_L$ gauge symmetry acts on left-handed fermion doublets and on the Higgs doublet:

$$Q_L^i, \quad L_L^i, \quad \Phi. \quad (2.15)$$

405 For an $SU(2)_L$ doublet $\chi(x)$, the local transformation is

$$406 \quad \chi(x) \rightarrow \chi'(x) = U_L(x)\chi(x), \quad U_L(x) = \exp \left[i\alpha^I(x) \frac{\tau^I}{2} \right], \quad (2.16)$$

407 where τ^I are the Pauli matrices and $I = 1, 2, 3$. The right-handed charged fermions u_R^i, d_R^i ,
408 and e_R^i are singlets under $SU(2)_L$ and therefore have $T_L^I = 0$.

409 The three gauge fields associated with $SU(2)_L$ are denoted by W_μ^I . They can be written
410 in matrix form as

$$411 \quad W_\mu(x) = W_\mu^I(x) \frac{\tau^I}{2}. \quad (2.17)$$

412 With the convention

$$413 \quad D_\mu = \partial_\mu - igW_\mu^I \frac{\tau^I}{2}, \quad (2.18)$$

414 where g is the $SU(2)_L$ coupling constant, the weak gauge field transforms as

$$415 \quad W_\mu(x) \rightarrow W'_\mu(x) = U_L(x)W_\mu(x)U_L^\dagger(x) - \frac{i}{g} (\partial_\mu U_L(x)) U_L^\dagger(x). \quad (2.19)$$

416 This transformation law is the non-Abelian generalisation of the Abelian gauge-field trans-
417 formation. The matrix structure reflects the fact that the $SU(2)_L$ generators do not commute.

418 The $SU(2)_L$ field-strength tensor is

$$419 \quad W_{\mu\nu}^I = \partial_\mu W_\nu^I - \partial_\nu W_\mu^I + g\epsilon^{IJK} W_\mu^J W_\nu^K. \quad (2.20)$$

420 The last term arises from the non-Abelian nature of $SU(2)_L$ and leads to self-interactions
421 among weak gauge bosons.

422 The $U(1)_Y$ gauge transformation

423 The $U(1)_Y$ gauge symmetry is the Abelian part of the electroweak gauge group. Its cor-
424 responding gauge charge is the weak hypercharge Y , which determines how each field
425 transforms under a local $U(1)_Y$ phase rotation. For a field $\psi(x)$ with hypercharge Y , the local
426 $U(1)_Y$ transformation is

$$427 \quad \psi(x) \rightarrow \psi'(x) = \exp[iY\beta(x)] \psi(x), \quad (2.21)$$

where $\beta(x)$ is a spacetime-dependent transformation parameter. The corresponding gauge field is the hypercharge field B_μ . With the convention

$$D_\mu = \partial_\mu - ig'YB_\mu, \quad (2.22)$$

the gauge field transforms as

$$B_\mu(x) \rightarrow B'_\mu(x) = B_\mu(x) + \frac{1}{g'}\partial_\mu\beta(x). \quad (2.23)$$

This transformation ensures that $D_\mu\psi$ transforms in the same way as ψ .

The hypercharge assignments in Table 2.1 are chosen so that the electric charge is obtained from Eq. (2.2). For the left-handed lepton doublet, $Y = -1/2$. The upper component has $T^3 = +1/2$ and therefore $Q = 0$, corresponding to the neutrino. The lower component has $T^3 = -1/2$ and therefore $Q = -1$, corresponding to the charged lepton. Similarly, the left-handed quark doublet has $Y = 1/6$, giving charges $+2/3$ and $-1/3$ for the up- and down-type components.

Because $U(1)_Y$ is Abelian, the corresponding field-strength tensor is

$$B_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu. \quad (2.24)$$

There is no self-interaction term involving two B_μ fields in $B_{\mu\nu}$. Although $U(1)_Y$ is Abelian, it should not be identified with the electromagnetic gauge symmetry $U(1)_{\text{em}}$. Before electroweak symmetry breaking, the corresponding gauge field is the hypercharge field B_μ rather than the photon field. The electromagnetic gauge symmetry appears only after electroweak symmetry breaking, when B_μ mixes with the neutral weak gauge field W_μ^3 to form the photon and the Z boson.

2.1.5 Covariant derivative and gauge field strengths

Combining the three gauge symmetries, the covariant derivative acting on a generic SM field can be written as

$$D_\mu = \partial_\mu - ig_s G_\mu^a T_C^a - ig W_\mu^I T_L^I - ig' Y B_\mu. \quad (2.25)$$

Here g_s , g , and g' are the gauge couplings of $SU(3)_C$, $SU(2)_L$, and $U(1)_Y$, respectively. The matrices T_C^a and T_L^I are the generators acting on the particular field. For a colour triplet, $T_C^a = \lambda^a/2$, while for a colour singlet $T_C^a = 0$. For an $SU(2)_L$ doublet, $T_L^I = \tau^I/2$, while

for an $SU(2)_L$ singlet $T_L^I = 0$. The value of Y is the hypercharge of the field, as listed in Table 2.1.

The gauge kinetic terms are constructed from the field-strength tensors:

$$G_{\mu\nu}^a = \partial_\mu G_\nu^a - \partial_\nu G_\mu^a + g_s f^{abc} G_\mu^b G_\nu^c, \quad (2.26)$$

$$W_{\mu\nu}^I = \partial_\mu W_\nu^I - \partial_\nu W_\mu^I + g \epsilon^{IJK} W_\mu^J W_\nu^K, \quad (2.27)$$

$$B_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu. \quad (2.28)$$

The tensors $G_{\mu\nu}^a$ and $W_{\mu\nu}^I$ contain non-linear terms because $SU(3)_C$ and $SU(2)_L$ are non-Abelian groups. By contrast, $B_{\mu\nu}$ has the Abelian form because $U(1)_Y$ has only one generator.

2.1.6 The Standard Model Lagrangian

The SM Lagrangian is constructed from the field content described above and is invariant under the local gauge transformations of $SU(3)_C \times SU(2)_L \times U(1)_Y$. Using the covariant derivative and field-strength tensors introduced in the previous subsections, the minimal SM Lagrangian can be written as the sum of gauge-boson kinetic terms, fermion kinetic terms, the Higgs-sector terms, and Yukawa interactions. Ignoring gauge-fixing terms, ghost terms, and the QCD θ -term, it is written schematically as

$$\mathcal{L}_{\text{SM}} = \mathcal{L}_{\text{gauge}} + \mathcal{L}_{\text{fermion}} + \mathcal{L}_{\text{Higgs}} + \mathcal{L}_{\text{Yukawa}}. \quad (2.29)$$

The four terms in Eq. (2.29) are fixed by Lorentz invariance, local gauge invariance, and renormalisability, once the SM field content is specified [5].

The gauge-boson kinetic terms are

$$\mathcal{L}_{\text{gauge}} = -\frac{1}{4} G_{\mu\nu}^a G^{a\mu\nu} - \frac{1}{4} W_{\mu\nu}^I W^{I\mu\nu} - \frac{1}{4} B_{\mu\nu} B^{\mu\nu}. \quad (2.30)$$

Here $G_{\mu\nu}^a$, $W_{\mu\nu}^I$, and $B_{\mu\nu}$ are the field-strength tensors of $SU(3)_C$, $SU(2)_L$, and $U(1)_Y$, respectively. The indices $a = 1, \dots, 8$ and $I = 1, 2, 3$ label the generators of $SU(3)_C$ and $SU(2)_L$, and repeated Lorentz and gauge indices are summed over. The common factor $-1/4$ gives the conventional canonical normalisation of the gauge-boson kinetic terms. With this normalisation, the quadratic parts of Eq. (2.30) describe the free propagation of the gluon, weak, and hypercharge gauge fields, while the non-Abelian parts of $G_{\mu\nu}^a$ and $W_{\mu\nu}^I$ generate self-interactions among the corresponding gauge bosons.

The fermion kinetic terms are

$$\mathcal{L}_{\text{fermion}} = \sum_{i=1}^3 \left[\bar{Q}_L^i i\gamma^\mu D_\mu Q_L^i + \bar{u}_R^i i\gamma^\mu D_\mu u_R^i + \bar{d}_R^i i\gamma^\mu D_\mu d_R^i + \bar{L}_L^i i\gamma^\mu D_\mu L_L^i + \bar{e}_R^i i\gamma^\mu D_\mu e_R^i \right]. \quad (2.31)$$

The sum runs over the three fermion generations. The fields Q_L^i and L_L^i are the left-handed quark and lepton doublets, while u_R^i , d_R^i , and e_R^i are right-handed singlets under $SU(2)_L$. The Dirac adjoint is defined as $\bar{\psi} = \psi^\dagger \gamma^0$, and γ^μ are the Dirac gamma matrices. The covariant derivative D_μ contains the gauge fields and couplings appropriate to the gauge quantum numbers of each fermion field. Therefore, Eq. (2.31) describes both the free propagation of fermions and their gauge interactions.

The Higgs-sector terms are

$$\mathcal{L}_{\text{Higgs}} = (D_\mu \Phi)^\dagger (D^\mu \Phi) - V(\Phi), \quad V(\Phi) = -\mu^2 \Phi^\dagger \Phi + \lambda (\Phi^\dagger \Phi)^2. \quad (2.32)$$

The first term is the kinetic term of the Higgs doublet and also contains the gauge interactions of the Higgs field through the covariant derivative. The function $V(\Phi)$ is the Higgs potential. With the convention used here, $\mu^2 > 0$ and $\lambda > 0$. The condition $\lambda > 0$ ensures that the potential is bounded from below, while the negative quadratic term causes the minimum of the potential to occur at a non-zero value of $\Phi^\dagger \Phi$. This leads to electroweak symmetry breaking through the Brout–Englert–Higgs mechanism [6–8].

The Yukawa interactions are

$$\mathcal{L}_{\text{Yukawa}} = -y_d^{ij} \bar{Q}_L^i \Phi d_R^j - y_u^{ij} \bar{Q}_L^i \tilde{\Phi} u_R^j - y_e^{ij} \bar{L}_L^i \Phi e_R^j + \text{h.c.}, \quad (2.33)$$

where $i, j = 1, 2, 3$ are generation indices, and y_u^{ij} , y_d^{ij} , and y_e^{ij} are complex Yukawa coupling matrices in flavour space. h.c. denotes the Hermitian conjugate. The conjugate Higgs doublet is defined as

$$\tilde{\Phi} = i\tau^2 \Phi^*, \quad (2.34)$$

where τ^2 is the second Pauli matrix. The field $\tilde{\Phi}$ transforms as an $SU(2)_L$ doublet with hypercharge opposite to that of Φ , allowing a gauge-invariant Yukawa interaction for up-type quarks. After electroweak symmetry breaking, the Yukawa interactions generate masses for the charged fermions. In the minimal SM, no right-handed neutrino field is included, and therefore no renormalisable neutrino Yukawa mass term appears in Eq. (2.33).

2.1.7 Electroweak symmetry breaking and the Higgs mechanism

The Higgs potential in Eq. (2.32) has a minimum at a non-zero value of $\Phi^\dagger \Phi$. For the convention used in Eq. (2.32), the minimum satisfies $\Phi^\dagger \Phi = v^2/2$, where $v = \sqrt{\mu^2/\lambda}$ is the Higgs vacuum expectation value and sets the electroweak scale. A convenient choice of vacuum expectation value is

$$\langle \Phi \rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix}. \quad (2.35)$$

The value of v is related to the Fermi constant by $v = (\sqrt{2}G_F)^{-1/2} \simeq 246 \text{ GeV}$ [5].

This vacuum breaks the electroweak symmetry according to

$$SU(2)_L \times U(1)_Y \rightarrow U(1)_{\text{em}}. \quad (2.36)$$

The unbroken symmetry is electromagnetism. This mass-generation mechanism for gauge bosons is the Brout–Englert–Higgs mechanism.

In unitary gauge, the Higgs doublet can be written as

$$\Phi(x) = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h(x) \end{pmatrix}, \quad (2.37)$$

where $h(x)$ is the physical Higgs boson field. Expanding the Higgs potential around the vacuum gives

$$V(\Phi) = V(\langle \Phi \rangle) + \frac{1}{2}m_h^2 h^2 + \lambda v h^3 + \frac{\lambda}{4}h^4, \quad (2.38)$$

where the physical Higgs-boson mass is

$$m_h^2 = 2\lambda v^2 = 2\mu^2. \quad (2.39)$$

Unlike the W and Z boson masses, which are fixed at tree level by the electroweak gauge couplings and the vacuum expectation value, the Higgs-boson mass depends on the quartic coupling λ . Therefore, the SM does not predict the numerical value of m_h before λ is fixed experimentally. The Higgs boson was observed by the ATLAS and CMS experiments in 2012 [9, 10]. The observed Higgs-boson mass is approximately 125 GeV, corresponding to $\lambda \simeq m_h^2/(2v^2) \simeq 0.13$ [11].

The masses of the electroweak gauge bosons arise from the Higgs kinetic term,

$$(D_\mu \Phi)^\dagger (D^\mu \Phi), \quad (2.40)$$

after substituting the Higgs vacuum expectation value. It is useful to define the charged weak fields as

$$W_\mu^\pm = \frac{1}{\sqrt{2}} (W_\mu^1 \mp iW_\mu^2). \quad (2.41)$$

The terms quadratic in the electroweak gauge fields are then

$$(D_\mu \langle \Phi \rangle)^\dagger (D^\mu \langle \Phi \rangle) = \frac{g^2 v^2}{4} W_\mu^+ W^{-\mu} + \frac{v^2}{8} (g W_\mu^3 - g' B_\mu)^2. \quad (2.42)$$

The first term is the mass term of the charged weak bosons. The second term shows that the neutral gauge fields W_μ^3 and B_μ mix and must be diagonalised.

The weak mixing angle is defined by

$$\cos \theta_W = \frac{g}{\sqrt{g^2 + g'^2}}, \quad \sin \theta_W = \frac{g'}{\sqrt{g^2 + g'^2}}. \quad (2.43)$$

The corresponding neutral mass eigenstates are

$$A_\mu = B_\mu \cos \theta_W + W_\mu^3 \sin \theta_W, \quad (2.44)$$

$$Z_\mu = -B_\mu \sin \theta_W + W_\mu^3 \cos \theta_W. \quad (2.45)$$

The field A_μ is identified with the photon, while Z_μ is the neutral weak boson. The photon remains massless because it is associated with the unbroken gauge symmetry $U(1)_{\text{em}}$.

The resulting gauge-boson masses are

$$m_W = \frac{1}{2} g v, \quad m_Z = \frac{1}{2} \sqrt{g^2 + g'^2} v = \frac{m_W}{\cos \theta_W}, \quad m_\gamma = 0. \quad (2.46)$$

Thus, the SM predicts the relation $m_W = m_Z \cos \theta_W$. Inserting $v \simeq 246$ GeV and the electroweak couplings gives $m_W \simeq 80$ GeV and $m_Z \simeq 91$ GeV. Therefore, the Higgs mechanism gives masses to the W and Z bosons while leaving the photon massless.

After electroweak symmetry breaking, the unbroken gauge symmetry is $U(1)_{\text{em}}$, whose gauge field is the photon field A_μ . At energies much below the electroweak scale, the electromagnetic interactions of charged fermions are described by quantum electrodynamics (QED).

For a charged fermion field ψ with electric charge Q , the QED Lagrangian can be written as

$$\mathcal{L}_{\text{QED}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}(i\gamma^\mu D_\mu - m)\psi, \quad (2.47)$$

where

$$D_\mu = \partial_\mu - ieQA_\mu, \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu. \quad (2.48)$$

Here e is the electromagnetic coupling constant, Q is the electric charge of the fermion in units of e , and $F_{\mu\nu}$ is the electromagnetic field-strength tensor. Thus, QED appears as the low-energy gauge theory associated with the unbroken $U(1)_{\text{em}}$ symmetry of the SM.

2.1.8 Fermion masses and flavour mixing

The Yukawa interactions generate fermion masses after electroweak symmetry breaking. When the Higgs field acquires its vacuum expectation value v , the Yukawa coupling matrices are converted into fermion mass matrices:

$$M_u^{ij} = \frac{v}{\sqrt{2}}y_u^{ij}, \quad M_d^{ij} = \frac{v}{\sqrt{2}}y_d^{ij}, \quad M_e^{ij} = \frac{v}{\sqrt{2}}y_e^{ij}. \quad (2.49)$$

Here M_u , M_d , and M_e are the mass matrices of the up-type quarks, down-type quarks, and charged leptons, respectively. The matrices y_u , y_d , and y_e are the corresponding dimensionless Yukawa coupling matrices, and the indices $i, j = 1, 2, 3$ label the fermion generations.

The physical fermion masses are obtained by diagonalising the mass matrices. In the quark sector, the up-type and down-type mass matrices are diagonalised by different rotations in flavour space. As a result, flavour mixing appears in the charged-current weak interaction. This mixing is described by the Cabibbo–Kobayashi–Maskawa (CKM) matrix [12, 13]. In the mass eigenstate basis, the charged-current interaction of quarks is

$$\mathcal{L}_{\text{CC}} = \frac{g}{\sqrt{2}}\bar{u}_L^i \gamma^\mu V_{ij}^{\text{CKM}} d_L^j W_\mu^+ + \text{h.c.} \quad (2.50)$$

Here g is the $SU(2)_L$ gauge coupling, u_L^i and d_L^j are the left-handed up-type and down-type quark mass eigenstates, and W_μ^+ is the charged weak gauge boson. The matrix element V_{ij}^{CKM} describes the strength of the charged-current transition from the down-type quark d^j to the up-type quark u^i . The matrices γ^μ are the Dirac gamma matrices, and the Hermitian-conjugate term describes the corresponding interaction involving W_μ^- . The CKM matrix is

essential for describing flavour-changing charged-current processes such as $b \rightarrow c\tau\nu$. Its complex phase also provides a source of CP violation in the quark sector of the SM.

In the minimal SM, neutrino masses are absent because no right-handed neutrino fields are included and no renormalisable neutrino Yukawa interaction can be written with the minimal field content. This provides one of the motivations for considering extensions of the SM.

2.2 Beyond the Standard Model

Although the SM has been tested with high precision and has successfully described a wide range of experimental observations, it is not a complete theory of nature. Several experimental and theoretical facts point to physics beyond the SM.

2.2.1 Gravity and the Planck scale

The SM describes the electromagnetic, weak, and strong interactions as quantum gauge field theories, but it does not include gravity. Gravity is instead described at macroscopic scales by general relativity. A complete theory of nature is therefore expected to provide a consistent quantum description of gravity and to explain its connection to the SM interactions.

The difficulty of connecting gravity with the SM appears in two related ways. First, quantum gravitational effects are expected to become important at the Planck scale, $M_{\text{Pl}} \sim 10^{19}$ GeV. This is much higher than the electroweak scale, $v \simeq 246$ GeV. This large separation of scales is closely related to the hierarchy problem discussed below.

Second, constructing a quantum theory of gravity is theoretically challenging. If general relativity is quantised perturbatively in the same way as ordinary quantum field theories, it is not renormalisable as a fundamental theory. This makes it difficult to combine gravity with the quantum-field-theoretic framework of the SM in a straightforward way. Several possible approaches to quantum gravity have been proposed, including string theory, loop quantum gravity, and asymptotic safety, although no complete theory has been experimentally confirmed.

2.2.2 The hierarchy problem

The electroweak scale is set by the Higgs vacuum expectation value, $v \simeq 246$ GeV, and is closely related to the masses of the weak gauge bosons and the Higgs boson. In the SM, the

Higgs-boson mass is determined at tree level by

$$m_h^2 = 2\lambda v^2, \quad (2.51)$$

where λ is the Higgs quartic coupling. The observed value $m_h \simeq 125$ GeV therefore corresponds to $\lambda \simeq 0.13$.

The hierarchy problem arises because the Higgs mass parameter is highly sensitive to physics at very high energy scales. If the SM is treated as an effective theory valid up to an ultraviolet cutoff scale Λ , loop corrections to the Higgs mass can be written schematically as

$$m_{h,\text{phys}}^2 = m_{h,0}^2 + \delta m_h^2, \quad (2.52)$$

where $m_{h,0}^2$ is the bare Higgs mass parameter and δm_h^2 denotes the quantum correction. At one-loop order, the correction contains terms quadratically sensitive to the cutoff scale,

$$\delta m_h^2 \sim \frac{\Lambda^2}{16\pi^2}, \quad (2.53)$$

If the cutoff scale is close to the Planck scale, $\Lambda \sim M_{\text{Pl}} \sim 10^{19}$ GeV, the magnitude of δm_h^2 is many orders of magnitude larger than the observed value $m_h^2 \sim (125 \text{ GeV})^2$. Therefore, the bare parameter $m_{h,0}^2$ and the quantum correction δm_h^2 must cancel each other with extremely high precision in order to leave the observed electroweak-scale Higgs mass. This required cancellation is known as fine-tuning.

The hierarchy problem is the question of why the electroweak scale remains so much smaller than a possible ultraviolet scale such as the Planck scale. It motivates many extensions of the SM, such as supersymmetry, composite Higgs models, and other scenarios in which new physics appears near the TeV scale and stabilises the Higgs mass against large ultraviolet corrections. More generally, the hierarchy problem provides one of the motivations for experimental searches for new particles at the TeV scale.

2.2.3 Neutrino masses

In the minimal SM, neutrinos are massless because no right-handed neutrino fields are included and no renormalisable neutrino mass term can be written using only the SM field content. However, neutrino oscillation experiments have shown that neutrinos change flavour as they propagate, which requires non-zero neutrino mass differences. Evidence for atmospheric neutrino oscillations was reported by the Super-Kamiokande experiment [14], and solar neutrino flavour transformation was established by the Sudbury Neutrino Obser-

vatory [15]. Therefore, neutrino masses provide direct experimental evidence for physics beyond the minimal SM.

2.2.4 Dark matter

Astrophysical and cosmological observations indicate the existence of non-luminous matter that cannot be explained by the known particles of the SM. Measurements of the cosmic microwave background, for example by the Planck satellite, are well described by a cosmological model containing a significant non-baryonic dark matter component [16]. The SM does not contain a suitable cold dark matter candidate. Although neutrinos are electrically neutral and weakly interacting, their small masses and relativistic behaviour in the early universe do not provide the required cold dark matter component.

2.2.5 Matter–antimatter asymmetry

The observable universe contains much more matter than antimatter. Sakharov identified three necessary conditions for dynamically generating such a baryon asymmetry: baryon-number violation, C and CP violation, and departure from thermal equilibrium [17]. The SM contains CP violation through the CKM matrix and baryon-plus-lepton number violation through non-perturbative electroweak effects. However, the amount of CP violation in the SM and the nature of the electroweak phase transition are not sufficient to explain the observed baryon asymmetry of the universe [18, 19]. This motivates new sources of CP violation and new dynamics in the early universe.

2.2.6 The strong CP problem

The most general QCD Lagrangian allows a CP-violating term of the form

$$\mathcal{L}_\theta = \theta \frac{g_s^2}{32\pi^2} G_{\mu\nu}^a \tilde{G}^{a\mu\nu}, \quad \tilde{G}^{a\mu\nu} = \frac{1}{2} \varepsilon^{\mu\nu\rho\sigma} G_{\rho\sigma}^a. \quad (2.54)$$

Here g_s is the strong coupling constant, and $G_{\mu\nu}^a$ is the gluon field-strength tensor of the $SU(3)_C$ gauge group, with the colour index $a = 1, \dots, 8$ labelling the eight gluon fields. The tensor $\tilde{G}^{a\mu\nu}$ is the dual field-strength tensor, defined using the totally antisymmetric Levi-Civita tensor $\varepsilon^{\mu\nu\rho\sigma}$. The parameter θ is a dimensionless QCD vacuum angle that controls the size of this CP-violating interaction. The factor $g_s^2/(32\pi^2)$ is a conventional normalisation.

The operator $G_{\mu\nu}^a \tilde{G}^{a\mu\nu}$ is gauge invariant and has mass dimension four, so it is allowed by the gauge symmetry and renormalisability of QCD. However, it violates CP symmetry. Therefore, a non-zero value of the effective QCD vacuum angle would lead to CP violation in the strong interaction.

This term would contribute to the electric dipole moment of the neutron. Experimental limits on the neutron electric dipole moment strongly constrain the effective QCD vacuum angle $\bar{\theta}$ [20]. The effective parameter $\bar{\theta}$ includes both the QCD vacuum angle θ and phases from the quark mass matrix. The absence of observable strong CP violation, despite the fact that such a term is allowed by the SM gauge symmetries, is known as the strong CP problem.

2.2.7 Flavour anomalies and lepton flavour universality

The flavour structure of the SM is mainly determined by the Yukawa matrices. After electroweak symmetry breaking, these matrices generate the fermion mass matrices; their eigenvalues determine the fermion masses, while the mismatch between the diagonalisation of the up- and down-type quark mass matrices gives rise to quark flavour mixing through the CKM matrix.

Although the SM accommodates the observed masses and mixing angles, it does not explain their hierarchical pattern or their origin. Precision measurements in flavour physics therefore provide sensitive probes of possible physics beyond the SM.

A key property of the SM is lepton flavour universality (LFU), which means that the electroweak gauge interactions couple equally to the three charged-lepton generations, apart from effects due to their different masses. Semileptonic B -meson decays provide precise tests of LFU by comparing decay rates involving different charged leptons. In charged-current transitions of the form $b \rightarrow c\ell\nu$, the commonly studied ratios are

$$R(D^{(*)}) = \frac{\mathcal{B}(\bar{B} \rightarrow D^{(*)} \tau \bar{\nu}_\tau)}{\mathcal{B}(\bar{B} \rightarrow D^{(*)} \ell \bar{\nu}_\ell)}, \quad \ell = e, \mu. \quad (2.55)$$

These observables compare decays involving a τ lepton with the corresponding decays involving light charged leptons.

Neutral-current transitions of the form $b \rightarrow s\ell^+\ell^-$ can be tested through ratios such as

$$R(K^{(*)}) = \frac{\mathcal{B}(B \rightarrow K^{(*)} \mu^+ \mu^-)}{\mathcal{B}(B \rightarrow K^{(*)} e^+ e^-)}. \quad (2.56)$$

These ratios compare final states with muon pairs and electron pairs. Recent measurements of $R(K)$ and $R(K^*)$ are compatible with SM predictions [21, 22]. However, measurements of

$R(D)$ and $R(D^*)$ continue to show tensions with SM predictions. The HFLAV CKM 2025 average gives a combined discrepancy of about 3.8σ with respect to the SM prediction [23] as shown in Figure 2.1. These observables remain important motivations for models that couple differently to different lepton generations.

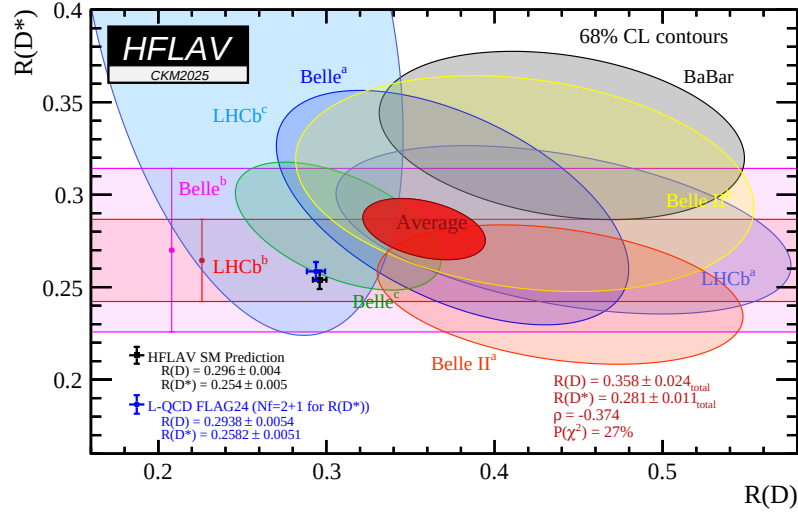


Fig. 2.1 Average of $R(D^*)$ and $R(D)$ [23] measured by the BaBar [24, 25], Belle [26–29], Belle-II [30, 31], and LHCb [32–34] experiments.

2.3 Possible Solutions to the Remaining Problems of the Standard Model

Many extensions of the SM have been proposed to address one or more of the problems described above. This section briefly introduces several representative ideas: supersymmetry, axions, and leptiquarks. These models are not mutually exclusive, and in many cases they address different aspects of the SM's limitations.

2.3.1 Supersymmetry

Supersymmetry (SUSY) is a spacetime symmetry relating bosons and fermions [35, 36]. In supersymmetric theories, each SM particle is accompanied by a superpartner whose spin differs by one half unit. The fermionic superpartners of bosons are conventionally named by adding the suffix ‘-ino’, such as the gluino, wino, bino, and higgsino, while the bosonic superpartners of fermions are named by adding the prefix ‘s-’, such as squarks and sleptons.

If supersymmetry were exact, particles and their superpartners would have equal masses. Since no such superpartners have been observed, SUSY must be broken if it is realised in nature.

One of the main motivations for SUSY is the hierarchy problem. Loop corrections to the Higgs mass from SM particles can be cancelled by corresponding contributions from their superpartners, reducing the sensitivity of the electroweak scale to very high energy scales [37]. In models with conserved R -parity, supersymmetric particles are produced in pairs and the lightest supersymmetric particle (LSP) is stable [38]. If the LSP is electrically neutral and weakly interacting, such as a neutralino in many SUSY scenarios, it can provide a dark matter candidate [39, 40]. SUSY can also improve the unification of gauge couplings at high energy, since the additional superpartners modify the renormalisation-group evolution of the gauge couplings [41, 42]. However, no direct evidence for supersymmetric particles has yet been observed, and the simplest low-scale SUSY scenarios are increasingly constrained by collider searches [43].

2.3.2 Axions and the strong CP problem

The Peccei–Quinn mechanism provides a dynamical solution to the strong CP problem [44, 45]. It introduces a new global chiral symmetry, usually denoted as the Peccei–Quinn symmetry $U(1)_{\text{PQ}}$, which is spontaneously broken. The associated pseudo-Nambu–Goldstone boson is the axion [46, 47]. The axion field dynamically relaxes the effective value of $\bar{\theta}$ to zero, thereby explaining the absence of observable strong CP violation. In many models, axions or axion-like particles can also contribute to the dark matter density of the universe [48].

2.3.3 Leptoquarks

Leptoquarks provide another well-motivated class of particles beyond the Standard Model. They are hypothetical bosons that couple directly to a quark and a lepton, thereby connecting the quark and lepton sectors. Such particles appear naturally in several extensions of the SM, including models based on quark–lepton unification, such as the Pati–Salam framework [49].

Leptoquarks are particularly interesting because they can affect semileptonic processes. This makes them relevant to the flavour anomalies observed in several B -meson decay measurements, where the experimental results have shown tensions with the SM predictions in observables testing lepton-flavour universality.

Among the possible leptoquark scenarios, the vector singlet leptoquark U_1 is especially well motivated for charged-current anomalies such as $b \rightarrow c \tau \nu$. This motivates collider

⁷⁴⁷ searches for final states containing hadronically decaying τ leptons. The following chapter
⁷⁴⁸ introduces the leptoquark models in more detail and the leptoquark searches at the LHC.

Chapter 3

Leptoquark Search at the LHC

3.1 Leptoquark Models and Phenomenology

3.1.1 General properties of leptoquarks

Leptoquarks (LQs) are hypothetical colour-charged bosons that couple directly to a quark and a lepton and therefore mediate interactions connecting the quark and lepton sectors. Depending on their spin, leptoquarks can be classified as scalar leptoquarks with spin zero or vector leptoquarks with spin one. They appear in several theoretical frameworks, including grand unified theories such as $SU(5)$ and $SO(10)$, quark–lepton unification models such as the Pati–Salam model, composite models, and models motivated by flavour anomalies [50, 49, 51].

Leptoquarks are particularly interesting in the context of semileptonic flavour physics. Their tree-level exchange can generate effective four-fermion interactions involving two quark fields and two lepton fields. Such interactions can modify flavour observables in processes such as $b \rightarrow c\tau\nu$ and $b \rightarrow s\ell^+\ell^-$. In addition, the flavour structure of LQ couplings is model dependent and can be arranged to favour interactions with third-generation fermions. This feature is motivated by the distinctive role of the top quark, bottom quark, and τ -lepton in many flavour-motivated scenarios.

At the LHC, leptoquarks can be searched for directly. Since they carry colour charge, they can be pair-produced through QCD interactions. Additional production modes, such as single production or non-resonant exchange, are also possible, depending on the LQ coupling to SM quarks and leptons. Therefore, the LQ searches in ATLAS provide information complementary to low-energy flavour measurements: flavour observables probe virtual effects through precision measurements, while the search at pp collider probes direct LQ production or high-energy virtual exchange.

3.1.2 The simplified U_1 leptoquark model

Several leptoquark representations have been studied in connection with semileptonic B -meson anomalies, including the scalar singlet S_1 and the scalar doublet R_2 . In this thesis, however, the benchmark interpretation is based on the vector singlet leptoquark U_1 . The U_1 benchmark is well motivated because its left-handed coupling to quark and lepton doublets can generate the SM-like vector operator for $b \rightarrow c\tau\nu$. In addition, flavour structures with enhanced couplings to third-generation fermions naturally lead to final states involving τ -leptons, neutrinos, and jets.

The U_1 leptoquark transforms as $U_1^\mu \sim (3, 1, 2/3)$ under the SM gauge group. It is therefore a colour-triplet, weak-singlet vector boson with hypercharge $Y = 2/3$. Since it is an $SU(2)$ singlet, its electric charge is $Q = T^3 + Y = 2/3$.

A commonly used simplified-model Lagrangian [52] for U_1 is

$$\begin{aligned} \mathcal{L}_U = & -\frac{1}{2}U_{1\mu\nu}^\dagger U_1^{\mu\nu} + M_U^2 U_{1\mu}^\dagger U_1^\mu \\ & -ig_c(1 - \kappa_c)U_{1\mu}^\dagger T^a U_{1\nu} G^{a\mu\nu} - i\frac{2}{3}g_Y(1 - \kappa_Y)U_{1\mu}^\dagger U_{1\nu} B^{\mu\nu} \\ & + (U_1^\mu J_\mu + \text{h.c.}). \end{aligned} \quad (3.1)$$

Here U_1^μ is the vector leptoquark field and M_U is the leptoquark mass. The first term is the kinetic term, while the second term is the mass term. The field-strength tensor of U_1 is defined as

$$U_{1\mu\nu} = D_\mu U_{1\nu} - D_\nu U_{1\mu}, \quad (3.2)$$

with the covariant derivative

$$D_\mu = \partial_\mu - ig_c G_\mu^a T^a - i\frac{2}{3}g_Y B_\mu. \quad (3.3)$$

Here g_c and g_Y are the colour and hypercharge gauge couplings. The fields G_μ^a and B_μ are the gluon and hypercharge gauge fields, and $G^{a\mu\nu}$ and $B^{\mu\nu}$ are their corresponding field-strength tensors. The matrices T^a are the generators of $SU(3)$. For a colour-triplet field, they are given by $T^a = \lambda^a/2$, where λ^a are the Gell-Mann matrices and $a = 1, \dots, 8$.

The parameters κ_c and κ_Y control the interactions of the vector leptoquark with the SM gauge fields. $\kappa_c = \kappa_Y = 0$ corresponds to the Yang–Mills scenario, whereas $\kappa_c = \kappa_Y = 1$ corresponds to the minimal-coupling scenario [53]. The Yang–Mills scenario is adopted throughout this thesis.

The last term in Eq. (3.1) describes the interaction of U_1 with SM fermions through the current J_μ . It is given by

$$J_\mu = \frac{g_U}{\sqrt{2}} [\beta_L^{i\alpha} (\bar{q}_L^i \gamma_\mu \ell_L^\alpha) + \beta_R^{i\alpha} (\bar{d}_R^i \gamma_\mu e_R^\alpha)]. \quad (3.4)$$

Here $i = 1, 2, 3$ labels the quark generation and $\alpha = 1, 2, 3$ labels the lepton generation. q_L^i and ℓ_L^α are left-handed quark and lepton doublets, while d_R^i and e_R^α are right-handed down-type quark and charged-lepton singlets. The parameter g_U denotes the overall strength of the U_1 coupling to fermions. The matrices $\beta_L^{i\alpha}$ and $\beta_R^{i\alpha}$ are complex 3×3 flavour-coupling matrices for the left- and right-handed interactions, respectively. The strength of a given flavour coupling is therefore proportional to $g_U \beta_{L,R}^{i\alpha}$.

The down-quark and charged-lepton mass eigenstate basis is adopted for the $SU(2)$ multiplets:

$$q_L^i = \begin{pmatrix} V_{ji}^* u_L^j \\ d_L^i \end{pmatrix}, \quad \ell_L^\alpha = \begin{pmatrix} \nu_L^\alpha \\ e_L^\alpha \end{pmatrix}, \quad (3.5)$$

where V is the CKM matrix.

Expanding the left-handed current from Eq. 3.4 gives

$$\bar{q}_L^i \gamma_\mu \ell_L^\alpha = V_{ji} \bar{u}_L^j \gamma_\mu \nu_L^\alpha + \bar{d}_L^i \gamma_\mu e_L^\alpha. \quad (3.6)$$

This expression shows that a single left-handed flavour coupling $\beta_L^{i\alpha}$ gives two related interactions. One involves an up-type quark together with a neutrino, while the other involves a down-type quark together with a charged lepton.

The simplified U_1 model allows general flavour couplings to different quark and lepton generations. Meanwhile, we focus on the charged-current $b \rightarrow c \tau \nu$ transition and the signatures containing a τ -lepton and a neutrino. Therefore, the couplings to second-generation leptons in our benchmark are set to zero, which is $\beta_L^{22} = \beta_L^{32} = 0$, and only the couplings involving third-generation leptons are retained. Finally, the flavour-coupling matrices are taken to be

$$\beta_L = \begin{pmatrix} 0 & 0 & \beta_L^{13} \\ 0 & 0 & \beta_L^{23} \\ 0 & 0 & \beta_L^{33} \end{pmatrix}, \quad \beta_R = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \beta_R^{33} \end{pmatrix}. \quad (3.7)$$

The third-generation coupling is normalised to $\beta_L^{33} = 1$, while g_U remains a free parameter controlling the overall fermionic coupling strength.

Within this benchmark, β_L^{23} controls the $s\tau$ and $c\nu_\tau$ coupling through the CKM factor V_{cs} , while β_L^{33} controls the $b\tau$ coupling and contributes $c\nu_\tau$ coupling as $V_{cb}\beta_L^{33}$.

The heavy U_1 field can be integrated out at the matching scale $\mu \sim M_U$. At tree level, its exchange generates four-fermion operators involving left- and right-handed quark and lepton fields [54]. The corresponding effective Lagrangian can be written as

$$\mathcal{L}_{\text{eff}}^U = -\frac{2}{v^2} \left[C_{LL}^{ij\alpha\beta} \mathcal{O}_{LL}^{ij\alpha\beta} + C_{RR}^{ij\alpha\beta} \mathcal{O}_{RR}^{ij\alpha\beta} + \left(C_{LR}^{ij\alpha\beta} \mathcal{O}_{LR}^{ij\alpha\beta} + \text{h.c.} \right) \right]. \quad (3.8)$$

The four-fermion operators are defined as

$$\mathcal{O}_{LL}^{ij\alpha\beta} = (\bar{q}_L^i \gamma_\mu \ell_L^\alpha) (\bar{\ell}_L^\beta \gamma^\mu q_L^j), \quad (3.9)$$

$$\mathcal{O}_{RR}^{ij\alpha\beta} = (\bar{d}_R^i \gamma_\mu e_R^\alpha) (\bar{e}_R^\beta \gamma^\mu d_R^j), \quad (3.10)$$

$$\mathcal{O}_{LR}^{ij\alpha\beta} = (\bar{q}_L^i \gamma_\mu \ell_L^\alpha) (\bar{e}_R^\beta \gamma^\mu d_R^j). \quad (3.11)$$

The Wilson coefficients are

$$C_{LL}^{ij\alpha\beta} = C_U \beta_L^{i\alpha} (\beta_L^{j\beta})^*, \quad (3.12)$$

$$C_{RR}^{ij\alpha\beta} = C_U \beta_R^{i\alpha} (\beta_R^{j\beta})^*, \quad (3.13)$$

$$C_{LR}^{ij\alpha\beta} = C_U \beta_L^{i\alpha} (\beta_R^{j\beta})^*. \quad (3.14)$$

The overall coefficient is defined as

$$C_U = \frac{g_U^2 v^2}{4M_U^2}, \quad (3.15)$$

where $v \simeq 246$ GeV is the Higgs vacuum expectation value. The coefficient C_U sets the overall strength of the effective interaction and scales as g_U^2/M_U^2 , while the flavour dependence is encoded in the coupling matrices β_L and β_R .

The effective Lagrangian for the $b \rightarrow c\tau\nu$ transition can be written as:

$$\begin{aligned} \mathcal{L}_{\text{eff}}^{b \rightarrow c\tau\nu} = & -\frac{4G_F}{\sqrt{2}} V_{cb} \left[(1 + C_{LL}^c) (\bar{c}_L \gamma_\mu b_L) (\bar{\tau}_L \gamma^\mu \nu_{\tau L}) \right. \\ & \left. - 2C_{LR}^c (\bar{c}_L b_R) (\bar{\tau}_R \nu_{\tau L}) \right], \end{aligned} \quad (3.16)$$

where G_F is the Fermi constant. The coefficient C_{LL}^c modifies the SM-like left-handed vector interaction, while C_{LR}^c is associated with the scalar operator arising from the combination of left- and right-handed fermion couplings of the U_1 leptoquark.

For the flavour structure introduced above, the corresponding coefficients are

$$C_{LL}^c = C_U \frac{(\beta_L^{33})^*}{V_{cb}} (V_{cd}\beta_L^{13} + V_{cs}\beta_L^{23} + V_{cb}\beta_L^{33}), \quad (3.17)$$

$$C_{LR}^c = C_U \frac{(\beta_R^{33})^*}{V_{cb}} (V_{cd}\beta_L^{13} + V_{cs}\beta_L^{23} + V_{cb}\beta_L^{33}), \quad (3.18)$$

Using the normalisation $\beta_L^{33} = 1$ and neglecting the small first-generation coupling β_L^{13} , the coefficients reduce to¹

$$C_{LL}^c \simeq C_U \left(1 + \frac{V_{cs}}{V_{cb}} \beta_L^{23} \right), \quad (3.19)$$

$$C_{LR}^c \simeq C_U (\beta_R^{33})^* \left(1 + \frac{V_{cs}}{V_{cb}} \beta_L^{23} \right). \quad (3.20)$$

3.1.3 Low-energy fit and benchmark scenarios

The Wilson coefficients C_{LL}^c and C_{LR}^c introduced above can be constrained using measurements of semileptonic charged-current B -meson decays. Ref. [54] performed a combined fit to low-energy $b \rightarrow c \tau \nu$ observables and determines the preferred region in the (C_{LL}^c, C_{LR}^c) plane. The fit result is shown in Fig. 3.1. The Wilson coefficients in this figure are evolved to a common reference scale of 1 TeV to facilitate comparison with high-energy observables. The best-fit point is $(C_{LL}^c, C_{LR}^c) = (0.05, -0.02)$, and the preferred region is compatible with both pure-left-handed ($\beta_R^{33} = 0$) and sizeable right-handed ($\beta_R^{33} = -1$) scenarios. Although the sign of β_R^{33} affects the low-energy relation between C_{LL}^c and C_{LR}^c , the collider predictions used in this thesis are identical for $\beta_R^{33} = +1$ and -1 . The new-physics amplitudes with different fermion chiralities do not interfere, and the amplitude proportional to a product of left- and right-handed couplings does not interfere with the SM amplitude. The collider interpretation therefore depends only on $|\beta_R^{33}|$.

Pure-left-handed scenario. For $\beta_R^{33} = 0$, Eq. 3.20 gives $C_{LR}^c = 0$. Therefore, the U_1 leptoquark contribution to $b \rightarrow c \tau \nu$ is described only by the left-handed vector operator, as given by Eq. 3.19. This is the primary benchmark scenario considered in this thesis.

¹The small first-generation coupling $\beta_L^{13} = (V_{td}^*/V_{ts}^*)\beta_L^{23} \sim 0.2\beta_L^{23}$.

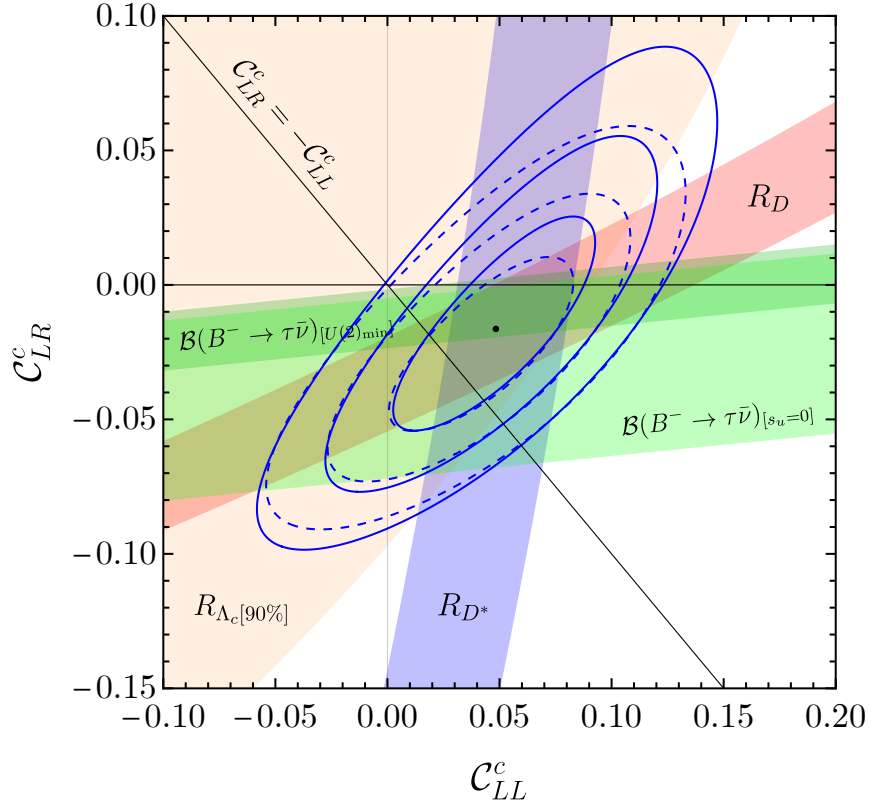


Fig. 3.1 The result in (C_{LL}^c, C_{LR}^c) plane obtained from the fit to low-energy $b \rightarrow c \tau \nu$ observables. The blue solid ellipses correspond to the 1, 2 and 3 σ contours. The blue dotted lines are obtained including also the constraint from $\mathcal{B}(B_u^- \rightarrow \tau \bar{\nu})$. The black point indicates the best-fit point of the fit, $(C_{LL}^c, C_{LR}^c) = (0.05, -0.02)$. The coloured bands show the $\Delta\chi^2 = 1$ regions preferred by the individual observables. The pure-left-handed scenario ($\beta_R^{33} = 0$) and the equal magnitude right-handed scenario ($\beta_R^{33} = -1$) are given by the black solid lines [54].

The corresponding B -anomaly-favoured region can be obtained from the intersection of the low-energy fit with the line $C_{LR}^c = 0$ in Fig. 3.1.

Right-handed scenario. The second benchmark is obtained by allowing a sizeable right-handed third-generation coupling. For $\beta_R^{33} = -1$, the relation between the Wilson coefficients becomes

$$C_{LR}^c = -C_{LL}^c. \quad (3.21)$$

This benchmark therefore corresponds to the diagonal line in Fig. 3.1, where the right-handed third-generation coupling has the same magnitude as the left-handed coupling.

The two scenarios are described by the parameter space (M_U, g_U, β_L^{23}) with different fixed values of β_R^{33} . Particular emphasis is placed on β_L^{23} , since charged-current low-energy data alone do not provide a stringent constraint on this coupling. For fixed M_U and g_U , β_L^{23} has a strong impact on C_{LL}^c because its contribution is enhanced by the CKM ratio V_{cs}/V_{cb} .

Indeed, one can constrain the size of β_L^{23} by considering possible ultraviolet (UV) completions of the simplified U_1 model. While the simplified model parametrises the relevant interactions directly in terms of the couplings β_L^{ij} , the UV completion provides a more fundamental description of their origin. One example is the “4321” model, which is based on the extended gauge group $\mathcal{G}_{4321} = SU(4) \times SU(3)' \times SU(2)_L \times U(1)'$. In the flavour structure considered in this model, the U_1 couplings are non-universal, with $|\beta_L^{23}| < |\beta_L^{33}|$ [52].

Additional constraints can be introduced when considering $\Delta F = 2$ observables. These are processes in which a quark-flavour quantum number changes by two units, including neutral-meson mixing such as $B_s - \bar{B}_s$, $K - \bar{K}$, and $D - \bar{D}$ mixing. In a given UV completion, the flavour-symmetry breaking that generates non-universal U_1 couplings may also be transmitted to the neutral-current sector, inducing flavour-changing neutral currents (FCNC) and consequently additional contributions to $\Delta F = 2$ processes. The strong experimental constraints on neutral-meson mixing therefore restrict the allowed flavour-violating structure of the UV model and indirectly constrain β_L^{23} .

Combining such considerations of the UV flavour structure with constraints from $\Delta F = 2$ observables, Ref. [54] identifies

$$\beta_L^{23} \in [0.06, 0.16] \quad (3.22)$$

as a theoretically motivated range for experimental searches. **However, this constraint should not be interpreted as a model-independent experimental bound on β_L^{23} .** Additional particles, flavour symmetries, or cancellation mechanisms may suppress flavour-

changing transitions and thereby allow larger values of β_L^{23} . Therefore, large values of β_L^{23} remain a possibility within the U_1 model, and it is important to test this complementary region directly.

3.1.4 Leptoquark searches at the LHC

Leptoquarks can be searched for at the LHC through several production modes, including pair production, resonant single production, and non-resonant t -channel exchange. Pair production is dominated by QCD interactions. For a fixed κ choice, its cross section depends strongly on M_U but weakly on the couplings g_U and $\beta^{i\alpha}$. Pair-production searches therefore provide relatively robust constraints on the leptoquark mass.

ATLAS and CMS have performed searches for pair-produced leptoquarks coupling to third-generation fermions in final states containing τ -leptons, b -jets, and, depending on the decay mode, missing transverse momentum [55, 56]. For the vector U_1 , the mass limit extends to approximately 1.5–1.8 TeV as shown in Fig. 3.3. The limit also depends on the choice of the U_1 -gluon coupling, such as the Yang–Mills or minimal-coupling scenario, as well as on the decay pattern determined by the assumed fermionic couplings.

For single production, the cross section also depends strongly on the leptoquark–fermion couplings. Searches for singly produced leptoquarks decaying into $b\tau$ have been performed in the $b\tau\tau$ final state, where the leptoquark is produced in association with a τ -lepton as shown in Fig. 3.2a [55, 57]. In addition, virtual t -channel U_1 exchange can contribute to final states such as $\tau\tau$ or $\tau\nu$, enhancing the event yields in the high-mass region. ATLAS and CMS have therefore searched for non-resonant leptoquark contributions in high-mass $\tau\tau$ production [58, 59]. These searches can remain sensitive to masses beyond the reach of the direct pair-production searches, particularly for large g_U , since the sensitivity is approximately governed by g_U/M_U .

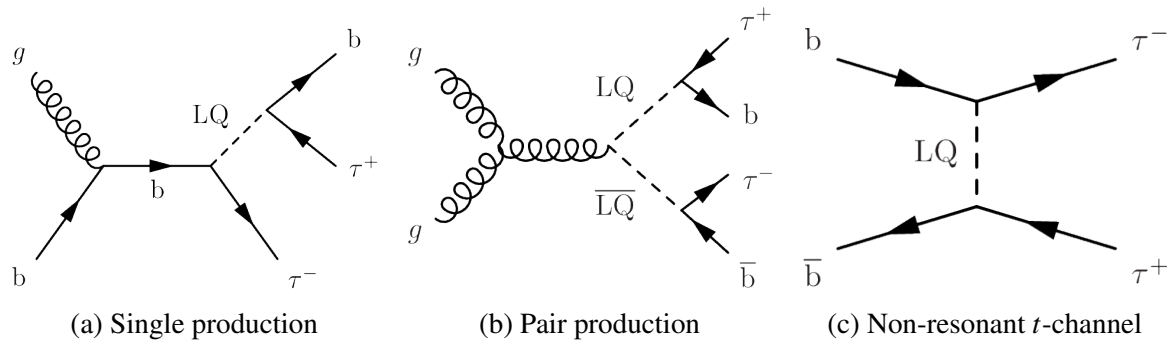


Fig. 3.2 Example Feynman diagrams of LQ production and exchange process for $\tau\tau(+b)$ searches.

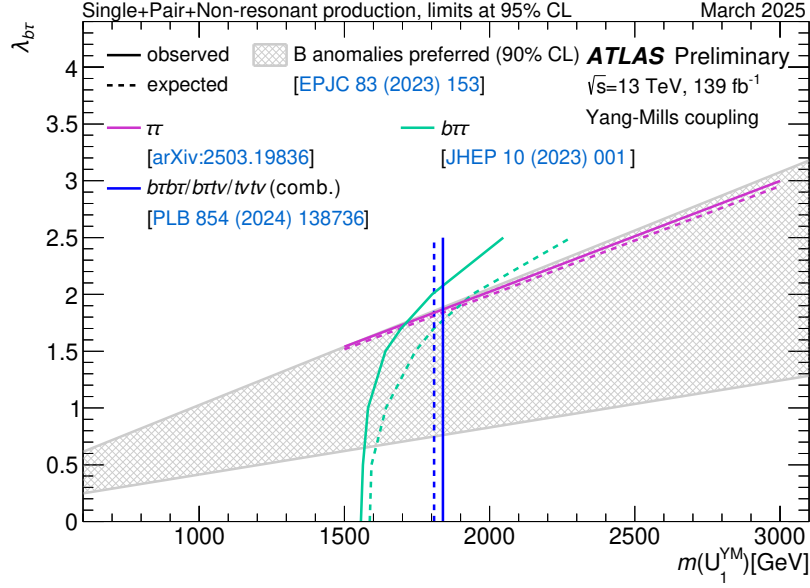


Fig. 3.3 Summary of the observed and expected 95% CL constraints on the U_1 vector-leptoquark coupling $\lambda_{b\tau}$ as a function of the leptoquark mass in the Yang–Mills coupling scenario, with $\beta_R^{33} = 0$. Pair-production searches dominate the direct mass constraint at lower masses, while single-production and $\tau\tau$ searches provide coupling-dependent sensitivity at larger masses. Here $\lambda_{b\tau} = g_U/\sqrt{2}$ for $\beta_L^{33} = 1$ [60].

On the other hand, the existing $\tau\tau(+b)$ searches generally assume β_L^{23} to be small or neglect its contribution, motivated by the flavour assumptions discussed in Sec. 3.1.3. Consequently, scenarios with a sizeable $c\nu_\tau$ coupling, and in particular the large- β_L^{23} region, have not been systematically explored at the LHC.

The $\tau\nu$ final state provides a complementary probe of the $b\tau$ and $c\nu_\tau$ coupling. CMS has performed an inclusive search for events with a τ_{had} and large missing transverse momentum, including an interpretation in terms of leptoquark exchange [61], but without an explicit b -jet requirement. This thesis presents a dedicated search in the $\tau\nu(+b)$ final state. Prior to the analysis, no dedicated ATLAS or CMS leptoquark search had targeted this topology. The $\tau\nu(+b)$ search presented here is therefore complementary to the existing $\tau\tau(+b)$ searches: it probes both the $b\tau$ and $c\nu_\tau$ couplings and extends the collider interpretation to large values of β_L^{23} . The corresponding benchmark and target signatures are introduced in the following subsection.

3.1.5 Signal model and target signatures

In this thesis, we focus on the simplified U_1 model introduced in Sec. 3.1.2 in the Yang–Mills scenario, corresponding to $\kappa_c = \kappa_Y = 0$ in Eq. 3.1. The third-generation coupling is normalised to $\beta_L^{33} = 1$, while the primary interpretation assumes a pure-left-handed scenario ($\beta_R^{33} = 0$). An additional interpretation including a right-handed coupling is presented with the final results. Therefore, the signal is characterised by (M_U, g_U, β_L^{23}) .

Figure 3.4 shows representative U_1 signal diagrams targeted by the analysis. Both resonant single production and non-resonant t -channel exchange are considered, and each is divided according to the presence or absence of a b -tagged jet. This results in four target signatures: resonant $\tau\nu + b$, non-resonant $\tau\nu + b$, resonant $\tau\nu + 0b$, and non-resonant $\tau\nu + 0b$.

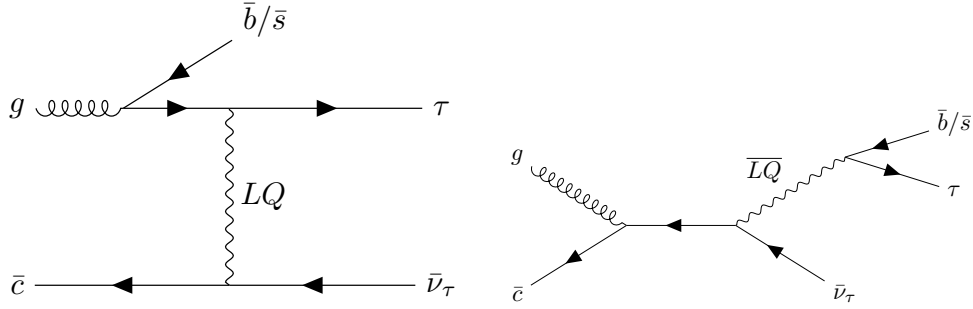


Fig. 3.4 Signal diagrams targeted in this analysis: non-resonant t -channel exchange (left) and resonant single production (right). The final-state jet can originate from a b - or s -quark.

The b -jet requirement provides strong discrimination against the dominant SM $\tau\nu + \text{jets}$ background, since genuine SM production of a $\tau\nu$ system in association with a b -quark is CKM suppressed [62]. The $\tau\nu + b$ categories are therefore particularly sensitive to the combination of the $b\tau$ and $c\nu_\tau$ interactions.

The zero- b categories extend the sensitivity to the large β_L^{23} region explored in this thesis. As β_L^{23} increases, the $s\tau$ interaction also becomes sizeable, producing signal events without a final-state b -quark. This contribution is particularly relevant for resonant production, where an on-shell U_1 can decay into $s\tau$. The four categories therefore provide complementary sensitivity to the different flavour and production configurations of the U_1 signal.

Chapter 4

LHC-ATLAS Experiment

4.1 Large Hadron Collider

The European Organisation for Nuclear Research (CERN) is an international particle-physics laboratory located near Geneva on the border between France and Switzerland. Since its foundation in 1954, CERN has developed and operated a chain of accelerators and experimental facilities to study the fundamental constituents of matter and their interactions [63]. The Large Hadron Collider (LHC) is the largest and highest-energy particle collider in the CERN accelerator complex. It is installed in the former Large Electron–Positron Collider (LEP) tunnel, which has a circumference of about 27 km and lies between approximately 45 m and 170 m below the surface [64, 65]. A schematic overview of the LHC ring and its interaction points is shown in Fig. 4.1.

The LHC was proposed as the next major collider after LEP. The first circulating beams were achieved in 2008, and the first physics run began in 2010. The discovery of a new boson compatible with the Standard Model (SM) Higgs boson by the ATLAS and CMS experiments in 2012 marked one of the central achievements of the LHC physics programme [9, 10]. Beyond Higgs-boson measurements and precision studies of SM processes, a major purpose of the LHC is to search for new particles and interactions beyond the SM.

The LHC collides protons at high centre-of-mass energies in order to extend the kinematic reach for the production of heavy particles. For two proton beams with equal beam energy E_{beam} , the proton–proton centre-of-mass energy is approximately $\sqrt{s} = 2E_{\text{beam}}$. However, the hard-scattering process in a proton–proton (pp) collision occurs between partons, namely quarks and gluons, which carry only fractions of the proton momenta. The partonic centre-of-mass energy therefore varies from event to event and is generally lower than the nominal pp collision energy. Increasing $\sqrt{s}$ consequently extends the accessible mass range for new heavy particles.

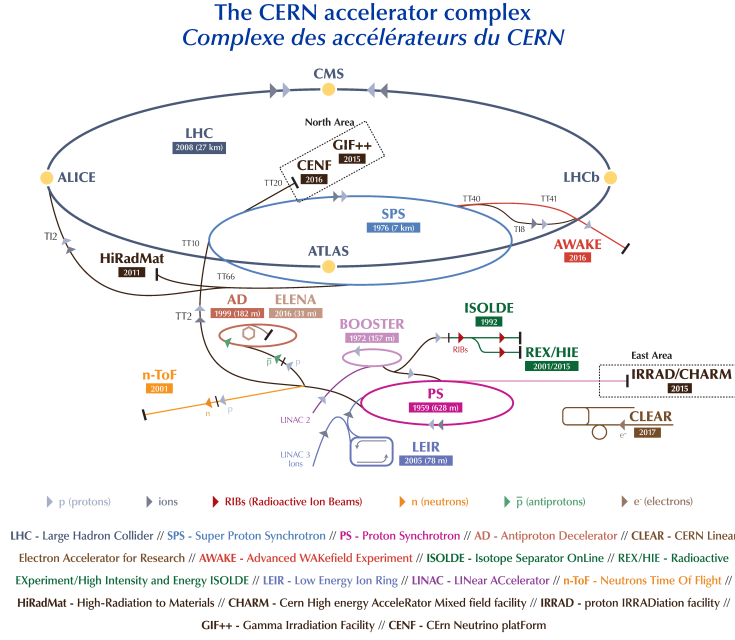


Fig. 4.1 Schematic layout of the CERN accelerator complex, including the LHC ring and its main interaction regions [66].

Another key machine parameter is the luminosity, which quantifies the collision rate per unit cross section. For a process with cross section σ , the production rate is

$$R = \mathcal{L}\sigma, \quad (4.1)$$

where $\mathcal{L}$ is the instantaneous luminosity. For two Gaussian bunches colliding with frequency f_{coll} , the luminosity can be expressed as [67]

$$\mathcal{L} = f_{\text{coll}} \frac{n_1 n_2}{4\pi\sigma_x^* \sigma_y^*} \mathcal{F}, \quad (4.2)$$

where n_1 and n_2 are the numbers of particles in the two colliding bunches, σ_x^* and σ_y^* are the RMS transverse beam sizes in the horizontal and vertical directions at the interaction point, and $\mathcal{F}$ is a geometrical reduction factor accounting for effects such as the crossing angle. The integrated luminosity,

$$\mathcal{L}_{\text{int}} = \int \mathcal{L} dt, \quad (4.3)$$

determines the total expected number of produced events by $N = \sigma \mathcal{L}_{\text{int}}$. High instantaneous luminosity increases the event rate, while high integrated luminosity improves the sensitivity to rare processes with small production cross sections.

Four major experiments are installed around the LHC ring. ATLAS, located at Point 1, and CMS, located at Point 5, are general-purpose detectors designed for a broad physics programme including Higgs-boson measurements, precision SM studies, and searches for new particles. ALICE, located at Point 2, is optimised for heavy-ion physics and the study of strongly interacting matter at high energy density. LHCb, located at Point 8, is designed for precision studies of heavy-flavour hadrons and flavour-changing processes [64]. The analysis presented in this thesis uses data recorded by the ATLAS detector.

The LHC physics programme is organised into run periods. Run 1 took place from 2010 to 2012, with pp collision data recorded at $\sqrt{s} = 7$ TeV and $\sqrt{s} = 8$ TeV. Run 2 took place from 2015 to 2018 at $\sqrt{s} = 13$ TeV. After standard data-quality selections, the full ATLAS Run 2 pp dataset corresponds to an integrated luminosity of $140.1 \pm 1.2 \text{ fb}^{-1}$, with a relative uncertainty of 0.83% [68]. Run 3 began in 2022 after the second long shutdown and operated at the higher centre-of-mass energy of $\sqrt{s} = 13.6$ TeV [69]. The Run 2 dataset is used for the leptoquark search presented in this thesis.

An important consequence of high-luminosity operation at the LHC is pile-up. The proton beams are divided into bunches, and a bunch crossing occurs when bunches from the two beams pass through the interaction point. Several pp interactions can occur in a single bunch crossing. The interaction corresponding to the hard-scattering process of interest is referred to as the hard-scatter interaction, while additional inelastic pp interactions in the same bunch crossing constitute in-time pile-up. Detector signals may also originate from interactions in neighbouring bunch crossings, depending on the detector response time. This contribution is referred to as out-of-time pile-up.

The mean number of inelastic pp collisions per bunch crossing is commonly denoted by μ . During Run 2, the nominal bunch spacing was 25 ns, corresponding to a maximum bunch-crossing frequency of 40 MHz. The luminosity-weighted mean value over the full Run 2 dataset was $\langle \mu \rangle = 33.7$, corresponding to approximately 34 interactions per bunch crossing [70].

Pile-up has an important impact on event reconstruction. It produces extra charged-particle tracks, calorimeter energy deposits, and reconstructed jets. These contributions can affect the reconstruction of the primary vertex, jet energies, τ_{had} candidates, flavour-tagged jets, and missing transverse momentum. Pile-up effects must therefore be modelled in simulated samples and mitigated through dedicated reconstruction and calibration procedures.

4.2 ATLAS experiment

4.2.1 Overview

ATLAS, which stands for A Toroidal LHC ApparatuS, is a detector located at Point 1 of the Large Hadron Collider. It was designed to exploit the full physics potential of the LHC, including precision measurements of SM processes, studies of the Higgs boson, and searches for new physics beyond the SM. The detector surrounds the nominal interaction point and provides nearly full solid-angle coverage, allowing the reconstruction of a wide range of final-state particles produced in proton–proton collisions.

The ATLAS detector is approximately 44 m long, 25 m in diameter, and has a total weight of about 7000 tonnes [71]. It has a cylindrical, forward–backward symmetric layout centred on the interaction point. A schematic view of the detector is shown in Figure 4.2.

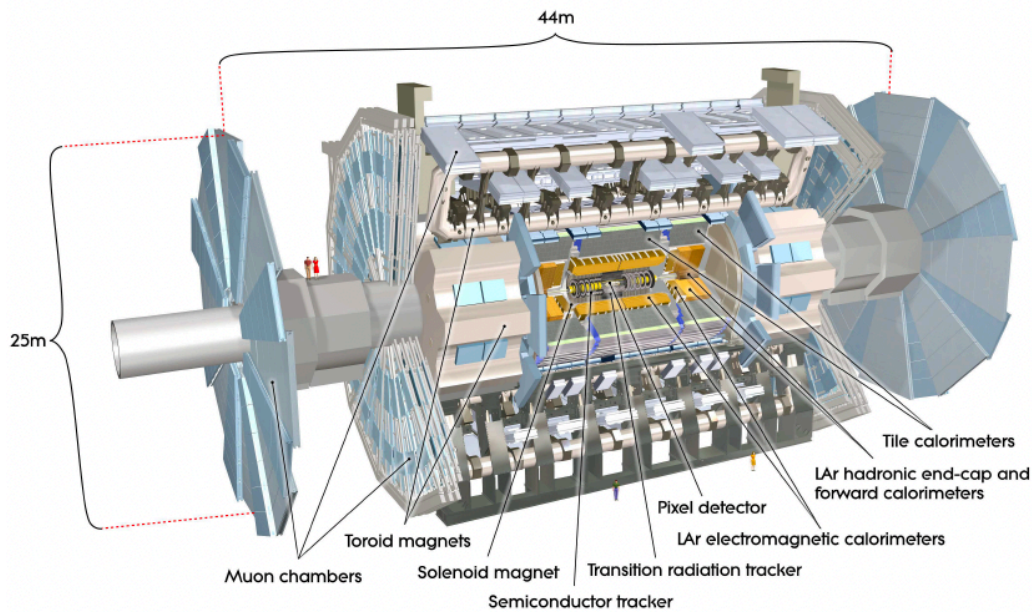


Fig. 4.2 Schematic view of the ATLAS detector and its subdetector systems [71].

Starting from the beam line and moving outwards, the main detector systems are the Inner Detector (ID), the calorimeters, and the Muon Spectrometer (MS):

- **Inner Detector (ID)** is immersed in a 2 T axial magnetic field provided by a thin superconducting solenoid and is used to reconstruct charged-particle trajectories close to the interaction point.
- **Calorimeters** surround the ID and measure the energies of electrons, photons, hadrons, τ_{had} candidates, and jets.

- ***Muon Spectrometer (MS)*** forms the outermost major detector system and measures muons in the magnetic field generated by large superconducting toroids.

The magnetic fields provided by the solenoid and toroids allow the momentum and charge of charged particles to be determined from the curvature of their trajectories. The trigger and data acquisition system selects potentially interesting events from the high LHC bunch-crossing rate for permanent storage and offline reconstruction.

The detector configuration described in this chapter corresponds primarily to Run 2. Several detector and trigger upgrades were installed for Run 3 to cope with higher luminosity conditions, but these upgrades are not used in the analysis presented in this thesis [72].

4.2.2 Coordinate system

ATLAS uses a right-handed coordinate system centred at the nominal interaction point. The z -axis is defined along the beam direction, the x -axis points from the interaction point towards the centre of the LHC ring, and the y -axis points upwards [71]. Cylindrical coordinates (r, ϕ) are used in the plane transverse to the beam, where r is the distance from the beam axis and ϕ is the azimuthal angle around the z -axis.

The polar angle θ is measured with respect to the positive z -axis. The pseudorapidity η is defined as

$$\eta = -\ln \tan \left(\frac{\theta}{2} \right). \quad (4.4)$$

The transverse momentum is defined as $p_T = p \sin \theta$, and the transverse energy is defined similarly. The angular separation between two reconstructed objects is commonly expressed as

$$\Delta R = \sqrt{(\Delta \eta)^2 + (\Delta \phi)^2}. \quad (4.5)$$

These quantities are used throughout this thesis to describe object kinematics, angular separations, overlap removal, and event selections. Figure 4.3 illustrates the coordinate system used in ATLAS.

4.2.3 Magnet system

The ATLAS magnet system consists of four large superconducting magnets: a central solenoid, one barrel toroid, and two end-cap toroids, as shown in Figure 4.4. The central solenoid surrounds the Inner Detector with a length of approximately 5.3 m and provides an

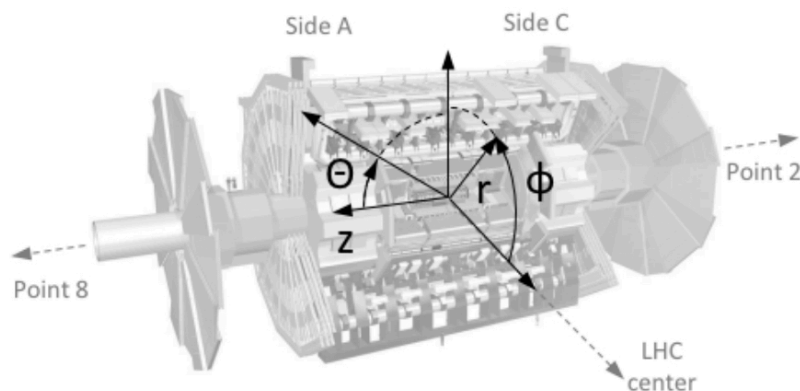


Fig. 4.3 The coordinate system used in ATLAS [73].

axial magnetic field of 2 T for charged-particle momentum measurements in the tracking system. The solenoid is designed to minimise the amount of inactive material in front of the electromagnetic calorimeter.

The toroid system provides magnetic fields of approximately 0.5 T and 1 T in the barrel and end-cap regions of the Muon Spectrometer, respectively. The barrel toroid surrounds the calorimeter system in the central region, while the two end-cap toroids provide bending power in the forward regions. The toroidal field bends muon trajectories, allowing their momenta to be measured using the precision tracking chambers of the MS. The air-core design of the toroids reduces multiple scattering and helps preserve the momentum resolution for high- p_T muons [71]. The combination of the solenoid and toroid systems allows charged-particle momenta to be measured independently in the ID and, for muons, in the MS.

4.2.4 Inner Detector

The Inner Detector (ID) is the innermost tracking system of ATLAS and provides tracking for charged particles within $|\eta| < 2.5$ [71]. It is immersed in the 2 T magnetic field of the central solenoid and records local position measurements, called hits, along their trajectories. During Run 2, it consisted of the pixel detector, the Semiconductor Tracker (SCT), and the Transition Radiation Tracker (TRT). The Insertable B-Layer (IBL) is installed before Run 2 as the innermost pixel layer [74]. Figure 4.5 shows a transverse view of the Inner Detector.

- *Insertable B-Layer (IBL) and Pixel detector*

The pixel detector provides high-granularity measurements close to the beam line. Its precise spatial resolution is essential for reconstructing tracks near the interaction point and for identifying displaced vertices from the decays of long-lived particles such

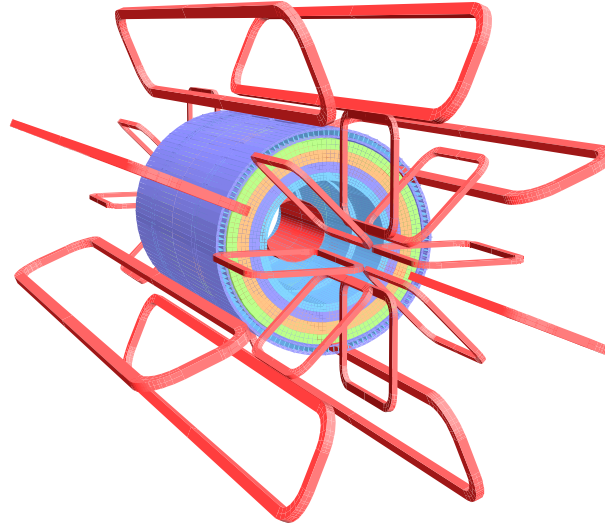


Fig. 4.4 Layout of the ATLAS superconducting magnet system, including the central solenoid, barrel toroid, and end-cap toroids [71].

1108 as b - and c -hadrons. The IBL further improves the impact-parameter resolution and
 1109 enhances the robustness of tracking and flavour tagging under high pile-up conditions.

1110 • ***Semiconductor Tracker (SCT)***

1111 The SCT surrounds the pixel detector and forms the intermediate precision-tracking
 1112 system of the ID. It is a silicon microstrip detector covering $|\eta| < 2.5$, consisting of four
 1113 cylindrical barrel layers and two end-caps with nine disks on each side. Each module
 1114 contains silicon strip sensors arranged back-to-back with a small stereo angle, allowing
 1115 two-dimensional position information to be reconstructed. The SCT typically provides
 1116 four space-point measurements per track, improving pattern recognition, momentum
 1117 reconstruction, and vertex reconstruction when combined with the high-granularity
 1118 pixel measurements.

1119 • ***Transition Radiation Tracker (TRT)***

1120 The TRT is the outermost tracking subsystem of the ID and provides tracking informa-
 1121 tion at larger radii, covering approximately $|\eta| < 2.0$. It is based on thin straw-tube
 1122 drift chambers, which provide a large number of measurements along the trajectory
 1123 of each charged particle. Although the spatial precision of each TRT hit is lower than
 1124 that of the silicon detectors, the large number of straw hits and the longer lever arm
 1125 improve the momentum measurement and help stabilise track reconstruction.

1126 Together, these subsystems provide the tracking information needed in ATLAS.

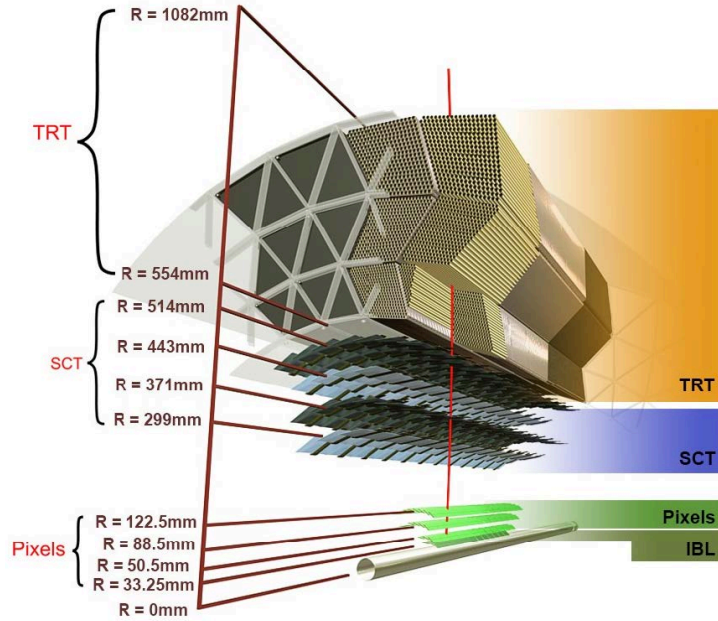


Fig. 4.5 Transverse view of the ATLAS Inner Detector, showing the IBL, pixel detector, SCT, and TRT subsystems [74].

4.2.5 Calorimeter

The ATLAS calorimeter system surrounds the Inner Detector and measures the energies and directions of particles [71]. It consists of electromagnetic (EM) and hadronic calorimeters, as shown in Figure 4.6. Specifically, it provides measurements of charged particles, photons and jets, and covers the region up to $|\eta| < 4.9$. It also provides the shower information to the Level-1 trigger system.

- *Electromagnetic (EM) calorimeter*

The EM calorimeter uses liquid argon (LAr) as the active medium and lead absorbers arranged in an accordion geometry, as shown in Figure 4.7. The accordion geometry provides continuous azimuthal coverage and fine segmentation of electromagnetic showers. The EM calorimeter is divided into a barrel section covering $|\eta| < 1.475$ and two end-cap sections covering $1.375 < |\eta| < 3.2$. It is used to reconstruct and identify electrons and photons and also contributes to the reconstruction of τ_{had} candidates, particularly through the measurement of electromagnetic energy deposits from neutral-pion decays, which don't leave the charged-particle tracks in the ID.

The EM calorimeter consists of a presampler followed by three longitudinal sampling layers, each segmented into readout cells, as illustrated in Figure 4.8. The presampler

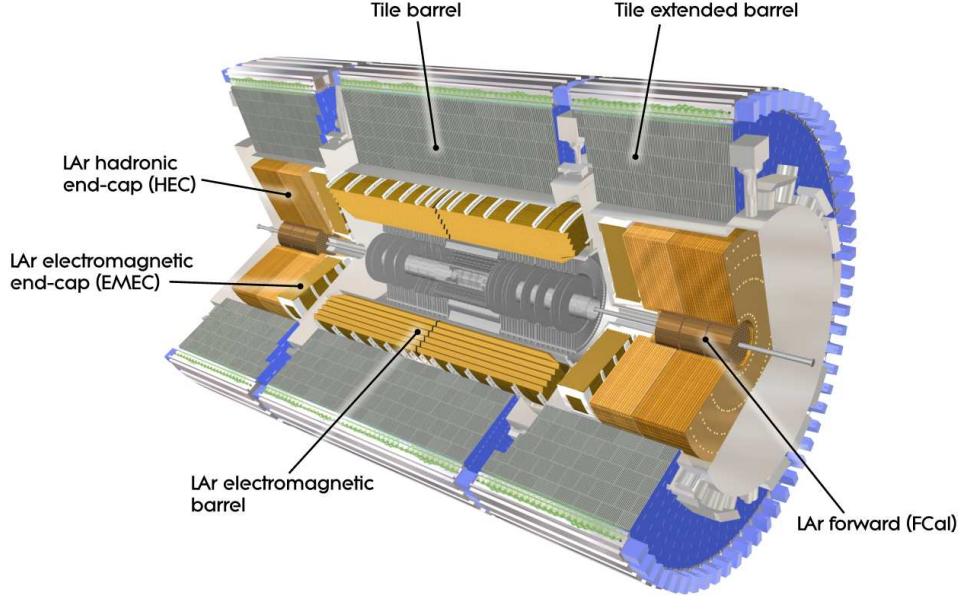


Fig. 4.6 Schematic view of the ATLAS calorimeter system [71].

is installed for $|\eta| < 1.8$ and is used to correct for energy lost in material upstream of the calorimeter. The first sampling layer has fine segmentation in η , which is useful for resolving the transverse shape of electromagnetic showers, while the middle layer collects the largest fraction of the shower energy. The back layer provides an additional measurement of the longitudinal shower development.

The energy resolution of the electromagnetic calorimeter can be parameterised as

$$\frac{\sigma_E}{E} = \frac{a}{\sqrt{E}} \oplus b \oplus \frac{c}{E}, \quad (4.6)$$

where a is the intrinsic sampling term, b is the constant term, and c represents the contribution from electronic and pile-up noise [75].

- **Hadronic calorimeter**

The hadronic calorimeter uses different structure for different η regions. In the region of $|\eta| < 1.7$, the Tile Calorimeter uses steel absorbers and scintillating tiles as the active medium. It measures the energies of hadrons and jets. In the end-cap and forward regions, liquid-argon calorimeters are used. The hadronic end-cap calorimeter consists of two wheels per end-cap, covers the range $1.5 < |\eta| < 3.2$, and uses copper absorbers. The forward calorimeter extends the coverage to $3.1 < |\eta| < 4.9$ and uses

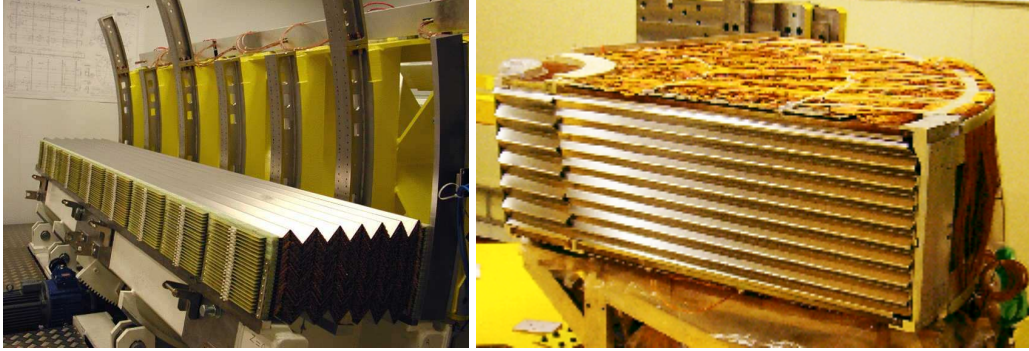


Fig. 4.7 Photograph of the EM barrel (left) and end-cap (right) LAr modules. The accordion absorber structure is visible [71].

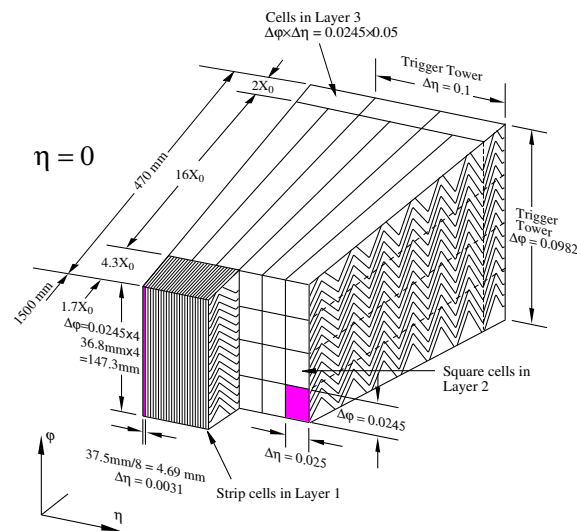


Fig. 4.8 Cut-away view of an ATLAS LAr electromagnetic calorimeter module, showing the longitudinal sampling layers and the segmentation of the readout cells [71].

1160 dense absorber materials suitable for the high-radiation environment close to the beam
1161 line.

1162 4.2.6 Muon Spectrometer

1163 The Muon Spectrometer (MS) forms the outermost major detector system of ATLAS. It is
1164 embedded in the magnetic field produced by the barrel and end-cap air-core toroid magnets.
1165 The bending of muon trajectories in this field is measured by precision chambers, and the
1166 track sagitta is used to determine the muon momentum. The MS provides precision muon
1167 measurements in the range $|\eta| < 2.7$, while the dedicated muon trigger chambers cover the
1168 region up to approximately $|\eta| < 2.4$ [71].

1169 During Run 2, the MS consisted of Monitored Drift Tubes (MDTs), Cathode Strip
1170 Chambers (CSCs), Resistive Plate Chambers (RPCs) and Thin Gap Chambers (TGCs). A
1171 view of these chambers is shown in Figure 4.9.

1172 • *Monitored Drift Tubes (MDTs)*

1173 The MDT chambers provide precision tracking over most of the MS acceptance. They
1174 consist of aluminium drift tubes with a diameter of about 30 mm, each containing a
1175 central sense wire. A muon crossing a tube ionises the gas, and the drift time of the
1176 electrons to the wire is measured to determine the track position. Several tube layers
1177 are grouped into chamber stations, providing precise measurements in the bending
1178 plane of the toroidal magnetic field. MDT chambers are used over most of the region
1179 $|\eta| < 2.7$, except for the innermost forward stations at $2.0 < |\eta| < 2.7$, where CSCs
1180 are used.

1181 • *Cathode Strip Chambers (CSCs)*

1182 The CSCs are multi-wire proportional chambers used in the innermost forward muon
1183 stations, where the particle rates are highest. They cover approximately $2.0 < |\eta| < 2.7$
1184 and provide precision measurements in the forward region. Their small drift distance
1185 and segmented cathode-strip readout allow them to operate efficiently in the high-rate
1186 environment close to the beam line.

1187 • *Resistive Plate Chambers (RPCs)*

1188 The RPCs provide fast muon trigger information in the barrel region, covering approx-
1189 imately $|\eta| < 1.05$. They are gaseous parallel-plate detectors operated in avalanche
1190 mode. Their fast time response allows the correct bunch crossing to be identified
1191 for each muon candidate and provides the Level-1 muon trigger in the barrel. The

RPCs also provide coordinate measurements complementary to the precision MDT chambers.

- *Thin Gap Chambers (TGCs)*

The TGCs are multi-wire proportional chambers designed to provide fast signals for the Level-1 trigger. They extend to approximately $|\eta| < 2.7$ and provide trigger coverage in the end-cap region up to $|\eta| < 2.4$. The TGC system identifies high- p_T muon candidates and assigns them to the correct bunch crossing. It also provides a second coordinate measurement, complementing the precision tracking information from MDTs and CSCs.

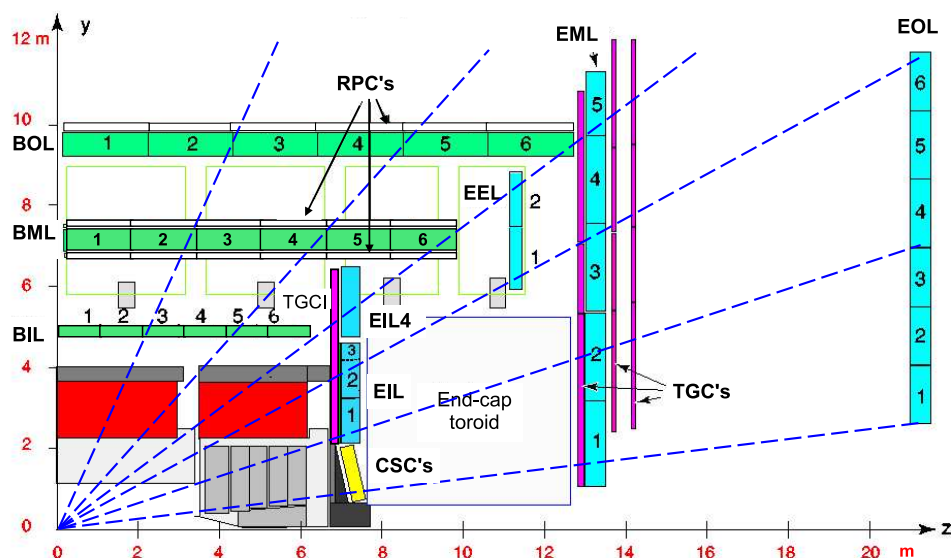


Fig. 4.9 Cross-section of the ATLAS Muon Spectrometer, showing the barrel and end-cap muon chambers. The green and cyan chambers represent the MDT system, while the dashed lines indicate infinite-momentum muon trajectories [71].

During the second long shutdown, the innermost end-cap muon stations were replaced by the New Small Wheels (NSWs) for Run 3 operation. This upgrade was introduced to maintain good muon trigger and reconstruction performance under higher luminosity and pile-up conditions. The NSW detectors are therefore not relevant to the Run 2 dataset analysed in this thesis.

4.2.7 Data Acquisition and Trigger System

At the LHC, proton bunches can cross at a rate of up to 40 MHz for a bunch spacing of 25 ns. This rate is far above the available detector readout and permanent-storage bandwidth. The ATLAS trigger and data acquisition (TDAQ) system is therefore designed to select potentially interesting events online and reduce the event rate to a manageable level. ATLAS uses a two-level trigger system consisting of a hardware-based Level-1 (L1) trigger and a software-based High-Level Trigger (HLT) [76].

Level-1 (L1) trigger

The L1 trigger is the first stage of the ATLAS online event selection. It is implemented in custom hardware and uses reduced-granularity detector information to make a fast decision with a latency of approximately $2.5 \mu\text{s}$. It reduces the input rate from up to 40 MHz to at most 100 kHz. The L1 trigger was based on information from the calorimeter and muon trigger systems, together with topological selections applied to L1 trigger objects. A schematic overview of the Run 2 TDAQ system is shown in Figure 4.10.

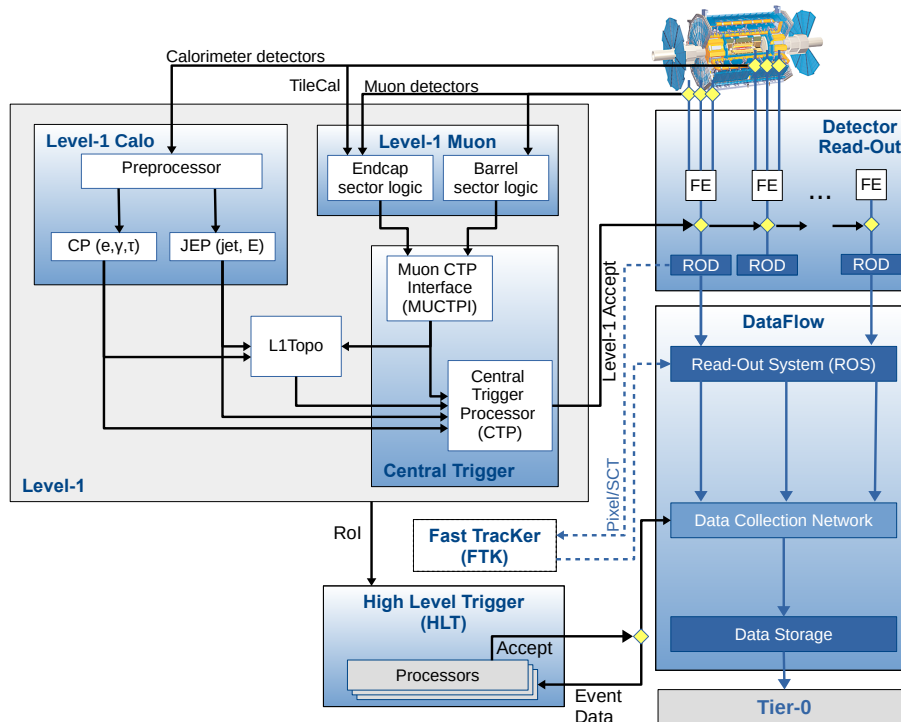


Fig. 4.10 Schematic overview of the ATLAS Run-2 trigger and data acquisition system [76].

• *L1 calorimeter trigger (L1Calo)*

L1Calo uses reduced-granularity information from the electromagnetic and hadronic calorimeters to reconstruct energy deposits within the Level-1 latency. Signals from neighbouring calorimeter cells are combined into trigger towers in the (η, ϕ) plane, and dedicated hardware algorithms search for local maxima in the transverse-energy distribution by comparing the energies of neighbouring towers. The same calorimeter information is also summed over larger regions to reconstruct jet-like energy deposits and global quantities such as the total and missing transverse energy.

• *L1 muon trigger (L1Muon)*

L1Muon identifies muon candidates using fast signals from the RPCs and the TGCs. Hits in different trigger-chamber layers are required to form spatial coincidence patterns consistent with a muon trajectory, while their timing information is used to assign the candidate to the correct bunch crossing. The amount of bending in the magnetic field is used to determine whether the candidate satisfies predefined transverse-momentum thresholds.

• *L1 topological trigger (L1Topo)*

L1Topo receives compact trigger-object information reconstructed by L1Calo and L1Muon, including their transverse momentum or energy and (η, ϕ) positions. Dedicated FPGA hardware combines these objects and calculates event-level topological quantities. And these quantities are compared with predefined selection requirements.

The trigger decisions from L1Calo, L1Muon, and L1Topo are then sent to the Central Trigger Processor (CTP), which combines them according to the predefined L1 trigger menu.

If at least one enabled trigger item is satisfied, the CTP issues a Level-1 Accept (L1A) signal for the corresponding bunch crossing. The L1A signal is distributed to the detector front-end electronics and initiates the transfer of the detector data for that bunch crossing from the front-end buffers to the readout system.

For accepted events, the L1 trigger also provides Regions of Interest (RoIs), which identify detector regions containing the trigger objects responsible for the L1 decision. These RoIs provide seed information for the HLT and allow detailed reconstruction to be performed initially in limited detector regions, reducing the required processing time.

High-Level Trigger

The HLT is the second stage of the ATLAS trigger system. It is implemented in a large computing farm and runs software algorithms using detector information with substantially

1253 finer granularity than is available at L1. The HLT reconstruction is guided by RoIs identified
1254 by the L1 trigger, while full-event reconstruction can also be used for signatures such
1255 as jets and missing transverse momentum. The HLT algorithms are designed to provide
1256 reconstruction performance close to that of the offline algorithms while remaining sufficiently
1257 fast for online operation. As a result, the HLT reduced the L1 output rate to an average
1258 recording rate of approximately 1 kHz for permanent storage.

1259 The trigger menu comprises the active L1 and HLT selections used during data taking with
1260 prescale factors, which are applied to selected trigger items to control their rates according to
1261 the instantaneous luminosity and available bandwidth.

Chapter 5

Data and Monte Carlo Simulations

Monte Carlo (MC) simulation is used to model the signal and Standard Model background processes and to predict their event yields and kinematic distributions. Event generation begins by sampling the incoming partons and their momentum fractions from proton PDFs and evaluating the matrix element for the hard-scattering process. The outgoing partons are then evolved through the parton shower and hadronisation, followed by the decays of unstable particles. Finally, pile-up interactions and the ATLAS detector response are simulated, after which the events are reconstructed with the same algorithms used for collision data.

This chapter describes the collision data and simulation samples used in this thesis. The Run 2 dataset is presented first, followed by the simulated signal and Standard Model background samples. The trigger selections are summarised at the end of the chapter.

5.1 Dataset selection

The search uses the full LHC Run 2 proton–proton collision dataset recorded by the ATLAS detector between 2015 and 2018 at a centre-of-mass energy of $\sqrt{s} = 13$ TeV. Only data recorded during stable-beam periods and passing the data-quality requirements are retained [70]. These periods are defined by the ATLAS good-runs lists (GRLs).

The selected dataset corresponds to an integrated luminosity of $140.1 \pm 1.2 \text{ fb}^{-1}$, with a relative uncertainty of 0.83% [68]. The integrated luminosity recorded in each year is summarised in Table 5.1.

The mean pile-up during Run 2 was $\langle\mu\rangle \simeq 34$, as described in Section 4.1.

Table 5.1 Integrated luminosity of the Run 2 proton–proton collision data [68].

Year	Integrated luminosity [fb^{-1}]
2015	3.24 ± 0.04
2016	33.40 ± 0.30
2017	44.63 ± 0.50
2018	58.79 ± 0.64
Total	140.07 ± 1.17

5.2 Signal samples

The signal samples implement the U_1 vector-leptoquark benchmark defined in Section 3.1.5. The Yang–Mills scenario with $\kappa_c = \kappa_Y = 0$ and $\beta_L^{33} = 1$ is used. The simulated signal covers both the pure-left-handed scenario with $\beta_R^{33} = 0$ and the right-handed scenario with $|\beta_R^{33}| = 1$. Apart from their coupling parameters, the same event-generation, parton-shower and detector-simulation settings are used for both scenarios.

Signal events are generated at leading order (LO) in QCD using MADGRAPH5_AMC@NLO 3.3.1 [77], interfaced to PYTHIA 8.308 [78]. The hard-scattering process produces a τ -lepton, a ν_τ and up to two additional partons, which can be gluons or u -, d -, s -, c - or b -quarks. The NNPDF3.0nlo PDF set [79] is used for the hard-scattering calculation.

The generation is performed in the five-flavour scheme, in which the b -quark is treated as an active parton in the proton. The proton PDFs therefore allow processes with an initial-state b -quark. All allowed U_1 diagrams contributing to this final state are retained in the matrix-element calculation. The resonant single-production and non-resonant t -channel processes shown in Figure 3.4 are therefore included coherently and cannot be separated at the generation stage. Other allowed contributions, including leptoquark pair production where kinematically accessible, are also retained.

PYTHIA subsequently simulates the parton shower and hadronisation. It also models the underlying event using the A14 tune [80] and the NNPDF2.3lo PDF set [81]. The CKKW–L merging procedure [82] prevents the same radiation from being included in both the matrix-element and parton-shower calculations. The Durham $k_\perp$ merging scale is set to 15 GeV and the D parameter to 0.4. Together, they determine whether an emission is treated as an additional matrix-element parton or as part of the shower.

For the pure-left-handed interpretation, dedicated signal samples are generated for 152 combinations of (M_U, g_U, β_L^{23}) satisfying

$$1.5 \text{ TeV} \leq M_U \leq 3.0 \text{ TeV}, \quad (5.1)$$

$$0.5 \leq g_U \leq 3.0, \quad (5.2)$$

$$0 \leq \beta_L^{23} \leq 2.2, \quad (5.3)$$

with $\beta_L^{33} = 1$ and $\beta_R^{33} = 0$. The lower boundary of the mass scan covers the region around the exclusion limits obtained from previous leptoquark pair-production searches [55, 56]. For $M_U > 3 \text{ TeV}$, the selected signal is dominated by non-resonant exchange, where the sensitivity is governed mainly by the ratio g_U/M_U . The primary mass scan is therefore restricted to $M_U \leq 3 \text{ TeV}$.

For the right-handed interpretation with $|\beta_R^{33}| = 1$, one baseline sample is generated at each of the six mass points $M_U \in \{1.5, 2.0, 2.5, 3.0, 4.0, 5.0\} \text{ TeV}$ with $g_U = 1$ and $\beta_L^{23} = 0.2$. Predictions at other values of g_U and β_L^{23} are obtained from these baseline samples using corrected MADGRAPH5_AMC@NLO internal event weights, rather than from separately generated samples at every parameter point. The correction includes an adjustment of the normalisation and a two-dimensional truth-level shape correction based on the leading-jet p_T and $m_{b\tau}$. This strategy avoids repeating event generation and detector simulation for every coupling configuration, thereby reducing the required computing resources.

The U_1 amplitude can interfere with the SM amplitude for production of the same $\tau\nu_\tau + \text{jets}$ final state. The total matrix element can be expressed as

$$|\mathcal{M}_{\text{SM}} + \mathcal{M}_{U_1}|^2 = |\mathcal{M}_{\text{SM}}|^2 + |\mathcal{M}_{U_1}|^2 + 2 \text{Re}(\mathcal{M}_{\text{SM}}\mathcal{M}_{U_1}^*). \quad (5.4)$$

The first term represents the SM contribution, the second represents the pure U_1 contribution, corresponding to the BSM-only component, and the third represents their interference. Unlike the BSM-only component, the interference term can be negative and can reduce the predicted yield relative to the SM-only expectation in some regions of phase space.

To separate the BSM-only and interference terms in Eq. (5.4), the two contributions are generated independently. The BSM-only component is selected by requiring exactly two new-physics couplings at the amplitude level, denoted by $\text{NP}==2$, whereas the interference component is isolated at the squared-matrix-element level by requiring $\text{NP}^2==2$. The interference sample therefore contains only the third term in Eq. (5.4).

All τ -lepton decay modes are retained in the signal simulation. Although the signal regions target τ_{had} candidates, leptonic decays can contribute to regions containing electrons or muons. The inclusive decay configuration allows this signal contamination to be evaluated.

The decays of bottom- and charm-hadrons are simulated with EVTGEN [83]. All signal samples used in this thesis are processed with the ATLAS fast detector simulation ATLFast3 [84], which provides a computationally efficient simulation of the calorimeter response.

5.3 Background samples

The simulated SM backgrounds considered in this analysis are $W + \text{jets}$, $Z + \text{jets}$, $t\bar{t}$, single-top and diboson production. The dominant contribution is $W \rightarrow \tau\nu + \text{jets}$ production. The QCD multijet background is negligible after event cleaning and is therefore neglected, as discussed in Section 7.1.2 and Appendix A.

The $W/Z + \text{jets}$ samples are generated with SHERPA 2.2.11 [85], except for $Z \rightarrow \tau\tau$, which uses version 2.2.14. Matrix elements include up to two additional partons at next-to-leading order (NLO) in QCD and up to five at LO. They are merged with the SHERPA parton shower [86] using the MEPS@NLO prescription [87–90].

Only a small fraction of inclusive $W \rightarrow \tau\nu$ events passes the high- E_T^{miss} requirement, leaving limited MC statistics. Two dedicated filtered samples are therefore used to reduce the MC statistical uncertainty in the high- E_T^{miss} tail. Their respective generator-level requirements are

$$\begin{aligned} 70 \text{ GeV} < E_T^{\text{miss}} < 150 \text{ GeV} \quad \text{and} \quad p_T^\tau < 150 \text{ GeV}, \\ E_T^{\text{miss}} > 150 \text{ GeV} \quad \text{or} \quad p_T^\tau > 150 \text{ GeV}. \end{aligned} \tag{5.5}$$

Top-quark pair and single-top production are generated at NLO accuracy in QCD using POWHEG BOX v2 [91–94], interfaced to PYTHIA 8.230 [95] for parton showering and hadronisation. The single-top samples include s -channel, t -channel and Wt production. The predicted $t\bar{t}$ yield is normalised to the inclusive production cross-section calculated at next-to-next-to-leading order (NNLO) in QCD with next-to-next-to-leading-logarithmic (NNLL) soft-gluon resummation [96]. A dilepton filter is applied to the $t\bar{t}$ sample, requiring both W bosons to decay leptonically. Diboson production is modelled with SHERPA 2.2.11 and 2.2.12.

The hard-scattering calculations use the NNPDF3.0nnlo PDF set for $W/Z + \text{jets}$ and NNPDF3.0nlo for $t\bar{t}$ and single-top production [79]. The underlying event is modelled using the A14 tune for PYTHIA and the default tune for SHERPA. For samples not generated with SHERPA, the decays of bottom- and charm-hadrons are modelled with EVTGEN [83].

The full ATLAS detector simulation based on GEANT4 [97, 98] is used for all background samples. The MC generator configurations are summarised in Table 5.2.

Table 5.2 Summary of the MC generator configurations used for the signal and SM background processes.

Process	Generator	ME order	Parton shower	PDF set	Tune
U_1 signal	MADGRAPH5_AMC@NLO 3.3.1	LO	PYTHIA 8	NNPDF3.0nlo	A14
$W(\rightarrow e\nu/\mu\nu/\tau\nu) + \text{jets}$	SHERPA 2.2.11	NLO	SHERPA	NNPDF3.0nnlo	SHERPA
$Z(\rightarrow ee/\mu\mu/\nu\nu) + \text{jets}$	SHERPA 2.2.11	NLO	SHERPA	NNPDF3.0nnlo	SHERPA
$Z(\rightarrow \tau\tau) + \text{jets}$	SHERPA 2.2.14	NLO	SHERPA	NNPDF3.0nnlo	SHERPA
$t\bar{t}$	POWHEG BOX v2	NLO	PYTHIA 8	NNPDF3.0nlo	A14
Single- t	POWHEG BOX v2	NLO	PYTHIA 8	NNPDF3.0nlo	A14
Diboson	SHERPA 2.2.11/2.2.12	NLO/LO	SHERPA	NNPDF3.0nnlo	SHERPA

5.4 Triggers

The ATLAS Run 2 trigger architecture was described in Chapter 4. Single-electron, single-muon and E_T^{miss} triggers are used in this analysis.

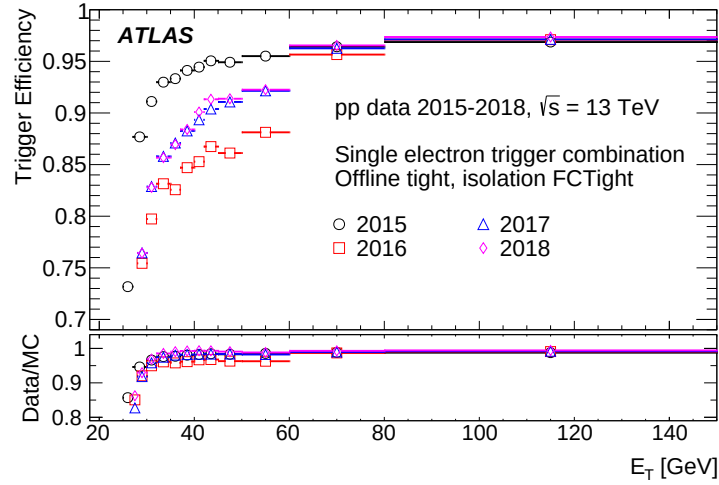
The signal region event selection uses the lowest-threshold unprescaled E_T^{miss} trigger, which is fully efficient for the offline requirement $E_T^{\text{miss}} > 200$ GeV applied in the signal regions and the corresponding validation regions. Single-electron and single-muon triggers are used in regions containing an electron or muon. The lowest online thresholds range from 20 to 26 GeV [99, 100].

Table 5.3 lists the trigger chains used in this analysis. In an HLT chain name, e, mu and xe represent an electron, a muon and E_T^{miss} , respectively. The following number gives the online threshold in GeV. The L1 suffix indicates the first-level trigger that seeds the chain. The electron labels lhloose, lhmedium and lhtight specify likelihood-based identification requirements, while nod0 indicates that transverse-impact-parameter information is omitted. The labels iloose, ivarloose and ivarmedium specify isolation requirements. For E_T^{miss} , mht denotes a calculation based on the negative vector sum of jet transverse momenta, whereas pufit denotes a calorimeter-based calculation with a fit-based pile-up correction [101].

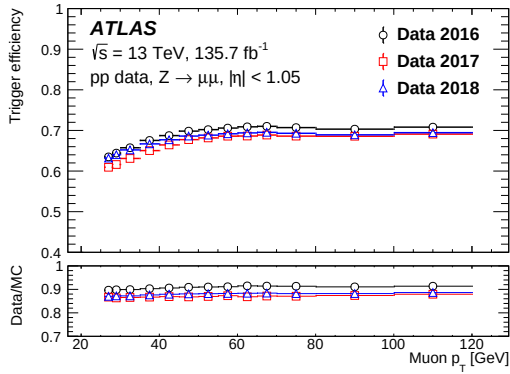
Figures 5.1 and 5.2 show representative Run 2 trigger-efficiency measurements. Differences in trigger efficiency between collision data and simulation are corrected using data-to-simulation efficiency scale factors.

Table 5.3 Single-electron, single-muon and E_T^{miss} trigger chains used in the Run 2 analysis. Single-lepton chains listed for the same flavour and year are combined with a logical OR, while the E_T^{miss} chain is selected according to the data-taking period.

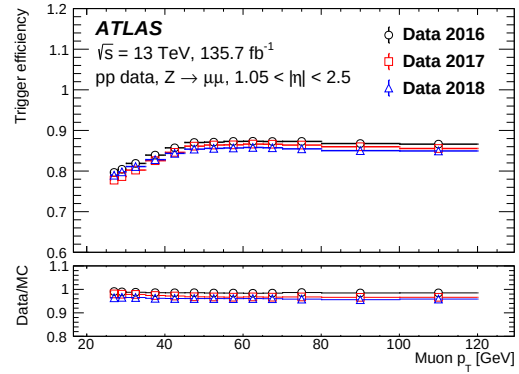
Trigger	Year	HLT chain
Single-electron	2015	HLT_e24_lhmedium_L1EM20VH
		HLT_e60_lhmedium
		HLT_e120_lhloose
	2016–2018	HLT_e26_lhtight_nod0_ivarloose
		HLT_e60_lhmedium_nod0
		HLT_e140_lhloose_nod0
Single-muon	2015	HLT_mu20_iloose_L1MU15
		HLT_mu40
	2016–2018	HLT_mu26_ivarmedium
		HLT_mu50
E_T^{miss}	2015	HLT_xe70_mht
	2016	HLT_xe90_mht_L1XE50
		HLT_xe100_mht_L1XE50
		HLT_xe110_mht_L1XE50
	2017	HLT_xe110_pufit_L1XE55
		HLT_xe110_pufit_L1XE50
	2018	HLT_xe110_pufit_xe70_L1XE50
		HLT_xe110_pufit_xe65_L1XE50



(a) Single-electron trigger combination.



(b) Single-muon triggers, barrel.



(c) Single-muon triggers, end-cap.

Fig. 5.1 Single-lepton trigger efficiencies measured in Run 2 $Z \rightarrow \ell\ell$ events. (a) Electron efficiency versus offline E_T for electrons with Tight identification and FCTight isolation [99]. (b,c) Muon efficiency versus offline p_T in the barrel and end-cap regions, respectively [100]. Electron uncertainties include statistical and systematic contributions, while muon uncertainties are statistical only.

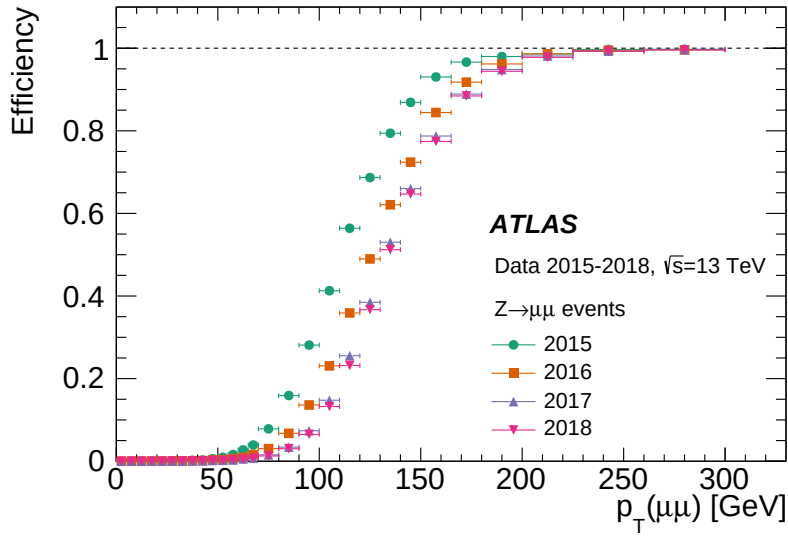


Fig. 5.2 Combined L1 and HLT E_T^{miss} trigger efficiencies in $Z \rightarrow \mu\mu$ events for each year of Run 2 [101]. Since muons are treated as invisible by these calorimeter-based triggers, the dimuon transverse momentum $p_T(\mu\mu)$ provides a reference for the missing transverse momentum measured by the trigger. The curves use the lowest unscaled trigger as it changes within each year. Error bars show statistical uncertainties.

Chapter 6

Object Reconstruction and Identification

τ_{had} candidates, electrons, muons and jets are used in this analysis. This chapter describes their reconstruction, identification and calibration, followed by their definitions in the analysis, overlap removal and the reconstruction of $E_{\text{T}}^{\text{miss}}$.

These objects are reconstructed and identified using detector measurements. Tracks are used to determine charged-particle momenta and production vertices, while energy deposits are measured in the calorimeters. The reconstructed energies and momenta are calibrated to correct biases in the detector response. Simulated event weights are further corrected using efficiency scale factors.

Two levels of object selection, referred to as baseline and signal, are used in this analysis. Baseline objects satisfy a looser set of requirements and are used for overlap removal and $E_{\text{T}}^{\text{miss}}$ reconstruction. Signal objects are selected from the baseline objects by applying tighter requirements and are used in the final event selection.

6.1 Primary vertex

A primary vertex (PV) is the reconstructed position of a pp collision. It is used as the reference for prompt-lepton selection, isolation, pile-up jet rejection and the track contribution to $E_{\text{T}}^{\text{miss}}$. Its position is determined from tracks originating from a common point. Several PVs can be reconstructed in an event because of pile-up. The vertex with the largest sum of squared transverse momenta of its tracks is selected as the PV of the hard scattering [102].

6.2 Tracks

6.2.1 Reconstruction

A track is reconstructed by fitting the trajectory of a charged particle to hits in the ID. Its momentum and charge are determined from the curvature in the magnetic field, while its position near the beam line is used to assign it to a production vertex. Track reconstruction is performed in the following stages [103]:

1. **Space-point formation.** Signals in neighbouring channels of a silicon sensor are grouped into clusters. The coordinates provided by Pixel clusters and SCT clusters are converted into three-dimensional space points.
2. **Seeding and track finding.** Combinations of space points are used to form track seeds. These seeds are extended through the silicon layers with a combinatorial Kalman filter, which is used to refine the track parameters.
3. **Ambiguity resolution.** Track candidates are ranked using their fit quality, momentum and hit variables. Meanwhile, candidates that share the same clusters are marked as “shared hits”. Candidates failing the track-quality requirements are then rejected to suppress duplicate and fake tracks.
4. **TRT extension.** Tracks are extended with additional TRT hits. The extended track is retained if its quality score is higher than that of the original track reconstructed using only pixel and SCT hits.

6.2.2 Track parameters and quality

The fitted trajectory is described by five parameters, as illustrated in Figure 6.1:

- d_0 : distance in the x – y plane between the beam line and the perigee point ¹.
- z_0 : the z coordinate of the perigee point relative to the reference plane at $z = 0$.
- ϕ and θ : the azimuthal and polar angles of the momentum, respectively.
- q/p : the electric charge divided by the momentum magnitude.

Two quantities are used to test whether a track is originating from the primary vertex:

¹The perigee point is the point on the track closest to the beam line.

- **Transverse impact-parameter significance**, d_0/σ_{d_0} : the transverse distance from the beam line in units of its uncertainty σ_{d_0} .
- **Projected longitudinal distance**, $\Delta z_0 \sin \theta = (z_0 - z_{\text{PV}}) \sin \theta$: the longitudinal distance between the track perigee and the primary vertex. Here, z_{PV} is the z coordinate of the primary vertex.

Small absolute values of both quantities suggest that the track can originate from the primary vertex. A large longitudinal distance can also indicate a track originating from another pp collision. Prompt electron and muon candidates are therefore selected by requiring small values of both quantities. And candidates with large values can originate from a displaced vertex, for example from a b - or c -hadron decay.

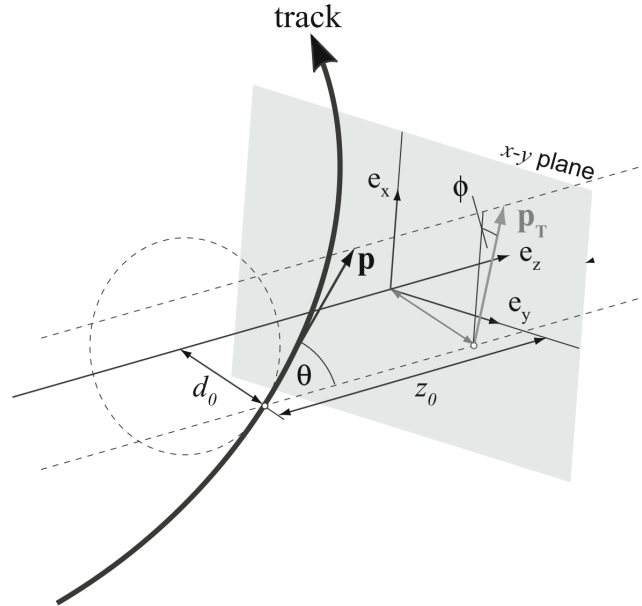


Fig. 6.1 Geometry of the track parameters. The grey plane represents the reference $z = 0$ plane [104].

6.3 Electrons

6.3.1 Reconstruction

The electron object is reconstructed from an ID track matched to an electromagnetic shower in the calorimeter. Bremsstrahlung photons emitted in the upstream material can spread the

shower energy over several clusters and alter the curvature of the electron track. Both effects are treated in the reconstruction [105]:

1. **Topo-cluster reconstruction.** Neighbouring cells are grouped into topological clusters according to their signal significance above the expected noise. Clusters with a substantial electromagnetic energy fraction are used as electron seeds.
2. **Track refitting and matching.** Electron-track candidates are refitted with a Gaussian-sum filter, in which a mixture of Gaussian components is used to describe radiative energy losses. The refitted tracks are extrapolated to the second layer of the EM calorimeter and matched to clusters using their differences in η and ϕ . If several tracks match the same cluster, the ΔR between each extrapolated track position and the cluster centre is used together with the pixel and SCT hits to rank the tracks.
3. **Supercluster formation.** Nearby clusters are added to the seed to recover energy from bremsstrahlung. Additional clusters sharing the matched electron track can be collected over a wider region in ϕ .

The cluster association is illustrated in Figure 6.2. The wider collection in ϕ helps recover energy separated from the electron by its bending in the magnetic field.

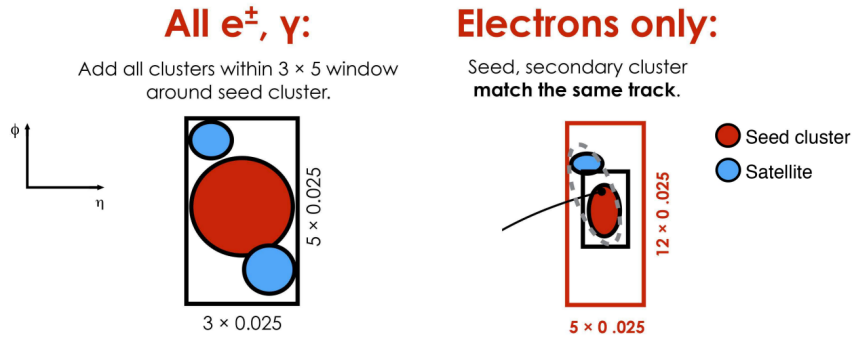


Fig. 6.2 Cluster association for electron superclusters. Seed and additional clusters are shown in red and blue, respectively. Geometrically nearby clusters are collected [105].

6.3.2 Identification

Electron candidates are distinguished from hadronic showers and converted photons using a likelihood discriminant [106]. Its inputs include the shower shapes in the EM calorimeter, hit variables, and the agreement between the track and calorimeter measurements. The Loose, Medium and Tight working points are defined with different likelihood-discriminant

thresholds and provide increasing background rejection. The Tight likelihood working point, denoted TightLH, is used for all electron candidates in this analysis. Its typical prompt-electron identification efficiency is about 80% [107].

6.3.3 Isolation

Electrons from hadron decays can pass the electron identification requirements. These electrons are usually accompanied by other particles from the same jet, while prompt electrons typically have smaller nearby track p_T and calorimeter E_T sums. Isolation requirements are therefore used to suppress this background.

Two isolation variables are calculated using tracks and calorimeter clusters within cones around the electron direction:

- **Track isolation** is performed using the p_T sum of tracks in the cone. At high electron p_T , other decay products can lie close to the electron due to the boost effect, so the cone radius is reduced.
- **Calorimeter isolation** follows the same principle as track isolation, but uses cluster E_T within a fixed cone of $\Delta R = 0.2$. The corrected sum estimates the E_T from other particles near the electron.

In the analysis, baseline electrons are required to pass Loose_VarRad isolation while signal electrons must additionally satisfy HighPtCaloOnly. The Loose_VarRad working point requires the track and calorimeter isolation sums to be below 15% and 20% of the electron p_T , respectively. The HighPtCaloOnly working point requires the calorimeter isolation sum to be below $\max(0.015 p_T, 3.5 \text{ GeV})$. At high E_T , the prompt-electron efficiencies are about 99% for Loose_VarRad and 90–95% for HighPtCaloOnly [107].

6.3.4 Calibration

The electron energy is calibrated for losses in upstream material, energy outside the cluster and variations in detector response. $Z \rightarrow ee$ events are used for the calibration. Reconstruction, identification and isolation efficiencies are measured with the tag-and-probe method: one electron (the tag) is required to pass strict identification cuts, while the other is used as the probe. After background subtraction, the efficiency is the fraction of probes that pass the selection. For a selection efficiency ε , the scale factor is defined as

$$\text{SF} = \frac{\varepsilon_{\text{data}}}{\varepsilon_{\text{MC}}}. \quad (6.1)$$

The scale factors are then applied to simulated event weights for the electron working points used in this analysis.

6.4 Muons

6.4.1 Reconstruction

Muons are reconstructed using tracks in the ID and MS, together with their energy deposits in the calorimeters [108]. The main reconstruction categories are:

- **Combined muons** are reconstructed by matching and fitting ID and MS tracks, including the energy loss between the two tracking systems.
- **Segment-tagged muons** are reconstructed by matching an ID track to one or more local track segments in the MS when a complete MS track is unavailable.
- **Calorimeter-tagged muons** are reconstructed by matching an ID track to calorimeter deposits compatible with a minimum-ionising particle. Acceptance is recovered in regions with limited MS coverage.
- **MS-extrapolated muons** are reconstructed by extrapolating an MS track towards the interaction region without a matched ID track.

Combined muons are used in this analysis.

6.4.2 Identification

Muon identification tests whether the detector hits and the ID and MS momentum measurements are consistent with a single muon trajectory. It's important since muon candidates can also be produced by in-flight decays of charged mesons, such as pions and kaons. These decays can change the trajectory and momentum, leading to a disagreement between the momentum measurements. For muon candidates, this agreement is quantified by the q/p significance,

$$S_{q/p} = \frac{|(q/p)_{\text{ID}} - (q/p)_{\text{MS}}|}{\sqrt{\sigma_{\text{ID}}^2 + \sigma_{\text{MS}}^2}}, \quad (6.2)$$

where $(q/p)_{\text{ID}}$ and $(q/p)_{\text{MS}}$ are the charge-to-momentum ratios determined from the ID and MS tracks, respectively. σ_{ID} and σ_{MS} denote the uncertainties. Large values indicate poor agreement and help reject incorrectly matched candidates.

In this analysis, baseline muons are required to pass Loose identification while signal muons must additionally satisfy the Medium. The Medium working point requires combined muons within the ID acceptance to satisfy $S_{q/p} < 7$ and, outside $|\eta| < 0.1$, to have at least two MS precision stations with at least three MDT or CSC hits each. The Loose working point applies the same requirements and additionally accepts segment-tagged and calorimeter-tagged candidates in $|\eta| < 0.1$, where MS coverage is limited. The typical prompt-muon efficiencies are about 99% for Loose and 97% for Medium.

6.4.3 Isolation

Muons from heavy-flavour decays are often accompanied by other particles inside a jet. They are suppressed using track and calorimeter isolation, calculated similarly to electron isolation, with the muon's own contribution excluded. Baseline muons are required to pass Loose_VarRad isolation, and signal muons must additionally satisfy Tight_VarRad. Both working points use a track-cone radius that decreases as the muon p_T increases and a fixed calorimeter cone of $\Delta R = 0.2$. The track and calorimeter isolation sums must be below 15% and 30% of the muon p_T for Loose_VarRad, respectively. These limits are reduced to 4% and 15% for Tight_VarRad, allowing less activity around the muon and rejecting more muons produced inside jets. The typical prompt-muon isolation efficiencies are about 95% for Loose_VarRad and 86% for Tight_VarRad [109].

6.4.4 Calibration

The muon momentum scale and resolution are calibrated using resonance decays, e.g., $Z \rightarrow \mu\mu$ and $J/\psi \rightarrow \mu\mu$ [110]. The reconstructed mass distributions constrain biases in the muon momentum measurement. The efficiency scale factors are measured with tag-and-probe samples as for electrons and applied to simulated event weights for the selected working points.

6.5 Jets

6.5.1 Reconstruction

Jets are reconstructed from the collimated hadrons produced through parton showering and hadronisation. In this analysis, the constituents used for jet clustering are first reconstructed with the particle-flow (PFlow) algorithm, which combines ID track momenta with calorimeter energy measurements [111]. The steps are:

1. Calorimeter topological clusters are initially calibrated for electromagnetic showers and matched to ID tracks.
2. The expected calorimeter energy deposited by the matched charged particles is subtracted from the clusters, avoiding double counting when the track momenta are used.
3. Vertex association is used to suppress charged-particle contributions from pile-up. The selected tracks and remaining calorimeter contributions form the PFlow constituents.

These PFlow constituents are then grouped into jets using the anti- k_t algorithm with $R = 0.4$ [112]. Pairs are selected for combination according to their angular separation and p_T . Their four-momenta are added, and the merged object is used in the next clustering step. Low- p_T constituents are preferentially clustered around nearby high- p_T constituents, so isolated jets have approximately circular boundaries with radius R .

6.5.2 Identification and pile-up rejection

Jets from the hard scattering are selected after jet cleaning and pile-up rejection.

- **Jet cleaning** is performed with requirements on calorimeter pulse shapes and the distribution of energy among calorimeter layers to reject noise and non-collision backgrounds. For example, a jet with $|\eta| < 2$ is rejected if more than 99% of its energy is deposited in a single calorimeter layer. The efficiency for collision jets exceeds 99.5% [113].
- **Pile-up rejection** uses the vertex origins and p_T of the jet tracks. The jet-vertex tagger (JVT) is constructed from two variables [114]. The first is based on the ratio of the p_T sum of tracks from the PV to that of tracks from all vertices. The second is the same PV-track p_T sum divided by the reconstructed jet p_T . Large values of both variables favour jets from the hard scattering. The neural-network JVT (NNJVT) is used in this analysis with the FixedEffPt working point [115], whose target efficiency for hard-scatter jets is 88–98% depending on the jet p_T [116].

6.5.3 Calibration

Jet energies are calibrated to correct the jet energy scale (JES) and improve the jet energy resolution (JER). The JES characterises the average energy response, while the JER quantifies the spread of reconstructed energies. The following corrections are applied [117]:

- 1589 1. **Pile-up subtraction** removes the estimated energy deposited by pile-up particles. The
 1590 corresponding correction to the jet p_T is estimated as ρA , where ρ is the event p_T
 1591 density and A is the jet area.
- 1592 2. **Energy and direction calibration** corrects the differences in energy and η between
 1593 reconstructed and matched particle-level jets. Particle-level jets are built from stable
 1594 simulated particles, excluding muons and neutrinos, and therefore provide a reference
 1595 without detector effects.
- 1596 3. **Global sequential calibration** corrects the remaining response differences between
 1597 jets with different particle compositions or shower shapes. Track multiplicity and
 1598 energy fractions in calorimeter layers are used to reduce these differences, improving
 1599 the resolution while preserving the average energy scale.
- 1600 4. **Residual in situ calibration** uses well-measured Z bosons and photons as references to
 1601 check the jet energy response. For higher- p_T jets, the vector sum of several calibrated
 1602 lower- p_T jets is used as the reference. The average ratio of jet p_T to reference p_T is
 1603 compared between data and simulation to determine the final correction, which is then
 1604 applied only to data.

1605 6.6 Jet flavour tagging

1606 Jet flavour tagging is used to distinguish jets originating from b -quarks (b -jets) from other jet
 1607 flavours.

1608 b -jets are identified through the displaced decays of b -hadrons, which have a lifetime of
 1609 about 1.5 ps. A boosted b -hadron can therefore travel several millimetres before decaying.
 1610 The decay tracks therefore typically have large impact parameters and originate from a
 1611 secondary vertex displaced from the PV. Charm hadrons produced in the b -hadron decay
 1612 can further decay at an additional displaced vertex. These features are used for jet flavour
 1613 tagging.

1614 The tagger uses the GN2v01 algorithm, which is based on a transformer neural network
 1615 trained on simulated events [118]. Its inputs include jet and track kinematics, impact
 1616 parameters, and hit variables. These variables are first encoded as a numerical vector for
 1617 each track. Within the transformer, each vector is updated through self-attention, using
 1618 information from the other tracks. Correlations between tracks can thereby be used to
 1619 recognise a common displaced origin without requiring explicitly reconstructed secondary
 1620 vertices as separate inputs. The network then calculates a weight for each track using its
 1621 updated vector and adds the weighted vectors together, which is called attention pooling.

The resulting vector is used for the main task of jet-flavour classification. It is also combined with the individual track vectors for two auxiliary tasks: identifying track origins and predicting whether pairs of tracks originate from the same vertex. The three tasks share the transformer and are trained jointly, helping it learn the decay structure. The network and its three tasks are illustrated in Figure 6.3.

The jet-flavour classifier produces four outputs for the b -jet, c -jet, hadronic- τ and light-jet hypotheses, where light jets originate from u , d or s quarks or gluons. These outputs are combined into a single discriminant, D_b , which compares the b -jet output with a weighted combination of the other three outputs. Larger values favour the b -jet hypothesis. The GN2v01 85% working point is used in this analysis, corresponding to a nominal b -jet tagging efficiency of 85%.

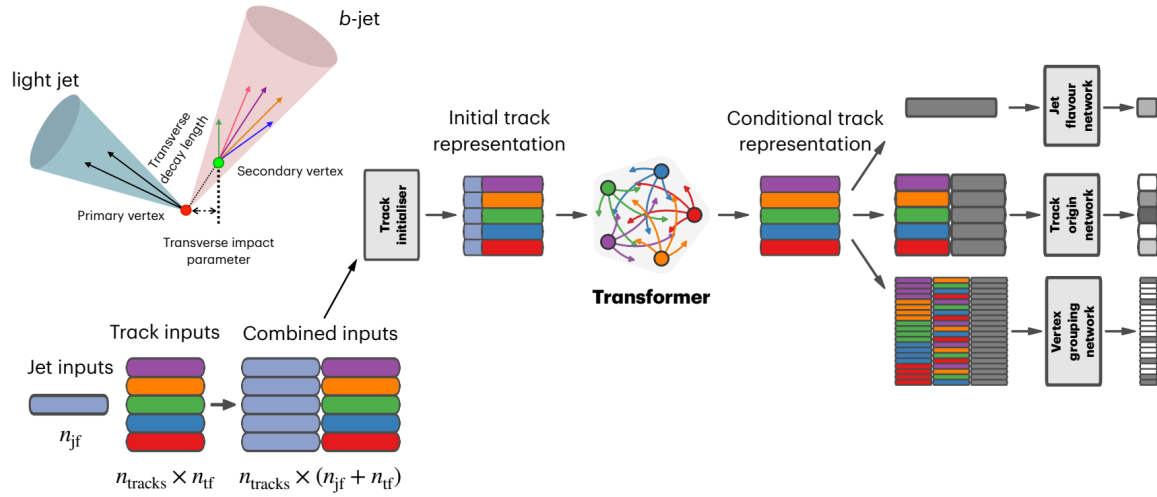


Fig. 6.3 GN2 flavour-tagging architecture. The jet variables are appended to the input vector of each track before processing by the track initialiser and transformer. The resulting track vectors and their weighted combination are passed to the jet-flavour, track-origin and vertex-grouping networks. The symbols n_{jf} , n_{tf} and n_{tracks} denote the numbers of jet variables, track variables and tracks, respectively [118].

6.6.1 Calibration

The b -jet tagging efficiency is measured in dileptonic $t\bar{t}$ events with an oppositely charged electron–muon pair and two jets. The c -jet mistagging efficiency, the probability for a c -jet to pass the b -tagging requirement, is measured in semileptonic $t\bar{t}$ events using charm jets from hadronic W decays. The corresponding light-jet mistagging efficiency is measured in Z +jets events with $Z \rightarrow ee$ or $Z \rightarrow \mu\mu$ decays.

These efficiencies are measured in jet p_T bins and compared with simulation to obtain flavour-dependent scale factors, as defined in Eq. 6.1. Corrections are then applied to the simulated event weights for both tagged and untagged jets.

6.7 Hadronic tau leptons

6.7.1 Reconstruction

τ leptons can decay both leptonically and hadronically. Approximately 65% of τ decays are hadronic, with the major modes containing charged and possibly neutral pions. In this analysis, only hadronically decaying tau leptons (τ_{had}) are reconstructed as tau candidates. The τ -neutrino is excluded from the reconstruction; the τ_{had} momentum therefore refers to the visible decay products.

τ_{had} candidates are first seeded by anti- k_t jets with $R = 0.4$. The production vertex is then chosen as the vertex with the largest fraction of the p_T sum of tracks within $\Delta R < 0.2$ of the seed axis. This vertex provides the reference for track association and impact parameter calculations. The tracks are then classified by a recurrent neural network (RNN) to select tau-decay tracks [119]. The number of selected tau-decay tracks defines the prong multiplicity: candidates with one or three tracks are called one-prong or three-prong candidates, respectively.

6.7.2 Identification

Hadronic tau decays are identified using an RNN discriminant to distinguish them from quark- and gluon-initiated jets. Track momenta and calorimeter shower properties are used as inputs, since tau decays typically have narrower showers and lower charged multiplicity than those jets. Track displacements and, for three-prong candidates, the significance of the reconstructed decay-vertex displacement are also included as inputs. Nearby activity is also included in the RNN inputs, so isolation information is used as part of the identification.

Baseline τ_{had} candidates are required to pass Loose RNN identification, and signal candidates must additionally satisfy Tight. The higher RNN-score threshold for Tight rejects more jets at the cost of τ_{had} efficiency. For reconstructed one-prong (three-prong) candidates, the target identification efficiencies are 85% (75%) for Loose and 60% (45%) for Tight [120].

Electrons can be misidentified as τ_{had} candidates because their narrow showers and tracks are similar to the signature of a one-prong tau decay. To suppress this background, both baseline and signal τ_{had} candidates are required to pass Loose electron rejection. This

working point requires a minimum score from a separate RNN using shower and track information, including the relation between calorimeter energy and track momentum. The τ_{had} efficiencies are approximately 95% and 98% for one-prong and three-prong candidates, respectively. The corresponding electron rejection efficiencies are approximately 99.7% and 98.8%.

6.7.3 Calibration

The τ_{had} energy scale (TES) calibration is derived from MC samples. In these samples, the generated visible tau p_T is used as the reference for correcting the reconstructed p_T . The visible p_T is reconstructed in two ways: from calorimeter energy deposits and shower directions, and from a combination of tracking and calorimeter measurements. In the latter, charged-hadron momenta are measured from track curvature in the magnetic field, while neutral particles, which leave no tracks, are measured through their calorimeter deposits. The calorimeter deposits associated with charged hadrons are subtracted to avoid double counting. These two measurements are combined in a weighted average. The averaged p_T is then corrected using a boosted regression tree (BRT) trained on these MC samples.

In addition to the momentum calibration, differences in τ_{had} reconstruction, identification and electron-rejection efficiencies between data and simulation must be corrected. For this purpose, data-to-simulation scale factors are applied to simulated event weights. In this analysis, for signal candidates passing Tight τ_{had} identification, the electron-veto scale factors for Medium τ_{had} identification are used because no corresponding factors were available for Tight.

6.8 Overlap removal

Different reconstructed objects can share the same detector measurements. The ΔR between two objects is used for the separation. An ordered overlap-removal procedure is therefore applied to resolve these ambiguities before event selection as follows:

1. **Electron–electron:** the lower- p_T electron is removed if two electrons share an ID track, so that a single track is not counted as two electrons.
2. **τ_{had} –electron:** a τ_{had} candidate within $\Delta R < 0.2$ of an electron is removed, since an electron shower and its track can also be reconstructed as a τ_{had} candidate.
3. **τ_{had} –muon:** a τ_{had} candidate within $\Delta R < 0.2$ of a muon is removed to reject muons misidentified as τ_{had} candidates.

- 1702 4. **Calorimeter-tagged muon–electron:** a calorimeter-tagged muon sharing the electron
1703 ID track is removed. The electron interpretation is preferred because the calorimeter
1704 tag has no independent MS confirmation.
- 1705 5. **Electron–muon:** an electron sharing an ID track with a remaining muon is removed.
- 1706 6. **Jet–electron:** a jet within $\Delta R < 0.2$ of an electron is removed because the electron’s
1707 EM shower can also be reconstructed as a jet.
- 1708 7. **Electron–jet:** an electron within $\Delta R < 0.4$ of a remaining jet is removed to suppress
1709 electrons from hadron decays inside jets.
- 1710 8. **Jet–muon:** a jet with fewer than three tracks is removed if the muon is ghost-associated
1711 with it.²
- 1712 9. **Muon–jet:** a muon within $\Delta R < 0.4$ of a remaining jet is removed to suppress muons
1713 from hadron decays inside jets.
- 1714 10. **Jet– τ_{had} :** a jet within $\Delta R < 0.2$ of a remaining τ_{had} candidate is removed, so that the
1715 visible tau-decay products are not also counted as a separate jet.

1716 6.9 Missing transverse momentum

1717 The incoming partons at the LHC carry little transverse momentum, so the vector sum of the
1718 transverse momenta of all the particles is expected to be approximately zero. An imbalance
1719 in the visible transverse momentum can therefore indicate undetected particles, such as
1720 neutrinos or weakly interacting BSM particles. In this analysis, the neutrino produced in the
1721 LQ process and the tau-decay neutrino can together give rise to sizeable missing transverse
1722 momentum ($\vec{p}_T^{\text{miss}}$). $\vec{p}_T^{\text{miss}}$ is reconstructed as the negative vector sum of the calibrated visible
1723 contributions, with its magnitude denoted by E_T^{miss} [121]:

$$1724 \quad \vec{p}_T^{\text{miss}} = - \left(\sum_e \vec{p}_T^e + \sum_\mu \vec{p}_T^\mu + \sum_{\tau_{\text{had}}} \vec{p}_T^{\tau_{\text{had}}} + \sum_j \vec{p}_T^j + \vec{p}_T^{\text{soft}} \right), \quad (6.3)$$

$$1725 \quad E_T^{\text{miss}} = \left| \vec{p}_T^{\text{miss}} \right|. \quad (6.4)$$

1726 Two types of contribution are included in the reconstruction:

²Ghost association means that a copy of the muon’s ID track is made, keeping its direction but reducing its momentum to a negligible value. This “ghost” is clustered together with the normal jet inputs. If the clustering algorithm places the ghost in a particular jet, the muon is associated with that jet.

- **hard term:** The electron, muon, τ_{had} and jet contributions selected for $\vec{p}_T^{\text{miss}}$ reconstruction. The jet contribution is selected using the Tight $\vec{p}_T^{\text{miss}}$ working point to suppress pile-up jets. This requires jet $p_T > 20$ GeV for $|\eta| < 2.5$ and $p_T > 30$ GeV for $2.5 \leq |\eta| < 4.5$.
- **track soft term:** The vector sum of transverse momenta of selected PV tracks not used in the hard term.

6.10 Object definitions in the analysis

As shown in Table 6.1, the object definitions in the analysis are summarised.

Table 6.1 Baseline object definitions and additional signal requirements used in the analysis. A dash in the signal column indicates that the requirement depends on the analysis region.

Object	Requirement	Baseline	Signal
Electron	p_T	$> 10 \text{ GeV}$	—
	$ \eta $	$(\eta < 1.37) \text{ } (1.52 < \eta < 2.47)$	
	$ d_0 /\sigma_{d_0}$		< 5
	$ \Delta z_0 \sin \theta $		$< 0.5 \text{ mm}$
	Identification		TightLH
	Isolation	Loose_VarRad	HighPtCaloOnly
Muon	p_T	$> 10 \text{ GeV}$	—
	$ \eta $		< 2.5
	$ d_0 /\sigma_{d_0}$		< 3
	$ \Delta z_0 \sin \theta $		$< 0.5 \text{ mm}$
	Identification	Loose	Medium
	Isolation	Loose_VarRad	Tight_VarRad
τ_{had}	$ \eta $	$(\eta < 1.37) \text{ } (1.52 < \eta < 2.5)$	
	Tracks		1, 3
	Charge		$ Q = 1$
	Energy calibration		MVA
	p_T	$> 20 \text{ GeV}$	—
	Jet rejection	RNN, Loose	RNN, Tight
	Electron rejection		RNN, Loose
Jet	Algorithm		Anti- k_t
	Radius		$R = 0.4$
	Inputs		PFlow
	Cleaning		LooseBad
	p_T	$> 20 \text{ GeV}$	—
	$ \eta $	< 2.5	—
	NNJVT		FixedEffPt
	b -tagging	GN2v01, 85% b -jet efficiency	

Chapter 7

Event Selection

This chapter defines the signal regions (SRs) targeting the processes shown in Figure 3.4. The analysis variables and preselection are introduced first, followed by the optimisation of the one- b -jet regions, the definition of the zero- b -jet regions, and a study of $\tau\tau$ signal contamination in the SRs.

7.1 Preselection and variable definitions

7.1.1 Analysis variables

The transverse mass m_T of the τ_{had} and $\vec{p}_T^{\text{miss}}$ system is defined as

$$m_T = \sqrt{2p_T^{\tau_{\text{had}}} E_T^{\text{miss}} [1 - \cos \Delta\phi(\tau_{\text{had}}, E_T^{\text{miss}})]}. \quad (7.1)$$

here $\Delta\phi(\tau_{\text{had}}, E_T^{\text{miss}})$ is the separation in ϕ between the τ_{had} momentum and $\vec{p}_T^{\text{miss}}$. The variable m_T is used to probe the high-energy tail of the $\tau\nu$ final state, which is particularly sensitive to non-resonant leptoquark contributions [62].

The invariant masses $m_{b\tau_{\text{had}}}$ and $m_{j\tau_{\text{had}}}$ are defined as:

$$m_{b\tau_{\text{had}}}^2 = (p_b + p_{\tau_{\text{had}}})^2, \quad m_{j\tau_{\text{had}}}^2 = (p_{j_1} + p_{\tau_{\text{had}}})^2, \quad (7.2)$$

where p_b , p_{j_1} and $p_{\tau_{\text{had}}}$ are four-momenta, and j_1 is the jet with the largest p_T , so-called the leading jet. These two variables are used to probe the b -tagged and b -veto SRs, respectively.

7.1.2 Multijet suppression and preselection

Although the high E_T^{miss} threshold rejects most multijet events, some can still enter the SRs and CRs due to the large multijet cross section. These events typically contain a misidentified τ_{had} and fake E_T^{miss} due to the poor jet resolution. Therefore, further suppression is necessary.

Specifically, fake E_T^{miss} is produced when an energetic jet is mismeasured. In this case, $\vec{p}_T^{\text{miss}}$ points approximately towards that jet and it tends to be small compared with the p_T of the mismeasured jet. On the other hand, in signal events, $\vec{p}_T^{\text{miss}}$ from neutrinos has no such tendency. Figure 7.1 illustrates this difference. The minimum jet- $\vec{p}_T^{\text{miss}}$ separation in ϕ and the ratio $\frac{E_T^{\text{miss}}}{p_T^{\text{jet}(\min \Delta \phi)}}$ are therefore used as discriminating variables. Here $p_T^{\text{jet}(\min \Delta \phi)}$ is the p_T of the jet with the smallest $\Delta \phi(E_T^{\text{miss}}, p_T^{\text{jet}})$.

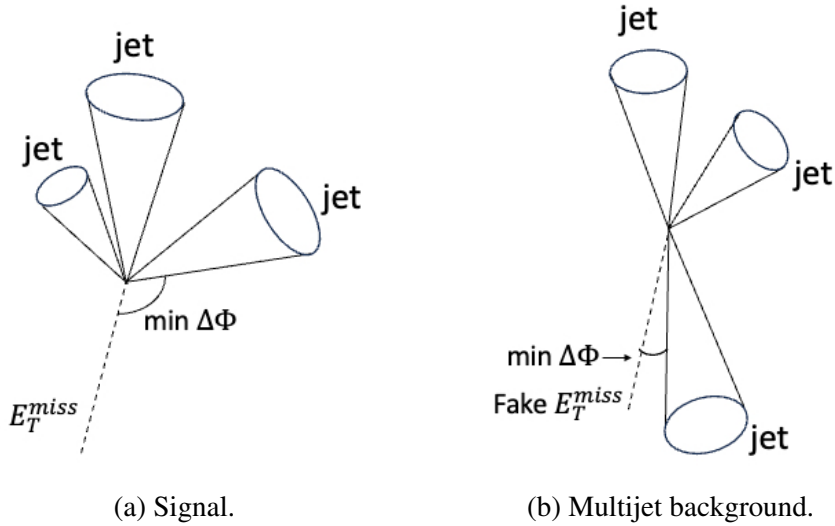


Fig. 7.1 Schematic diagrams of (a) E_T^{miss} from neutrinos in signal events and (b) fake E_T^{miss} from a mismeasured energetic jet in multijet events.

Events are retained if

$$\min \Delta \phi(E_T^{\text{miss}}, p_T^{\text{jet}}) > 0.4 \quad \text{or} \quad \frac{E_T^{\text{miss}}}{p_T^{\text{jet}(\min \Delta \phi)}} > 6. \quad (7.3)$$

The study used to determine the thresholds above is presented in detail in Appendix A.

In addition to the multijet suppression requirement, the following preselection requirements are applied before the SR optimisation:

- The lowest-threshold unscaled E_T^{miss} trigger to select events with large E_T^{miss} .
- $E_T^{\text{miss}} > 200$ GeV for full trigger efficiency.

- 1769 • Exactly one signal τ_{had} .
- 1770 • Signal electron/muon veto.
- 1771 • Exactly one b -tagged jet for the one- b -jet SR (SR1b).
- 1772 • At least one light-flavour jet and b -jet veto for the b -jet-veto SR (SR0b).

1773 7.2 SR1b optimisation and definition

1774 Event selection in SR1b is optimised using a grid scan of thresholds on variables that help
 1775 separate signal from background. SR1b-Res and SR1b-NonRes are optimised separately.

1776 7.2.1 Scan variables and criterion

1777 $p_{\text{T}}^{\tau_{\text{had}}}$ and $E_{\text{T}}^{\text{miss}}$ are considered first, since energetic τ leptons and neutrinos are expected in
 1778 the signal. The b -jet p_{T} is also used. An energetic b -jet from the leptoquark decay is expected
 1779 in the resonant process, whereas the b -jet is typically softer in the non-resonant process.
 1780 m_{T} is used to distinguish non-resonant leptoquark exchange from the SM background. A
 1781 tighter m_{T} threshold is therefore expected in SR1b-NonRes than in SR1b-Res. The invariant
 1782 mass $m_{b\tau_{\text{had}}}$ is used to probe resonant $U_1 \rightarrow b\tau$ decays. $\Delta\phi(\tau_{\text{had}}, E_{\text{T}}^{\text{miss}})$ is used to select the
 1783 back-to-back τ_{had} and $\vec{p}_{\text{T}}^{\text{miss}}$ configuration of the non-resonant signal, which helps suppress
 1784 the top-quark background. On the other hand, in SR1b-Res, small $\Delta\phi(\tau_{\text{had}}, E_{\text{T}}^{\text{miss}})$ values are
 1785 favoured due to the high p_{T}^b requirement. The jet multiplicity is used to suppress $t\bar{t}$ events
 1786 since they tend to have more jets than the signal events. The thresholds used for the grid scan
 1787 are summarised in Table 7.1.

Table 7.1 Variable scan range for the SR1b optimisation.

Variable	SR1b-Res	SR1b-NonRes
$\Delta\phi(\tau_{\text{had}}, E_{\text{T}}^{\text{miss}})$	$< \{0.3, 1.2, 2.4, 2.6, \pi\}$	$> \{0.3, 1.2, 2.4, 2.6\}$
$m_{b\tau_{\text{had}}} [\text{GeV}]$	$> \{0, 800\}$	N/A
$E_{\text{T}}^{\text{miss}} [\text{GeV}]$	$> \{200, 250, 300, 350, 400, 500, 600\}$	
$p_{\text{T}}^{\tau_{\text{had}}} [\text{GeV}]$	$> \{200, 250, 300, 350, 400, 500, 600\}$	
$m_{\text{T}} [\text{GeV}]$	$> \{100, 200, 300, 400, 500, 600, 700, 800, 900, 1000\}$	
$N_j - N_b$	$\leq \{1, 2, 3\}$	

1788 Note that only $m_{b\tau_{\text{had}}} > 0$ and 800 GeV are tested here. Since the signal mass resolution
 1789 is only about 50% and very few background events remain in the SRs, little improvement in
 1790 sensitivity is expected from a finer mass threshold scan.

Different signal benchmarks are selected for the optimisations of each signal region. The optimisation for SR1b-Res uses $(M_U, g_U, \beta_L^{23}) = (1.5 \text{ TeV}, 1.5, 0.6)$, where on-shell production is dominant. For SR1b-NonRes, the optimisation uses $(M_U, g_U, \beta_L^{23}) = (2.5 \text{ TeV}, 2.5, 1.0)$, where t -channel exchange is dominant. Interference is not included in the scan because its contribution is around 5–10% in the parameter range probed by the SR1b regions. The selected cut combination is therefore not expected to change significantly when interference is included.

The expected sensitivity of each cut combination is evaluated using the significance Z , calculated as follows [122]:

$$Z = \sqrt{2 \left[n \ln \left(\frac{n(b + \sigma^2)}{b^2 + n\sigma^2} \right) - \frac{b^2}{\sigma^2} \ln \left(\frac{b^2 + n\sigma^2}{b(b + \sigma^2)} \right) \right]}, \quad (7.4)$$

where s and b are the expected signal and background event yields, $n = s + b$, and the background uncertainty is assumed to be $\sigma = 0.3b$.

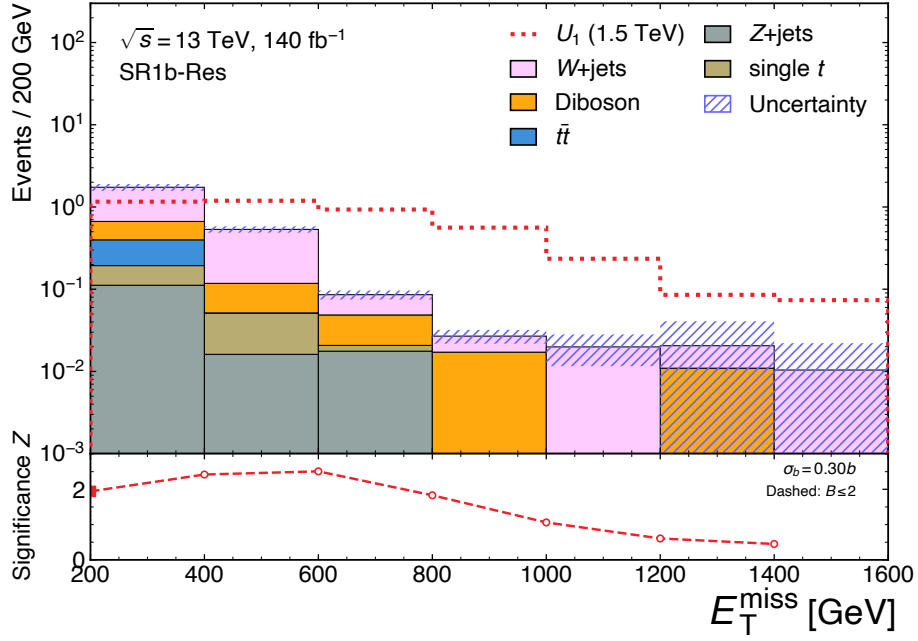
To obtain reliable results, the following two constraints are applied during the scan:

- Cut combinations with $b \leq 2$ are discarded.
- If a particular SM background (e.g. $W \rightarrow \tau\nu$) has a negative weighted yield, its contribution is set to zero.

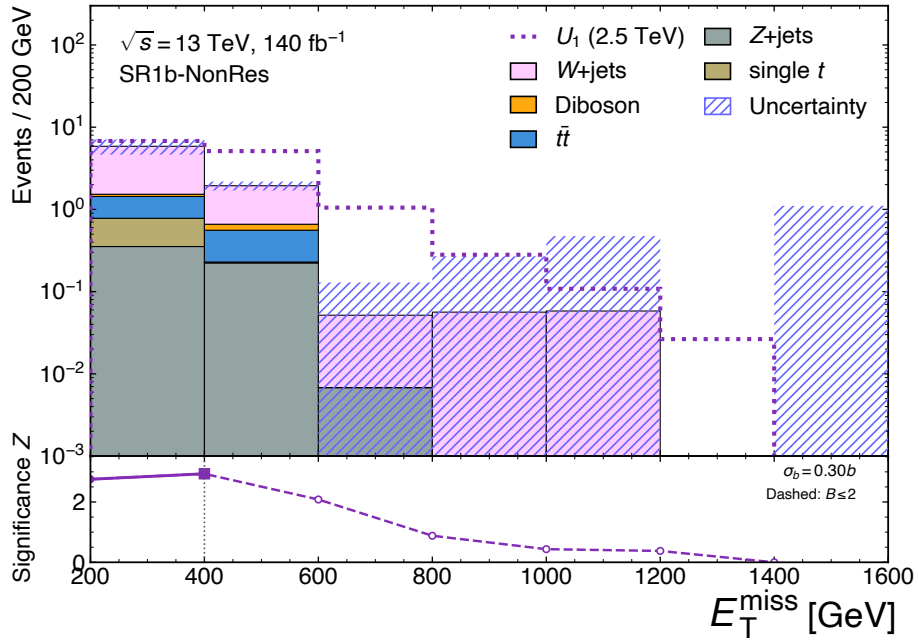
The first constraint is applied to avoid zero background estimates caused by limited MC statistics. It also ensures reliable estimates of the minor backgrounds. As a result, the MC statistical uncertainties are limited to 31% in the resonant SR and 40% in the non-resonant SR.

An example of the significance scan is shown in Figure 7.2, along with $N - 1$ E_T^{miss} distributions.¹ For each bin in the lower significance panels, Z is calculated using the signal and background yields summed above the corresponding threshold. Points with $b \leq 2$ are excluded from the cut choice. The largest Z is therefore obtained at 200 GeV for SR1b-Res and 400 GeV for SR1b-NonRes. A similar procedure is applied to the other variables.

¹For a selection with N requirements, an $N - 1$ distribution is obtained by removing the requirement on the plotted variable while keeping the other $N - 1$ requirements. Here, the E_T^{miss} requirement is removed.



(a) SR1b-Res



(b) SR1b-NonRes

Fig. 7.2 $N - 1$ E_T^{miss} distributions and the corresponding significance Z . Z is calculated from the signal and background yields above each threshold, with $\sigma = 0.30b$. Dashed segments of the Z curves indicate regions where $b \leq 2$.

7.2.2 SR1b-Res

The definition and cutflow for SR1b-Res are summarised in Table 7.2, using the resonant optimisation benchmark $(M_U, g_U, \beta_L^{23}) = (1.5 \text{ TeV}, 1.5, 0.6)$. The total event yield with $m_{b\tau_{\text{had}}} > 800 \text{ GeV}$ is used because the simulated background statistics at high $m_{b\tau_{\text{had}}}$ are insufficient for a fit to the mass distribution.

Table 7.2 Signal and background cutflow for SR1b-Res. The signal is evaluated at $(M_U, g_U, \beta_L^{23}) = (1.5 \text{ TeV}, 1.5, 0.6)$. The uncertainties are from MC statistics only.

Requirement	Signal	$t\bar{t}$	Single top	W+jets	Z+jets	Diboson	Total background
Baseline	35.2 ± 0.4	1043.0 ± 6.9	283.8 ± 3.4	2156.9 ± 16.1	369.2 ± 7.7	173.9 ± 2.7	4027.3 ± 19.7
$p_T^{\tau_{\text{had}}} > 200 \text{ GeV}$	19.9 ± 0.3	32.3 ± 1.3	11.8 ± 0.5	108.9 ± 9.4	30.4 ± 4.7	9.7 ± 0.6	193.1 ± 10.6
$E_T^{\text{miss}} > 200 \text{ GeV}$	19.9 ± 0.3	32.3 ± 1.3	11.8 ± 0.5	108.9 ± 9.4	30.4 ± 4.7	9.7 ± 0.6	193.1 ± 10.6
$m_T > 200 \text{ GeV}$	19.4 ± 0.3	25.1 ± 1.1	9.2 ± 0.4	79.2 ± 9.4	24.2 ± 4.7	6.9 ± 0.5	143.6 ± 10.6
$m_{b\tau_{\text{had}}} > 800 \text{ GeV}$	4.9 ± 0.1	0.4 ± 0.2	0.4 ± 0.1	2.8 ± 0.4	1.1 ± 0.2	0.5 ± 0.1	5.2 ± 0.5
$N_j - N_b \leq 3$	4.7 ± 0.1	0.3 ± 0.1	0.2 ± 0.1	2.5 ± 0.4	0.9 ± 0.2	0.4 ± 0.1	4.3 ± 0.5
$p_T^b > 250 \text{ GeV}$	4.2 ± 0.1	0.2 ± 0.1	0.1 ± 0.1	1.5 ± 0.2	0.2 ± 0.1	0.4 ± 0.1	2.4 ± 0.3

As a result, the expected signal and background yields are 4.2 and 2.4 events, respectively, giving $Z = 1.95$.

The background is dominated by $W(\rightarrow \tau\nu)$ +jets, which contributes approximately 64%. Approximately 16% is contributed by diboson production. The top-quark contribution consists of $t\bar{t}$ (8%) and single-top production (5%), with the latter mainly from the Wt process. These events are mainly from $t\bar{t} \rightarrow b\tau\nu\bar{b}\tau\nu$ or $Wt \rightarrow b\tau\nu\tau\nu$ when only one τ_{had} is selected and the other is misidentified as a jet. A smaller contribution is provided by $Z(\rightarrow \tau\tau)$ +jets, while the $Z(\rightarrow \nu\nu)$ +jets contribution is negligibly small.

The $N - 1$ distributions in SR1b-Res with relaxed kinematic requirements are shown in Figure 7.3. To reduce statistical fluctuations, the thresholds on the correlated variables m_T and $p_T^{\tau_{\text{had}}}$ are both lowered to 100 GeV.

For the 1.5 TeV benchmark, the $m_{b\tau_{\text{had}}}$ peak is shifted to about 900 GeV because the neutrino from the τ decay is not included in the reconstructed mass. The mass distribution is also broadened by the missing neutrino momentum and poor jet reconstruction resolution.

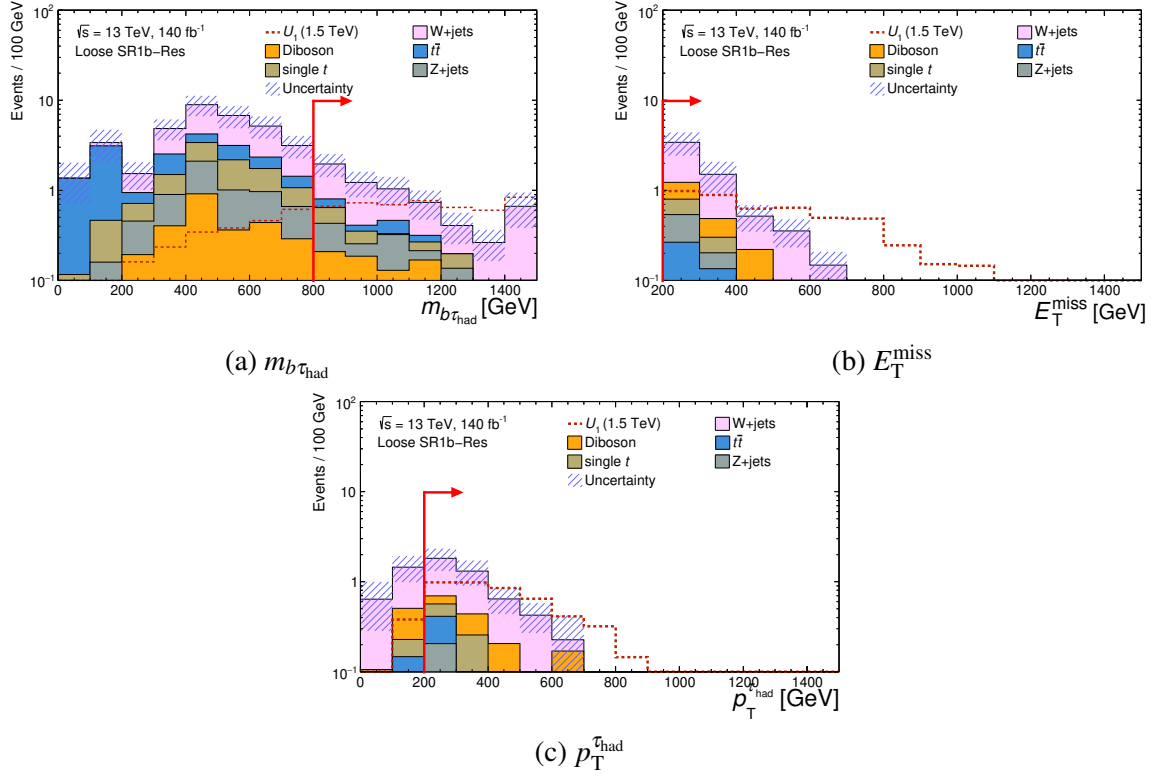


Fig. 7.3 $N-1$ distributions in the loose SR1b-Res region. The m_T and $p_T^{\tau_{\text{had}}}$ thresholds are reduced to 100 GeV. The red dashed line represents the signal at $(M_U, g_U, \beta_L^{23}) = (1.5 \text{ TeV}, 1.5, 0.6)$. Systematic uncertainties are also included. The red arrows indicate the final selection thresholds.

7.2.3 SR1b-NonRes

The definition and cutflow for SR1b-NonRes are summarised in Table 7.3, using the non-resonant benchmark $(M_U, g_U, \beta_L^{23}) = (2.5 \text{ TeV}, 2.5, 1.0)$. No $m_{b\tau_{\text{had}}}$ requirement is applied in the optimisation because the b -jet and τ_{had} do not reconstruct an on-shell leptoquark in the target topology. The total event yield is also used in this region.

Table 7.3 Signal and background cutflow for SR1b-NonRes. The signal is evaluated at $(M_U, g_U, \beta_L^{23}) = (2.5 \text{ TeV}, 2.5, 1.0)$. The uncertainties are from MC statistics only.

Requirement	Signal	$t\bar{t}$	Single top	W +jets	Z +jets	Diboson	Total background
Baseline	111.0 ± 4.2	1043.0 ± 6.9	283.8 ± 3.4	2156.9 ± 16.1	369.2 ± 7.7	173.9 ± 2.7	4027.3 ± 19.7
$p_T^{\tau_{\text{had}}} > 200 \text{ GeV}$	70.5 ± 3.8	32.3 ± 1.3	11.8 ± 0.5	108.9 ± 9.4	30.4 ± 4.7	9.7 ± 0.6	193.1 ± 10.7
$E_T^{\text{miss}} > 400 \text{ GeV}$	26.4 ± 2.3	2.1 ± 0.3	0.8 ± 0.1	9.0 ± 1.5	1.4 ± 0.3	1.8 ± 0.2	15.1 ± 1.6
$m_T > 600 \text{ GeV}$	24.0 ± 2.2	0.7 ± 0.2	0.3 ± 0.1	3.9 ± 1.5	0.8 ± 0.2	0.7 ± 0.2	6.4 ± 1.6
$\Delta\phi(\tau_{\text{had}}, E_T^{\text{miss}}) > 1.2$	23.8 ± 2.2	0.7 ± 0.2	0.3 ± 0.1	3.8 ± 1.5	0.8 ± 0.2	0.7 ± 0.2	6.4 ± 1.6
$N_j - N_b \leq 1$	17.2 ± 2.1	0.3 ± 0.1	0.1 ± 0.1	1.6 ± 1.4	0.4 ± 0.1	0.4 ± 0.2	2.8 ± 1.4
$50 < p_T^b < 250 \text{ GeV}$	6.6 ± 0.7	0.3 ± 0.1	0.0 ± 0.1	1.6 ± 0.5	0.2 ± 0.1	0.1 ± 0.1	2.1 ± 0.5

As a result, the expected signal and background yields are 6.6 and 2.1 events, respectively, giving $Z = 2.94$.

The background is also dominated by $W(\rightarrow \tau\nu)$ +jets, which contributes approximately 69%. Approximately 16% is contributed by $t\bar{t}$, followed by $Z(\rightarrow \tau\tau)$ +jets (8%) and diboson production (5%). The $Z(\rightarrow \tau\tau)$ +jets events pass the selection when only one τ_{had} is selected, with E_T^{miss} from the τ neutrinos. $Z(\rightarrow \nu\nu)$ +jets is still the main source of fake τ_{had} background. Multijet production is negligible after the cleaning and full SR requirements.

The yields and significances for both regions are summarised in Table 7.4.

Table 7.4 Signal and background yields for the two optimisation benchmarks in the final SR1b selections. The approximate significance is evaluated with $\sigma = 0.30b$.

SR	(M_U, g_U, β_L^{23})	S	B	Z
SR1b-Res	$(1.5 \text{ TeV}, 1.5, 0.6)$	4.262	2.439	1.95
SR1b-NonRes	$(2.5 \text{ TeV}, 2.5, 1.0)$	6.583	2.097	2.94

The $N - 1$ distributions in SR1b-NonRes with relaxed kinematic requirements are shown in Figure 7.4. To reduce statistical fluctuations, the m_T , E_T^{miss} and $p_T^{\tau_{\text{had}}}$ thresholds are each lowered by 100 GeV, while the baseline requirement $E_T^{\text{miss}} > 200 \text{ GeV}$ is retained.

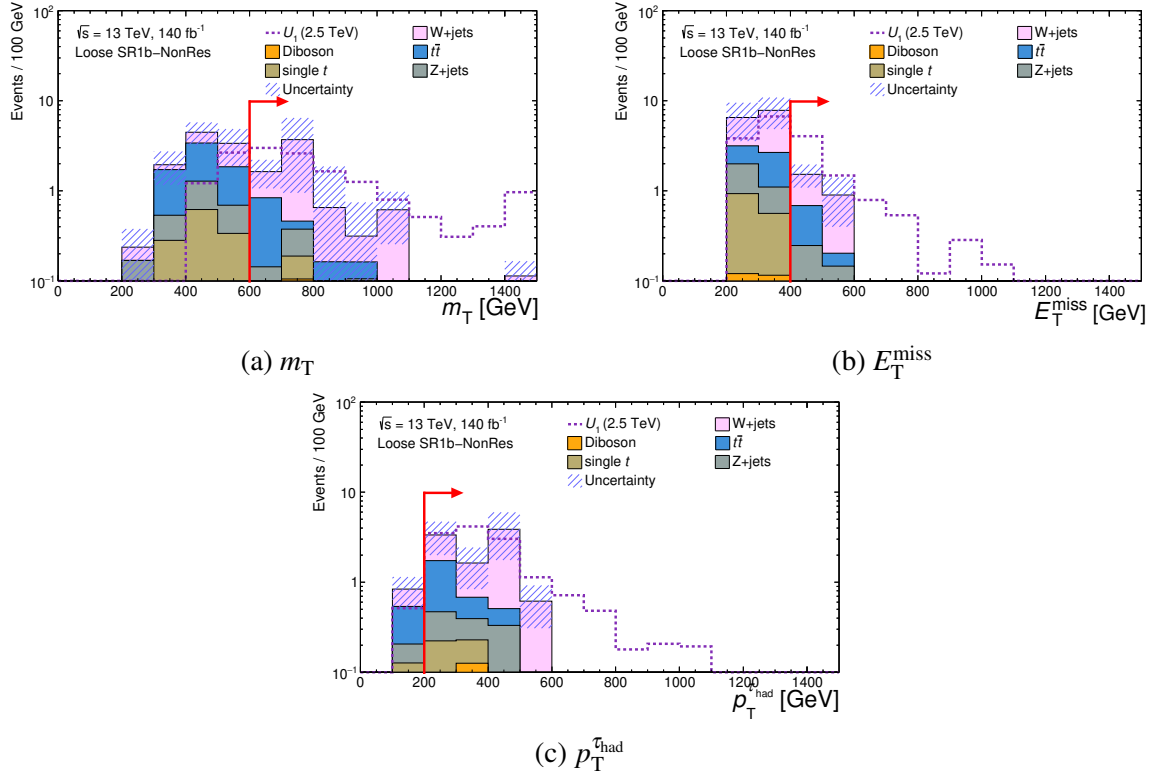


Fig. 7.4 $N-1$ distributions in the loose SR1b-NonRes region. The m_T , E_T^{miss} and $p_T^{\tau_{\text{had}}}$ thresholds are reduced to 500 GeV, 300 GeV and 100 GeV, respectively. The baseline $E_T^{\text{miss}} > 200$ GeV is retained. The purple dashed line represents the signal at $(M_U, g_U, \beta_L^{23}) = (2.5 \text{ TeV}, 2.5, 1.0)$. Systematic uncertainties are also included. The red arrows indicate the final selection thresholds.

7.3 b -veto signal region (SR0b)

The SR0b regions are defined based on the SR1b optimisation results, targeting on-shell $U_1 \rightarrow s\tau$ decays to extend the sensitivity to large β_L^{23} . The corresponding signal topologies were introduced in Section 3.1.5. In particular, the event selection is retained except that the b -jet requirement is replaced by a requirement on a non- b -tagged jet. The p_T^b requirement is applied to the leading jet instead. The final definitions of all SRs are summarised in Table 7.5.

Table 7.5 Common baseline requirements and final signal-region selections. A dash denotes no additional requirement. The jet multiplicity includes b -tagged jets.

Baseline requirements				
Trigger	Lowest-threshold unprescaled E_T^{miss} trigger			
$N_{\tau_{\text{had}}}$	1			
N_ℓ	0			
Multijet suppression	$\min \Delta\phi(E_T^{\text{miss}}, p_T^{\text{jet}}) > 0.4$ or $\frac{E_T^{\text{miss}}}{p_T^{\text{jet}(\min \Delta\phi)}} > 6$			
Requirement	Resonant		Non-resonant	
	SR1b-Res	SR0b-Res	SR1b-NonRes	SR0b-NonRes
N_b	1	0	1	0
p_T^b [GeV]	> 250	–	$50 < p_T^b < 250$	–
$p_T^{j_1}$ [GeV]	–	> 250	–	$50 < p_T^{j_1} < 250$
$p_T^{\tau_{\text{had}}}$ [GeV]			> 200	
E_T^{miss} [GeV]	> 200			> 400
m_T [GeV]	> 200			> 600
$m_{b\tau_{\text{had}}}$ [GeV]	> 800	–	–	–
$m_{j\tau_{\text{had}}}$ [GeV]	–	> 800	–	–
$\Delta\phi(\tau_{\text{had}}, E_T^{\text{miss}})$	–			> 1.2
N_j		≤ 4		≤ 2

The flavour of the leading jet is identified in the truth-level study using the parton label assigned to the jet (PartonTruthLabelID), where 3, 4 and 21 denote s quarks, c quarks and gluons, respectively. The distributions at $(M_U, g_U, \beta_L^{23}) = (1.5 \text{ TeV}, 1.5, 0.6)$ are shown in Figure 7.5. In SR0b-Res, approximately 30% of the events contain a leading s jet and 60% contain a leading c jet. The s -jet contribution is produced through single leptoquark production followed by $U_1 \rightarrow s\tau$ decays, which are enhanced at large β_L^{23} . For the c -jet contribution, a high- p_T c jet is produced in an on-shell $U_1 \rightarrow c\nu_\tau$ decay, as illustrated in Figure 7.6. In SR0b-NonRes, approximately 33% of the events are produced through non-resonant $\tau\nu$ production with a c jet from initial-state radiation (ISR), while 11% contain an ISR s jet. The remaining events are mainly produced through $sc \rightarrow \tau\nu_\tau$ with an ISR gluon jet.

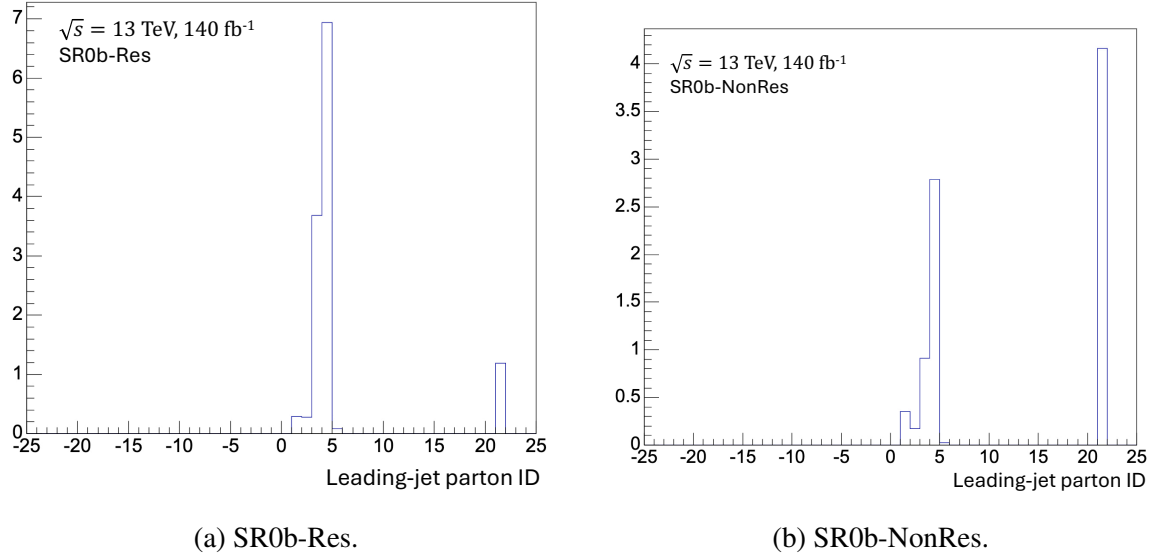


Fig. 7.5 Leading-jet parton-ID distributions in SR0b-Res and SR0b-NonRes, at the signal point $(M_U, g_U, \beta_L^{23}) = (1.5 \text{ TeV}, 1.5, 0.6)$. The parton IDs 3, 4 and 21 denote s quarks, c quarks and gluons, respectively.

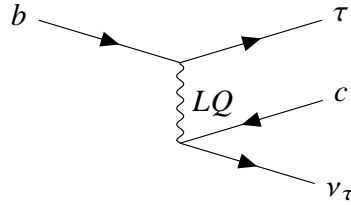


Fig. 7.6 Signal diagram contributing to SR0b-Res through an on-shell $U_1 \rightarrow c \nu_\tau$ decay.

The interference contribution in SR0b is evaluated at each signal parameter point. Since it is not CKM-suppressed, a larger destructive interference contribution is expected. The total signal yield is found to be reduced by approximately 20% at $M_U = 1.5$ TeV and by about 30–40% at higher masses. However, it remains positive over the parameter region probed by this analysis. The supplementary E_T^{miss} scan in Appendix B shows that a higher E_T^{miss} threshold can suppress this contribution. This improvement can be used in the next LQ search with the full Run 3 dataset. The $N - 1$ distributions for SR0b-Res and SR0b-NonRes are shown in Figures 7.7 and 7.8, respectively.

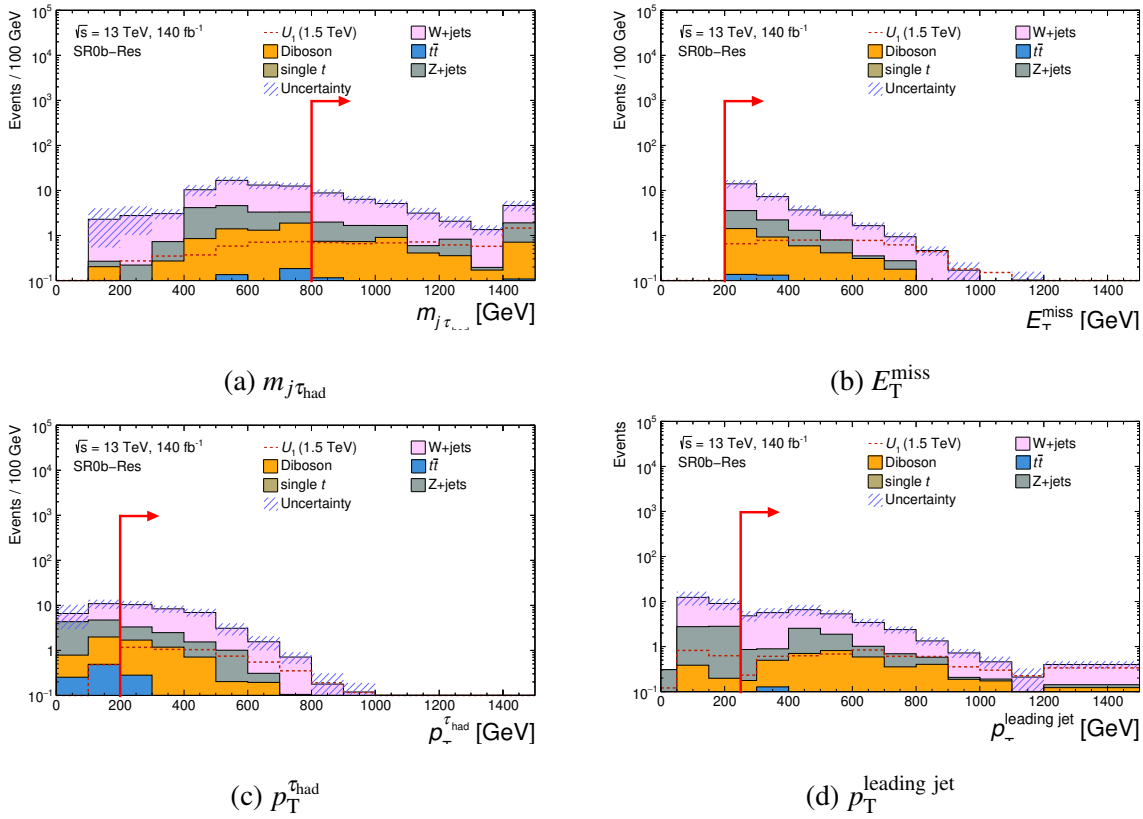


Fig. 7.7 $N - 1$ distributions in SR0b-Res. The baseline $E_T^{\text{miss}} > 200$ GeV requirement is kept. The red dashed line represents the signal at $(M_U, g_U, \beta_L^{23}) = (1.5 \text{ TeV}, 1.5, 0.6)$. Systematic uncertainties are also included. The red arrows indicate the final selection thresholds.

The SR0b is dominated by $W \rightarrow \tau\nu$ +jets, since vetoing b -jets removes much of the top-quark background while retaining W production with light-flavour jets. Specifically, W +jets contributes approximately 76% in SR0b-Res and 82% in SR0b-NonRes. In the resonant region, diboson production contributes about 10%, $Z \rightarrow \nu\nu$ +jets about 8%, and $Z \rightarrow \tau\tau$ +jets about 5%. In the non-resonant region, the corresponding contributions are approximately

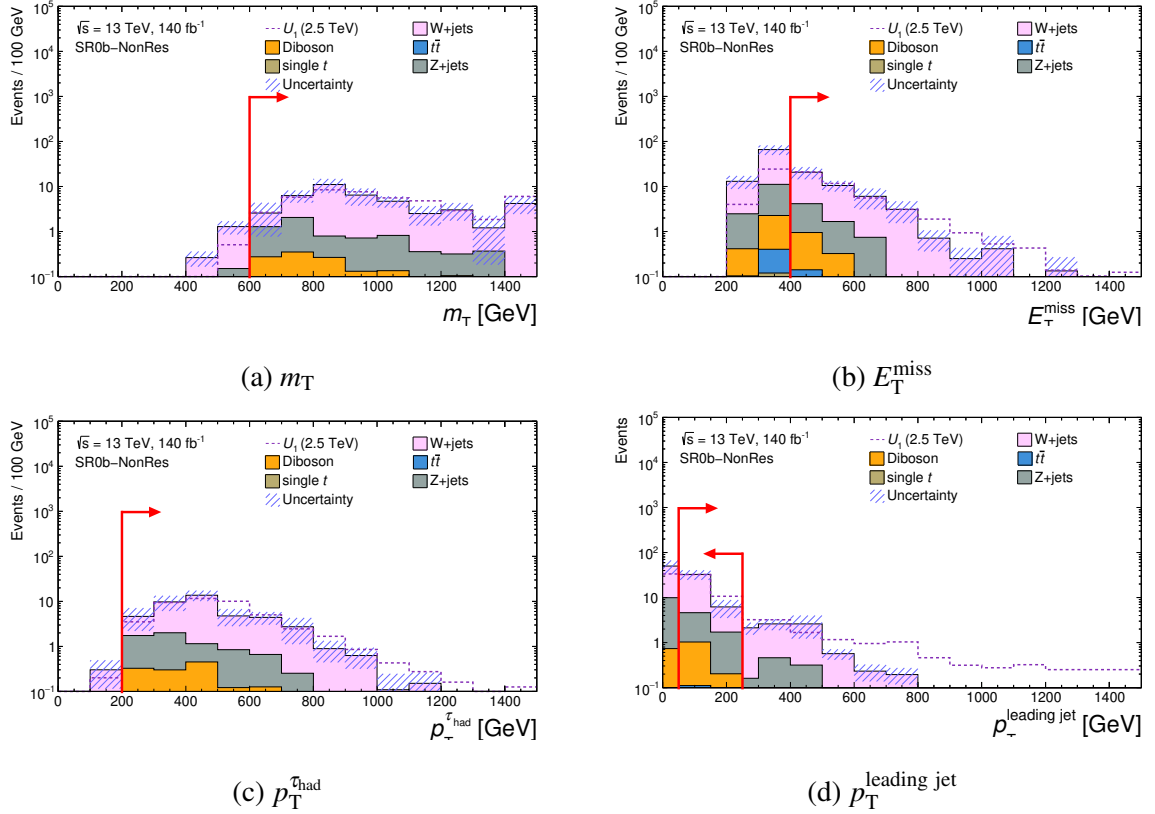


Fig. 7.8 $N-1$ distributions in SR0b-NonRes. The baseline $E_T^{\text{miss}} > 200$ GeV requirement is kept. The purple dashed line represents the signal at $(M_U, g_U, \beta_L^{23}) = (2.5 \text{ TeV}, 2.5, 1.0)$. Systematic uncertainties are also included. The red arrows indicate the final selection thresholds.

1881 3%, 6% and 8%. The $Z \rightarrow \nu\nu$, $Z \rightarrow \tau\tau$ and diboson events have a similar topology to that in
1882 SR1b, while the top-quark background contribution is negligible.

7.4 $\tau\tau$ signal contamination

Perhaps this section should be moved to the appendix?

The signal contamination $\tau\tau$ production is evaluated in this section. Although two τ -leptons are produced in the final state, an event can satisfy the single- τ_{had} requirement when only one is reconstructed and selected. Neutrinos from the τ decays produce missing momentum.

The study uses simulated $\tau\tau$ samples generated with $\beta_L^{33} = 1$, $\beta_R^{33} = 0$ and $\beta_L^{23} = 0$. Their yields are compared with those of the available $\tau\nu + b$ samples near the expected exclusion limit, at $(M_U, g_U, \beta_L^{23}) = (1.8 \text{ TeV}, 3.0, 0.6)$ and $(3.0 \text{ TeV}, 2.0, 1.4)$. The corresponding $\tau\tau$ samples use $g_U = 2.8$ and 2.1 . Since the $\tau\tau$ samples are generated at $\beta_L^{23} = 0$, a fair comparison with the $\tau\nu + b$ samples at the same β_L^{23} is not directly possible. To address this, the $\tau\tau$ production cross section is recalculated with MADGRAPH5_AMC@NLO at the corresponding β_L^{23} , and the yields of the $\tau\tau$ samples at $\beta_L^{23} = 0$ are corrected by the cross-section ratios to $\beta_L^{23} = 0.6$ and 1.4 . Specifically, the cross-section ratios are $\sigma_{\beta_L^{23}=0.6}/\sigma_{\beta_L^{23}=0} = 2.7$ and $\sigma_{\beta_L^{23}=1.4}/\sigma_{\beta_L^{23}=0} = 30$. This procedure therefore provides a sensitivity study while retaining the acceptance and kinematic shapes of the available samples. For simplicity, the signal theory uncertainty is not included in this study.

The $\tau\nu$ yields and estimated $\tau\tau$ contributions are summarised in Table 7.6. The estimated contamination from $\tau\tau$ production is of order 35% of the nominal $\tau\nu$ yield in SR1b, and several percent in SR0b.

Table 7.6 Signal yields from $\tau\nu$ and $\tau\tau$ production in the four SRs. The parameter values (M_U, g_U, β_L^{23}) are given for each sample. The $\tau\tau$ yields are shown at $\beta_L^{23} = 0$, with the yields scaled to $\beta_L^{23} = 0.6$ and 1.4 , respectively, given in parentheses.

Region	$\tau\nu$	$\tau\nu$	$\tau\tau$	$\tau\tau$
	(1.8 TeV, 3.0, 0.6)	(3.0 TeV, 2.0, 1.4)	(1.8 TeV, 2.8, 0)	(3.0 TeV, 2.1, 0)
SR1b-Res	21.0	2.2	2.5 (6.75)	0.03 (0.9)
SR1b-NonRes	11.8	3.3	1.3 (3.51)	0.04 (1.2)
SR0b-Res	28.9	7.5	0.8 (2.16)	0.01 (0.3)
SR0b-NonRes	43.5	21.4	1.3 (3.51)	0.04 (1.2)

In the further study, the expected exclusion limit on the signal strength is improved by approximately 8–14%, depending on M_U . Since the signal strength scales as g_U^4 and $(\beta_L^{23})^2$, the improvement in the coupling limit is much less significant. The expected improvement is about 5% at $M_U = 1.8 \text{ TeV}$ and about 2% at $M_U = 3.0 \text{ TeV}$. Therefore, the sensitivity is slightly underestimated if the contamination from $\tau\tau$ production is not included, but the

1908 improvement in the coupling limit is not significant and is mostly covered by the signal
1909 theory uncertainties.

Chapter 8

Background Estimation

The background estimation uses both simulated samples and data. The simulated samples are used to predict the background yields in the SRs, while data are used to correct mismodelling of the backgrounds. Since the fake- τ_{had} background is small and the signal topology is relatively easy to identify, a simple cut-and-count method is used for background estimation, with dedicated normalisation factors (NFs) for the dominant backgrounds. Therefore, dedicated control regions (CRs) and validation regions (VRs) are defined to estimate the NFs and validate the background predictions, respectively. This chapter presents the background estimation strategy, the CR and VR definitions, and the validation of the background predictions after the background-only fit¹.

8.1 Background estimation strategy

The estimation strategy for each background is summarised below:

- $W + \text{jets}$ is estimated using NFs constrained by data in CRW, where one light leptonic (electron or muon) and no τ_{had} or b -jets are required. The $0b \rightarrow 1b$ extrapolation is tested in VRW, where one light lepton, no τ_{had} and one b -jet are required.
- The top-quark background is estimated using NFs constrained by data in CRTop, where one τ_{had} , one light lepton and at least one b -jet are required. The prediction is tested in VRTop, where one τ_{had} , no light leptons and two b -jets are required.

¹The background-only fit is performed by fitting background in the CRs with the background-only hypothesis.

- $Z \rightarrow \nu\nu$ with a jet misidentified as a τ_{had} is estimated directly from simulation. A conservative uncertainty is assigned based on the study of $Z \rightarrow \ell\ell$ events with a fake- τ_{had} in Appendix C.
- $Z \rightarrow \tau\tau$ and diboson production are estimated directly from simulation without dedicated CRs. Their predicted yields are allowed to vary within conservative uncertainties in the fit.
- The multijet background is negligible after the full SR selections, including the suppression described in Section 7.1.2, as checked in Appendix A.

Four CRs are therefore defined in this analysis: CRW-Res and CRW-NonRes for $W + \text{jets}$, and CRTop-Res and CRTop-NonRes for the top-quark background, corresponding to the normalisation factors λ_W^{Res} , $\lambda_W^{\text{NonRes}}$, λ_t^{Res} and $\lambda_t^{\text{NonRes}}$, respectively. They are defined as follows:

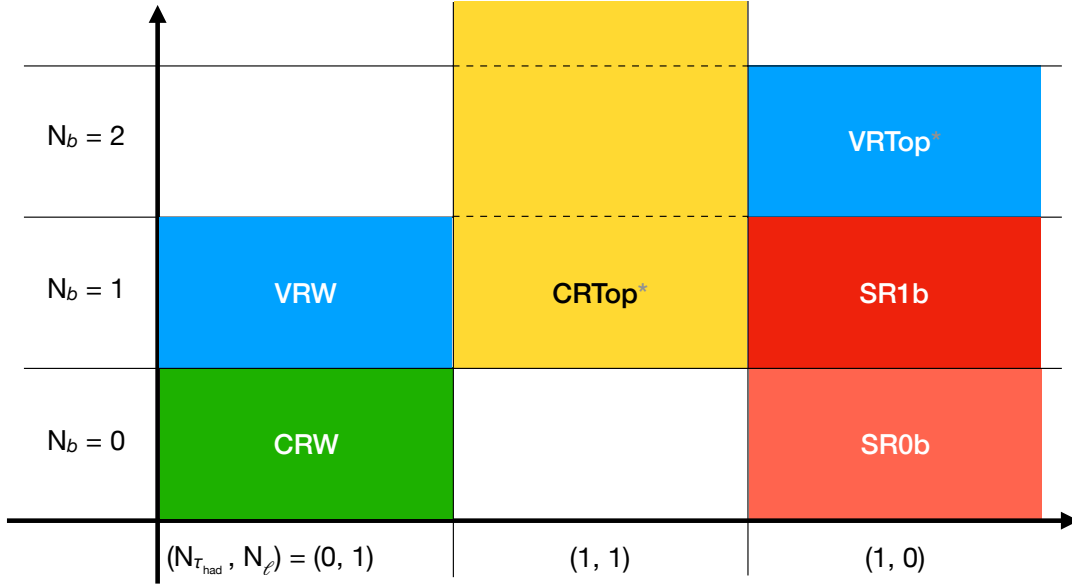
$$\lambda_{bg} = \frac{N_{bg}^{\text{CR,data}}}{N_{bg}^{\text{CR,MC}}}. \quad (8.1)$$

where $N_{bg}^{\text{CR,data}}$ and $N_{bg}^{\text{CR,MC}}$ are the observed and the predicted background event yields in the CR, respectively. The independent NFs allow different normalisation in the resonant and non-resonant regions, which probe different p_T ranges. The VRs do not participate in the fit. Residual mismodelling of the small contribution from the top-quark background in CRW can be partly absorbed by a shift in λ_W , and vice versa. The SR, CR and VR definitions are summarised in Figure 8.1.

8.2 $W + \text{jets}$

The CRW regions are defined by replacing the τ_{had} lepton in the SR selection with an light lepton, targeting the $W \rightarrow \ell\nu$ process. The b -jet veto suppresses the top-quark background and increases the $W + \text{jets}$ event purity. Single-lepton triggers are used, as listed in Table 5.3. This avoids differences in the E_T^{miss} trigger response between the electron and muon channels, since the muon momentum is not included in the trigger-level E_T^{miss} calculation. m_T^ℓ and $\Delta\phi(\ell, E_T^{\text{miss}})$ are calculated using the light lepton as a proxy for the τ_{had} lepton, and the leading jet is used as a proxy for the b -jet. The other selections are the same as the SRs.

The definitions of the CRW and VRW regions are summarised in Table 8.1. After the background-only fit, the $W + \text{jets}$ purity is approximately 95% in CRW-NonRes and 82% in CRW-Res. Diboson events are expected to contribute approximately 14% in CRW-Res. The additional neutrino in a hadronic τ decay introduces kinematic differences in the $1\ell \rightarrow 1\tau_{\text{had}}$



* For CRTop and VRTop, looser cuts for tau p_T , mEt and mT are applied for statistics

Fig. 8.1 SR, CR and VR definitions in terms of the τ_{had} , light-lepton and b -jet multiplicities. Separate resonant and non-resonant regions are defined for each.

1960 extrapolation. The prediction is further tested in the lower-mass $1\tau_{\text{had}}0\ell 1b$ regions described
 1961 in Section D.1, which check the extrapolation to events with both a τ_{had} and a b -jet.

1962 Signal contamination is checked with samples including both hadronic and leptonic τ
 1963 decays at parameter points $(M_U, g_U, \beta_L^{23}) = (1.5 \text{ TeV}, 1.5, 0.2)$ and $(2.5 \text{ TeV}, 1.5, 1.0)$ for
 1964 the resonant and non-resonant CRs, respectively. The contamination is found to be below
 1965 1% in both regions.

1966 The CRW $N - 1$ distributions after the background-only fit are shown separately for the
 1967 electron and muon channels in Figures 8.2–8.5. There are 243 electron and 219 muon events
 1968 in CRW-Res, and 474 electron and 398 muon events in CRW-NonRes. No clear slope is
 1969 observed in the data/bkg. ratios, except in the jet- p_T distribution in Fig. 8.5(d). The average
 1970 effect of this jet- p_T dependence in each region is expected to be absorbed by the independent
 1971 $W + \text{jets}$ normalisation factors.

1972 The difference between the data/bkg. ratios of the electron and muon channels is consis-
 1973 tent with statistical fluctuations, as shown in Appendix E. The two channels are therefore
 1974 combined.

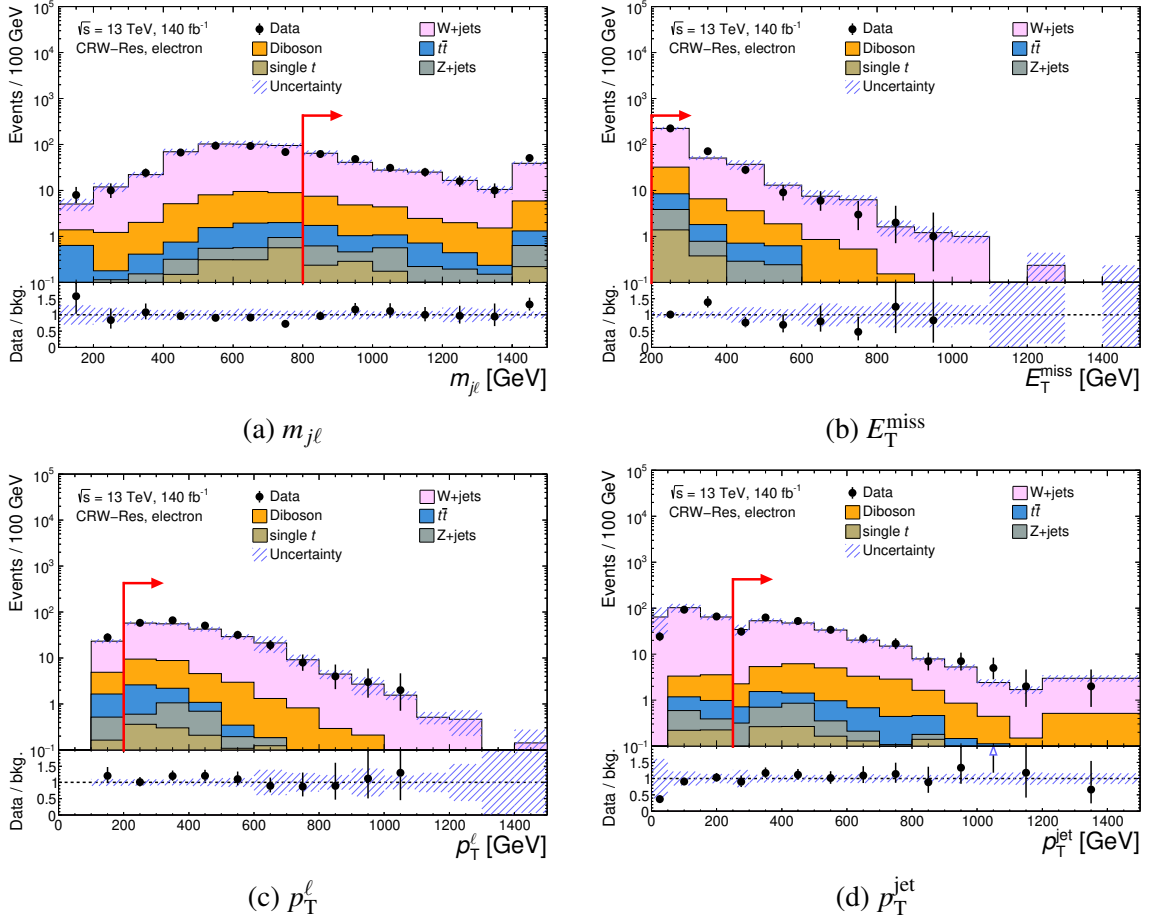


Fig. 8.2 $N - 1$ distributions in CRW-Res for the e channel after the background-only fit. The hatched bands show the combined statistical and systematic uncertainties. The bottom panels show the ratio of data to background prediction. Red arrows indicate variable thresholds for the CRW regions. The last bin includes overflow.

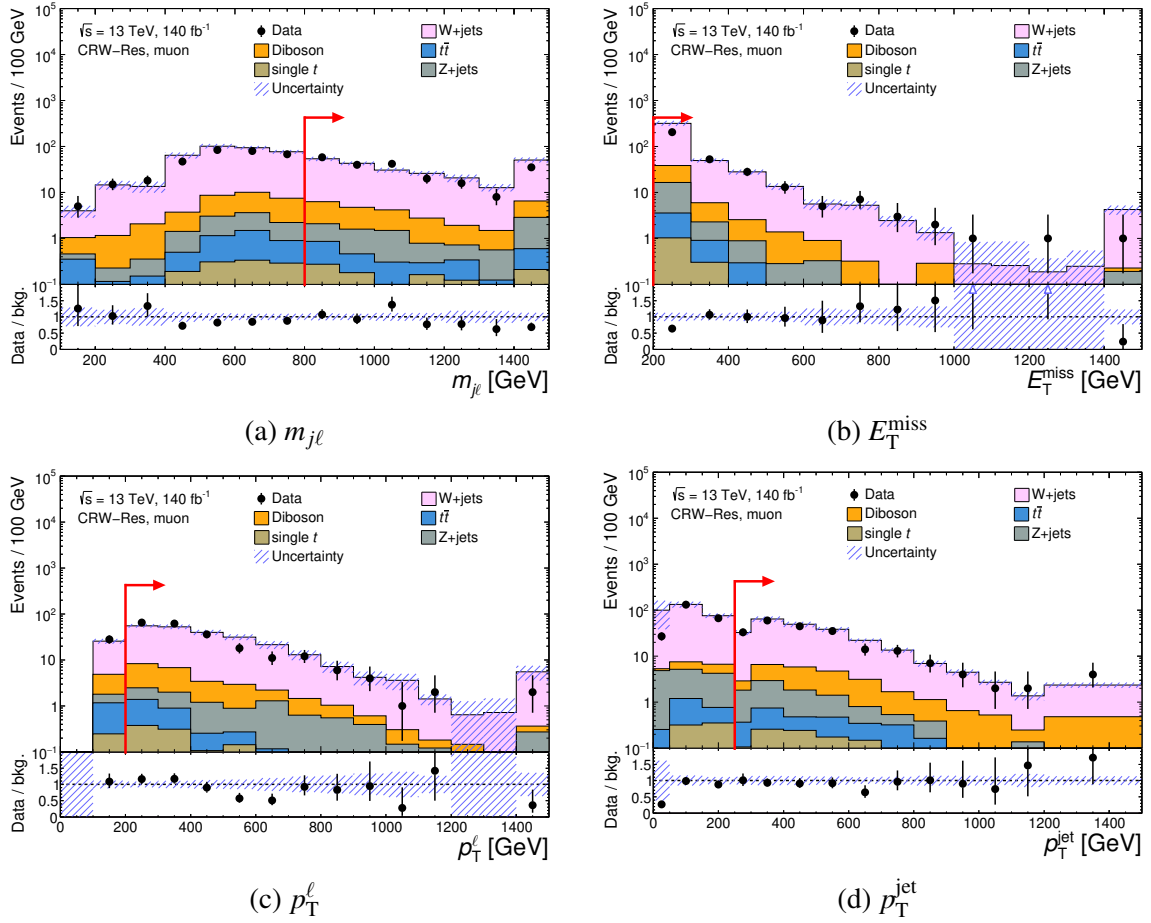


Fig. 8.3 $N - 1$ distributions in CRW-Res for the μ channel after the background-only fit. The hatched bands show the combined statistical and systematic uncertainties. The bottom panels show the ratio of data to background prediction. Red arrows indicate variable thresholds for the CRW regions. The last bin includes overflow.

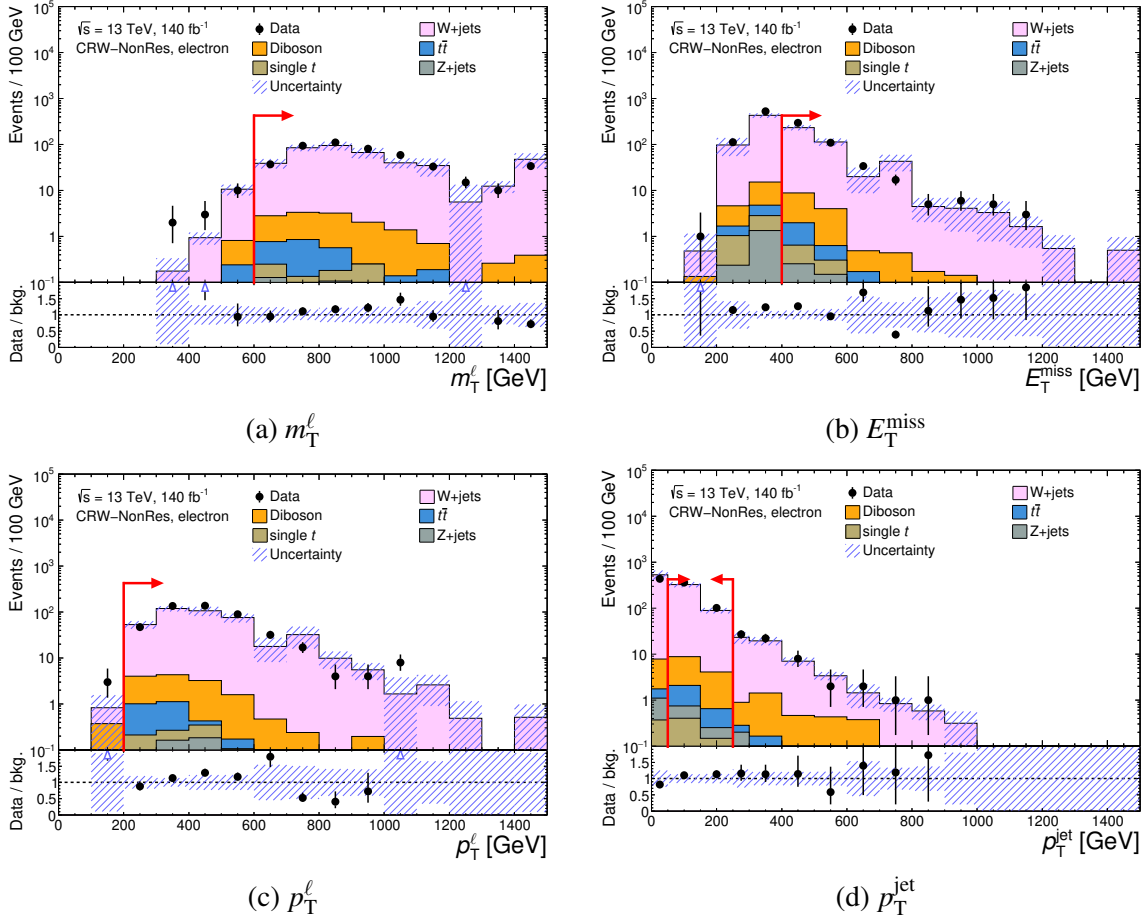


Fig. 8.4 $N - 1$ distributions in CRW-NonRes for the e channel after the background-only fit. The hatched bands show the combined statistical and systematic uncertainties. The bottom panels show the ratio of data to background prediction. Red arrows indicate variable thresholds for the CRW regions. The last bin includes overflow.

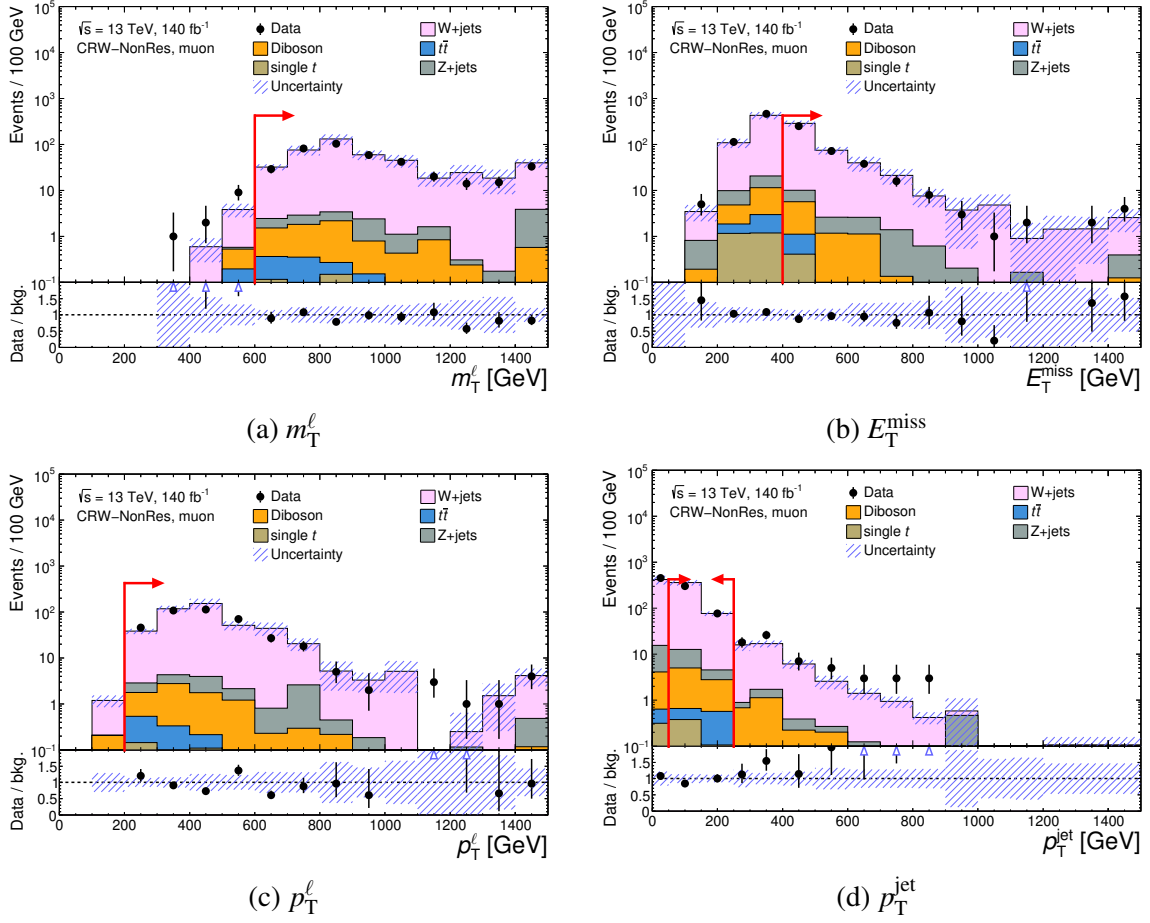


Fig. 8.5 $N - 1$ distributions in CRW-NonRes for the μ channel after the background-only fit. The hatched bands show the combined statistical and systematic uncertainties. The bottom panels show the ratio of data to background prediction. Red arrows indicate variable thresholds for the CRW regions. The last bin includes overflow.

Table 8.1 Event selections for CRW and VRW. Differences from SRs are shown in bold. The selected jet is the leading jet in CRW and the b -jet in VRW.

Requirement	CRW and VRW selections	
Trigger	Lowest-unprescaled single-lepton trigger	
τ_{had} multiplicity	$N_{\tau_{\text{had}}} = 0$	
b -jet multiplicity	$N_b = 0$ in CRW, $N_b = 1$ in VRW	
Light-lepton multiplicity	$N_\ell = 1$	
Multijet suppression	$\min \Delta\phi(E_T^{\text{miss}}, p_T^{\text{jet}}) > 0.4$ or $\frac{E_T^{\text{miss}}}{p_T^{\text{jet}(\min \Delta\phi)}} > 6$	
	Resonant	Non-resonant
p_T^{jet} [GeV]	> 250	$[50, 250]$
$\Delta\phi(\ell, E_T^{\text{miss}})$	N/A	> 1.2
E_T^{miss} [GeV]	> 200	> 400
p_T^ℓ [GeV]	> 200	> 200
m_T^ℓ [GeV]	> 200	> 600
$m_{j\ell}$ or $m_{b\ell}$ [GeV]	> 800	N/A
N_j	≤ 4	≤ 2

8.3 Top-quark background

The CRTop regions are defined by replacing the light-lepton veto in the SR selection with exactly one light lepton to correct the modelling through $t\bar{t} \rightarrow b\tau\nu\bar{b}\ell\nu$ and $Wt \rightarrow b\tau\nu\ell\nu$ processes. Since few top-quark background events pass the high- m_T requirement, single-lepton triggers are used to allow a lower E_T^{miss} threshold than in the SRs and increase the CR yields.

To further increase the CR yields, at least one b -jet is required instead of exactly one. The shape comparison in Appendix F supports combining the one- and two- b -jet samples. The $p_T^{\tau_{\text{had}}}$ and m_T thresholds are also lowered, and no $m_{b\tau_{\text{had}}}$ requirement is applied. The leading b -jet is used to calculate the b -related variables in case multiple b -jets are selected. The other selections are the same as the ones in SRs.

The CRTop definition is summarised in Table 8.2. The top-quark background purity is approximately 90% in both CRTop regions. Signal contamination is negligible because the additional light lepton is not expected in the signal topology.

Table 8.2 Event selections for CRTop. Differences from SRs are shown in bold.

Requirement	CRTop selections	
Trigger	Lowest-unprescaled single-lepton trigger	
τ_{had} multiplicity	$N_{\tau_{\text{had}}} = 1$	
b -jet multiplicity	$N_b \geq 1$	
Light-lepton multiplicity	$N_\ell = 1$	
Multijet suppression	$\min\Delta\phi(E_T^{\text{miss}}, p_T^{\text{jet}}) > 0.4$ or $\frac{E_T^{\text{miss}}}{p_T^{\text{jet}(\min\Delta\phi)}} > 6$	
p_T^ℓ [GeV]	> 28	
E_T^{miss} [GeV]	> 100	
$p_T^{\tau_{\text{had}}}$ [GeV]	> 100	
m_T [GeV]	> 100	
$m_{b\tau_{\text{had}}}$ [GeV]	N/A	
	Resonant	Non-resonant
p_T^b [GeV]	> 250	[50, 250]
$\Delta\phi(\tau_{\text{had}}, E_T^{\text{miss}})$	N/A	> 1.2
N_j	≤ 4	≤ 2

The VRTop regions are used to test the extrapolation from CRTop to final states without a light lepton, as in the SRs. Exactly two b -jets are required to enrich the top-quark background and avoid overlap with SR1b. The leading b -jet is used to calculate the b -jet kinematic variables. The $p_T^{\tau_{\text{had}}}$ and $m_{b\tau_{\text{had}}}$ thresholds are lowered to gain more statistics. The selections are summarised in Table 8.3.

Table 8.3 Event selections for VRTop. Differences from SRs are shown in bold.

Requirement	VRTop selections	
Trigger	Lowest-unprescaled E_T^{miss} trigger	
τ_{had} multiplicity	$N_{\tau_{\text{had}}} = 1$	
b -jet multiplicity	$N_b = 2$	
Light-lepton multiplicity	$N_\ell = 0$	
Multijet suppression	$\min \Delta\phi(E_T^{\text{miss}}, p_T^{\text{jet}}) > 0.4$ or $\frac{E_T^{\text{miss}}}{p_T^{\text{jet}(\min \Delta\phi)}} > 6$	
	Resonant	Non-resonant
p_T^b [GeV]	> 250	[50, 250]
$\Delta\phi(\tau_{\text{had}}, E_T^{\text{miss}})$	N/A	> 1.2
E_T^{miss} [GeV]	> 200	> 200
$p_T^{\tau_{\text{had}}}$ [GeV]	> 100	> 100
m_T [GeV]	> 100	> 200
$m_{b\tau_{\text{had}}}$ [GeV]	> 400	N/A
N_j	≤ 4	≤ 2

The CRTop $N - 1$ distributions after the background-only fit are shown separately for the electron and muon channels in Figures 8.6–8.9. There are 30 electron and 33 muon events in CRTop-Res, and 240 electron and 228 muon events in CRTop-NonRes. Similar behaviour is observed in the two lepton channels with no clear slope in the data/bkg. ratios. The electron and muon channels are therefore combined.

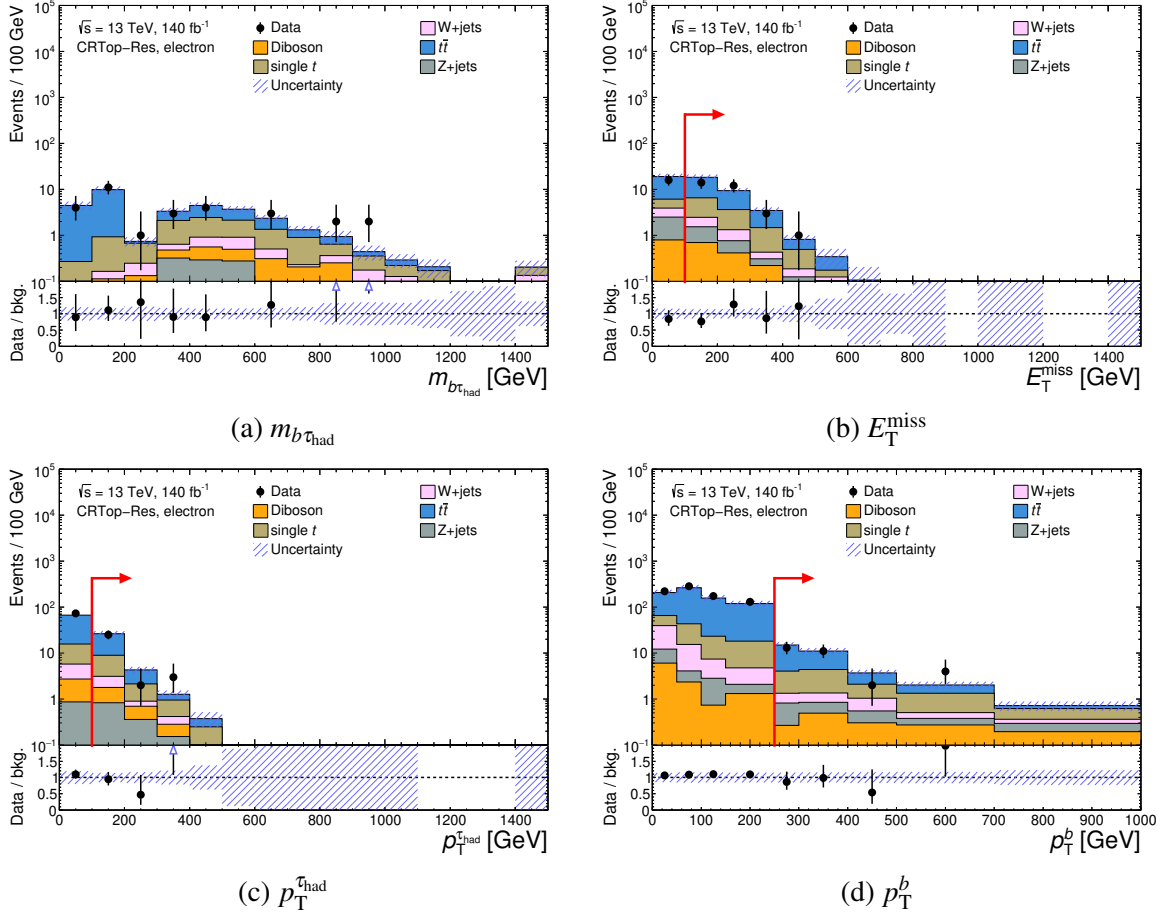


Fig. 8.6 $N-1$ distributions in CRTop-Res for the e channel after the background-only fit. The hatched bands show the combined statistical and systematic uncertainties. The bottom panels show the ratio of data to background prediction. Red arrows indicate variable thresholds for the CRTop regions. The last bin includes overflow.

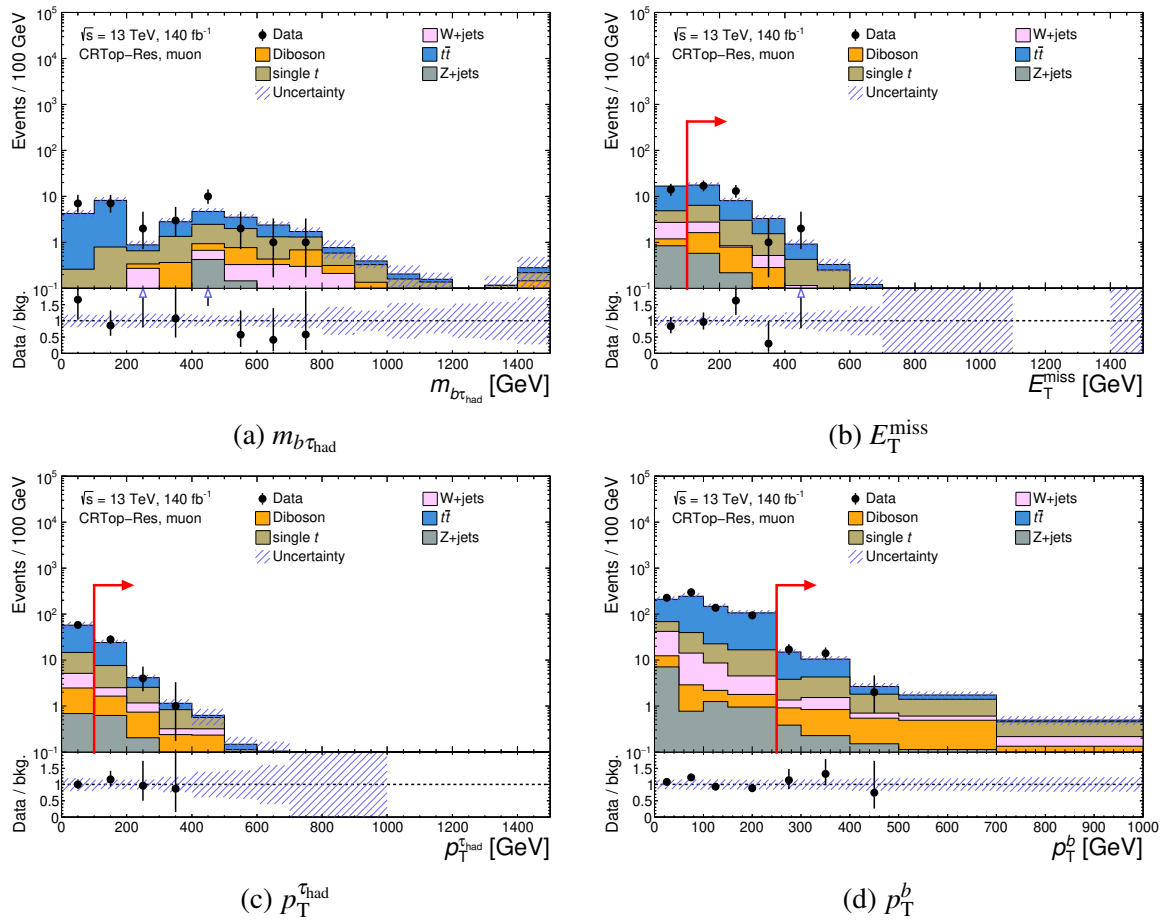


Fig. 8.7 $N - 1$ distributions in CRTop-Res for the μ channel after the background-only fit. The hatched bands show the combined statistical and systematic uncertainties. The bottom panels show the ratio of data to background prediction. Red arrows indicate variable thresholds for the CRTop regions. The last bin includes overflow.

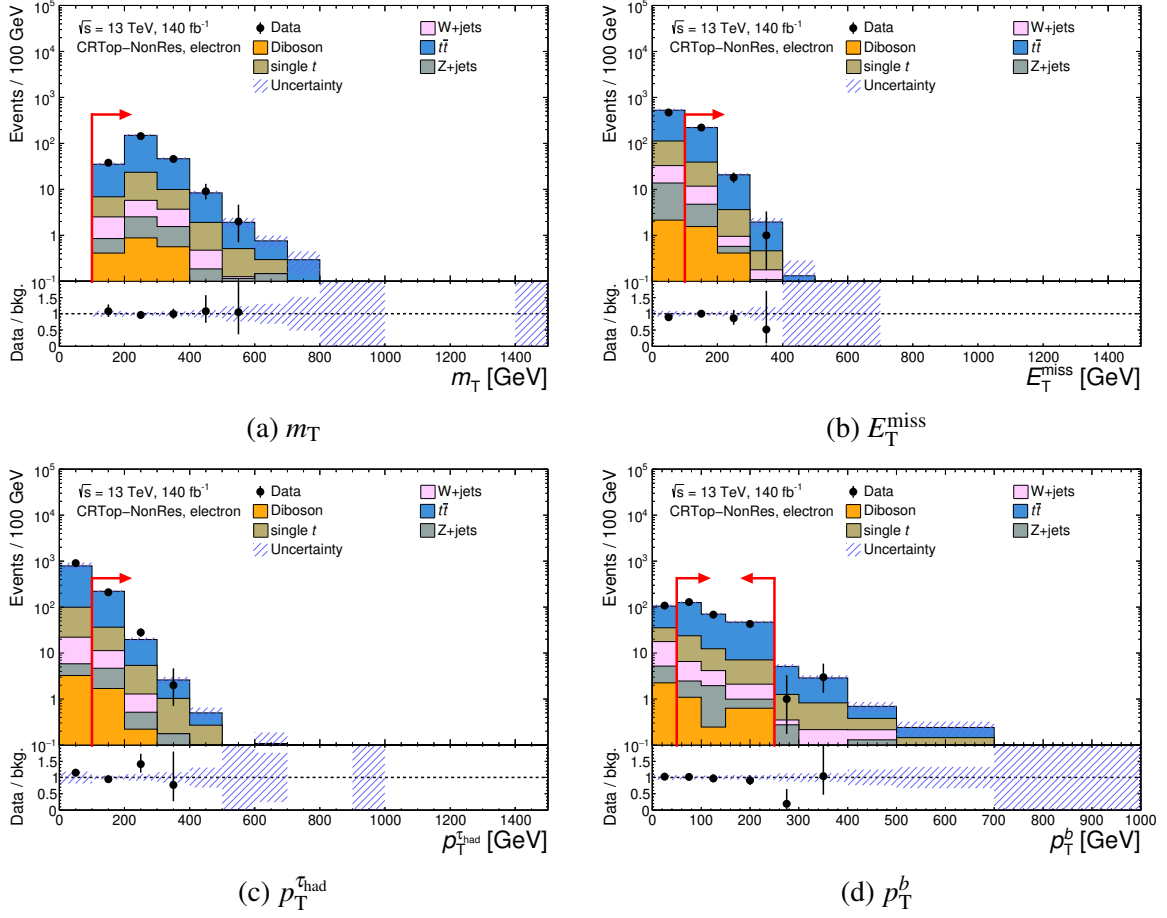


Fig. 8.8 $N - 1$ distributions in CRTop-NonRes for the e channel after the background-only fit. The hatched bands show the combined statistical and systematic uncertainties. The bottom panels show the ratio of data to background prediction. Red arrows indicate variable thresholds for the CRTop regions. The last bin includes overflow.

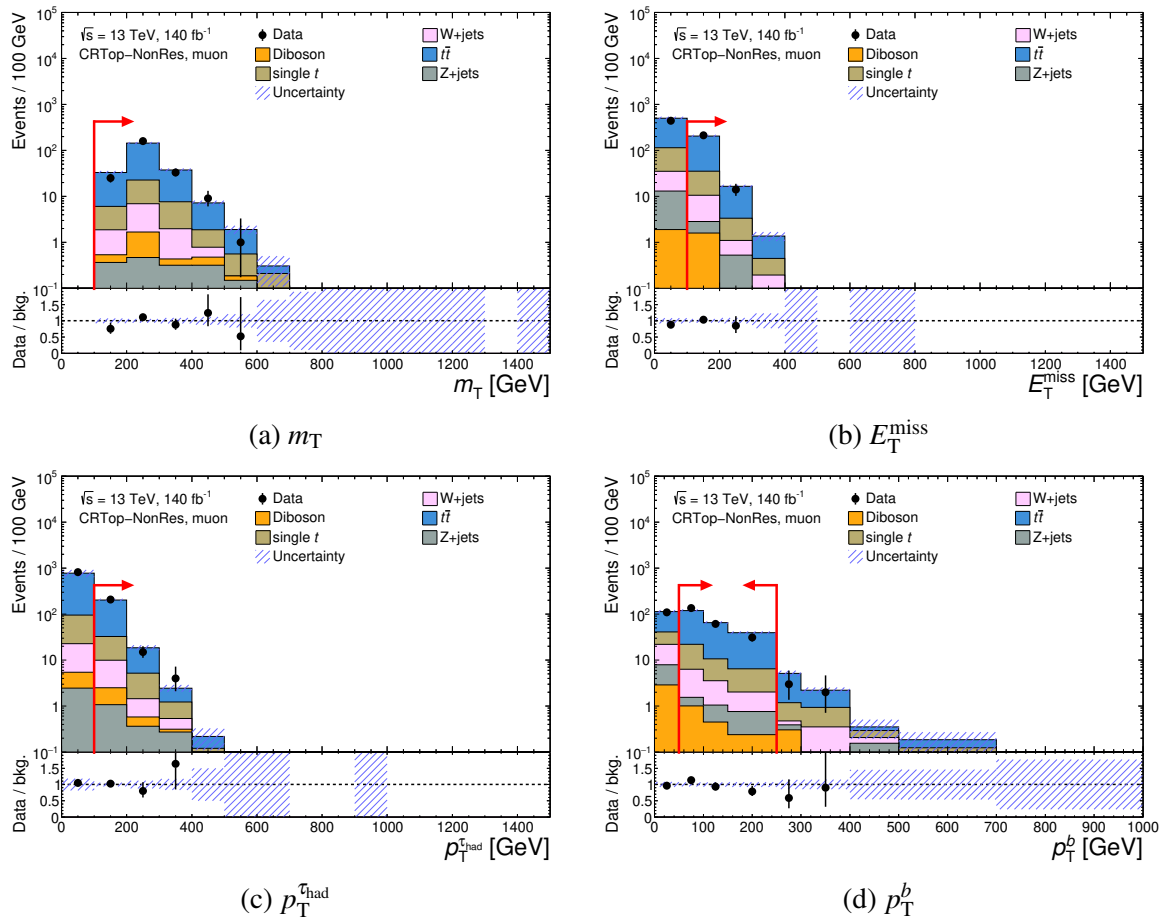


Fig. 8.9 $N - 1$ distributions in CRTop-NonRes for the μ channel after the background-only fit. The hatched bands show the combined statistical and systematic uncertainties. The bottom panels show the ratio of data to background prediction. Red arrows indicate variable thresholds for the CRTop regions. The last bin includes overflow.

8.4 Background-only fit and validation

The background-only fit is performed on the CRs to obtain the background normalisation factors. Both statistical and systematic uncertainties are included in the fit. The fit model and systematic uncertainties are described in detail in Section 10.1 and Chapter 9, respectively. The four fitted NFs are summarised in Table 8.4. The difference between λ_W^{Res} and $\lambda_W^{\text{NonRes}}$ is consistent with the expectation discussed in Section 8.2. The background predictions are tested in the VRs defined in Figure 8.1.

Table 8.4 Background normalisation factors from the background-only fit on CRs. Both statistical and systematic uncertainties are included in the fit.

Background	Resonant	Non-resonant
λ_W	1.21 ± 0.11	0.89 ± 0.08
λ_t	0.76 ± 0.16	0.81 ± 0.16

8.4.1 VRW

As in SRs, the background in VRW is dominated by $W + \text{jets}$, which contributes approximately 66.5% (68.4%) in the resonant (non-resonant) region after the background-only fit. Top-quark and diboson production contribute approximately 22.7% (28.6%) and 9.6% (2.9%), respectively.

Signal contamination is checked using samples introduced in Section 8.2. The contribution from leptonic τ decays is $(1.3 \pm 0.1)\%$ in VRW-Res and $(0.7 \pm 0.1)\%$ in VRW-NonRes, while the contribution from hadronic τ decays is negligible.

After the background-only fit, the VRW-Res yield agrees with the prediction within about 1σ , while a deviation of about 2σ is observed in the VRW-NonRes region. The post-fit $N - 1$ distributions are shown in Figures 8.10 and 8.11. Checks with finer E_T^{miss} bins and different E_T^{miss} thresholds show that the excess is localised in $425 < E_T^{\text{miss}} < 450$ GeV. And the neighbouring E_T^{miss} regions show better agreement between data and prediction. No indication of a local detector problem is found in the (η, ϕ) distributions of the leptons and b -jets. Good agreement is also found in the additional low-mass VRs. These results identify no specific modelling problem and are consistent with the excess being a statistical fluctuation. More details are presented in Appendix D.

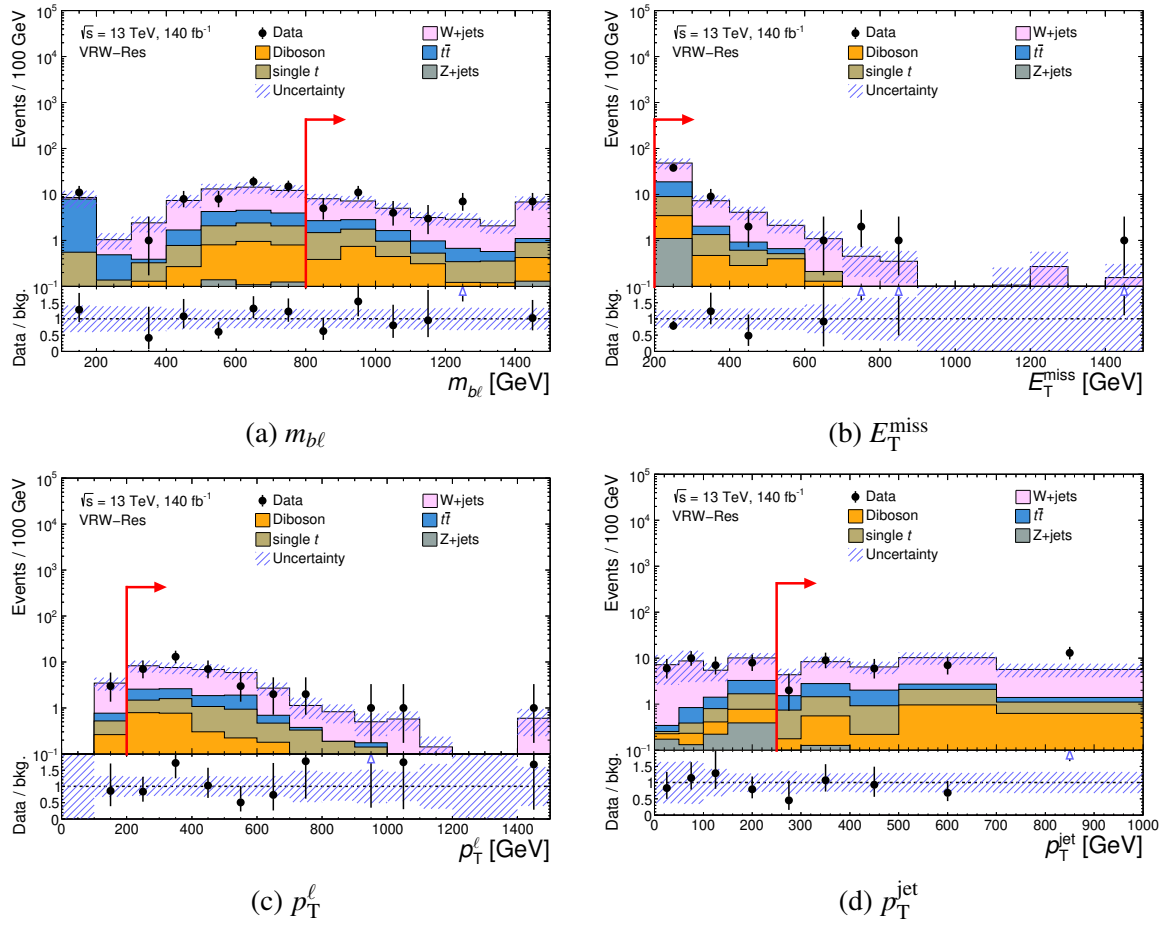


Fig. 8.10 $N - 1$ distributions in VRW-Res for the combined e and μ channels after the background-only fit. The hatched bands show the combined statistical and systematic uncertainties. The bottom panels show the ratio of data to background prediction. Red arrows indicate variable thresholds for the VRW regions. The last bin includes overflow.

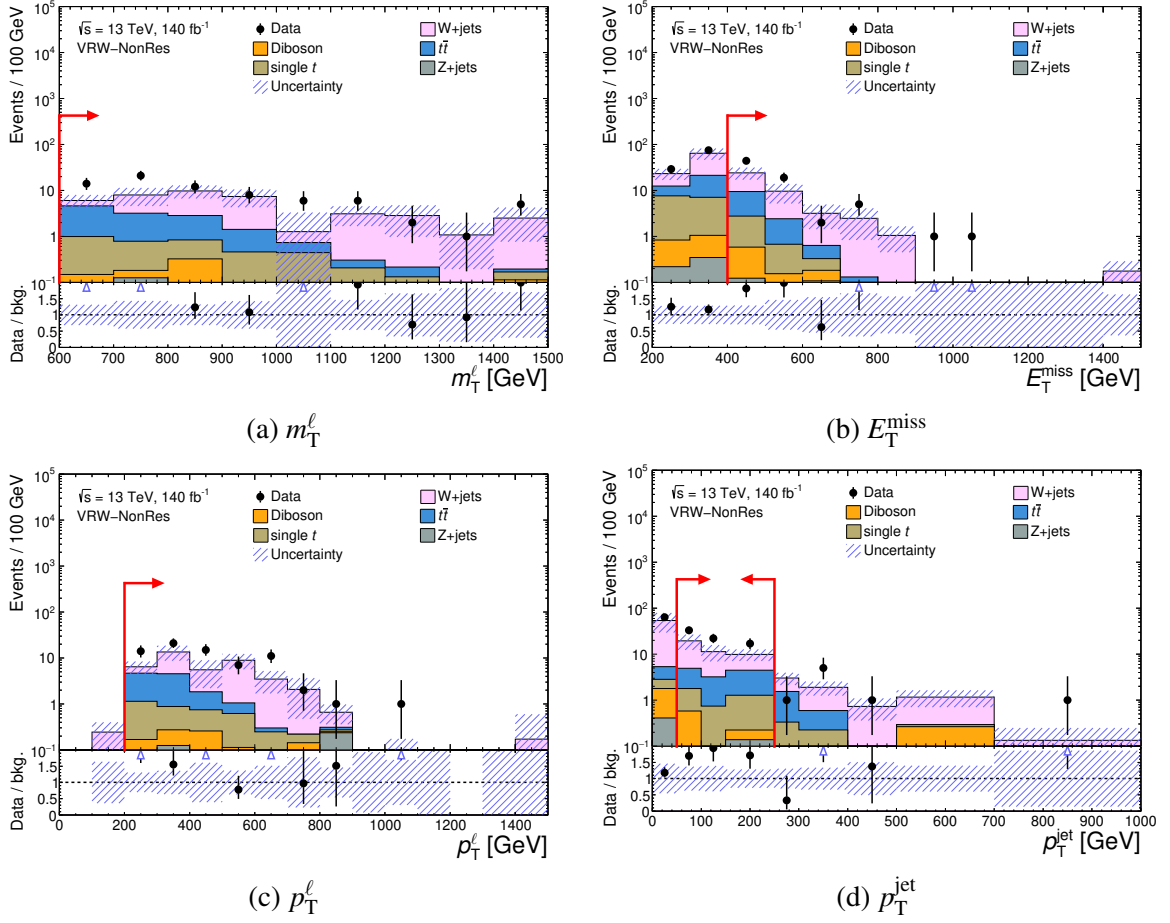


Fig. 8.11 $N - 1$ distributions in VRW-NonRes for the combined e and μ channels after the background-only fit. The hatched bands show the combined statistical and systematic uncertainties. The bottom panels show the ratio of data to background prediction. Red arrows indicate variable thresholds for the VRW regions. The last bin includes overflow.

8.4.2 VRTop

The background in VRTop is dominated by top-quark production with fractions of approximately 67% and 86% in the resonant and non-resonant regions, respectively. Specifically, single-top and $t\bar{t}$ production contribute approximately 35% and 32% of the total background in the resonant region, respectively, while $t\bar{t}$ dominates the non-resonant region with a contribution of approximately 80%. Signal contamination is checked using the same samples and benchmarks as in VRW, which is $(1.2 \pm 0.1)\%$ in VRTop-Res and $(0.6 \pm 0.03)\%$ in VRTop-NonRes.

After the background-only fit, the expected yields are 13.0 ± 3.9 and 30.1 ± 8.1 in VRTop-Res and VRTop-NonRes, compared with 19 and 30 observed events, respectively. The observed yields agree with the predictions within about 1σ in both regions. The post-fit $N - 1$ distributions are shown in Figures 8.12 and 8.13. The agreement supports the extrapolation from CRTop to final states without a light lepton.

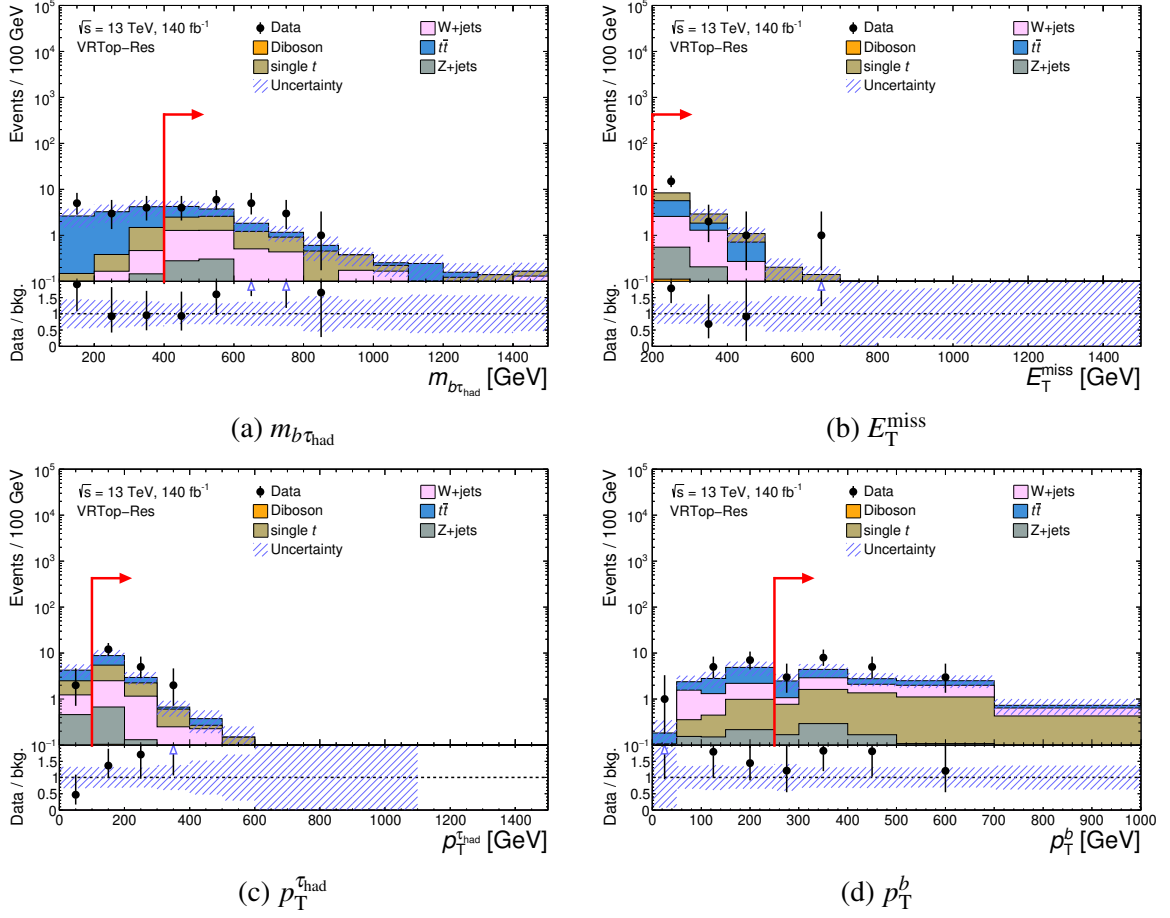


Fig. 8.12 $N - 1$ distributions in VRTop-Res after the background-only fit. The hatched bands show the combined statistical and systematic uncertainties. The bottom panels show the ratio of data to background prediction. Red arrows indicate variable thresholds for the VRTop regions. The last bin includes overflow.

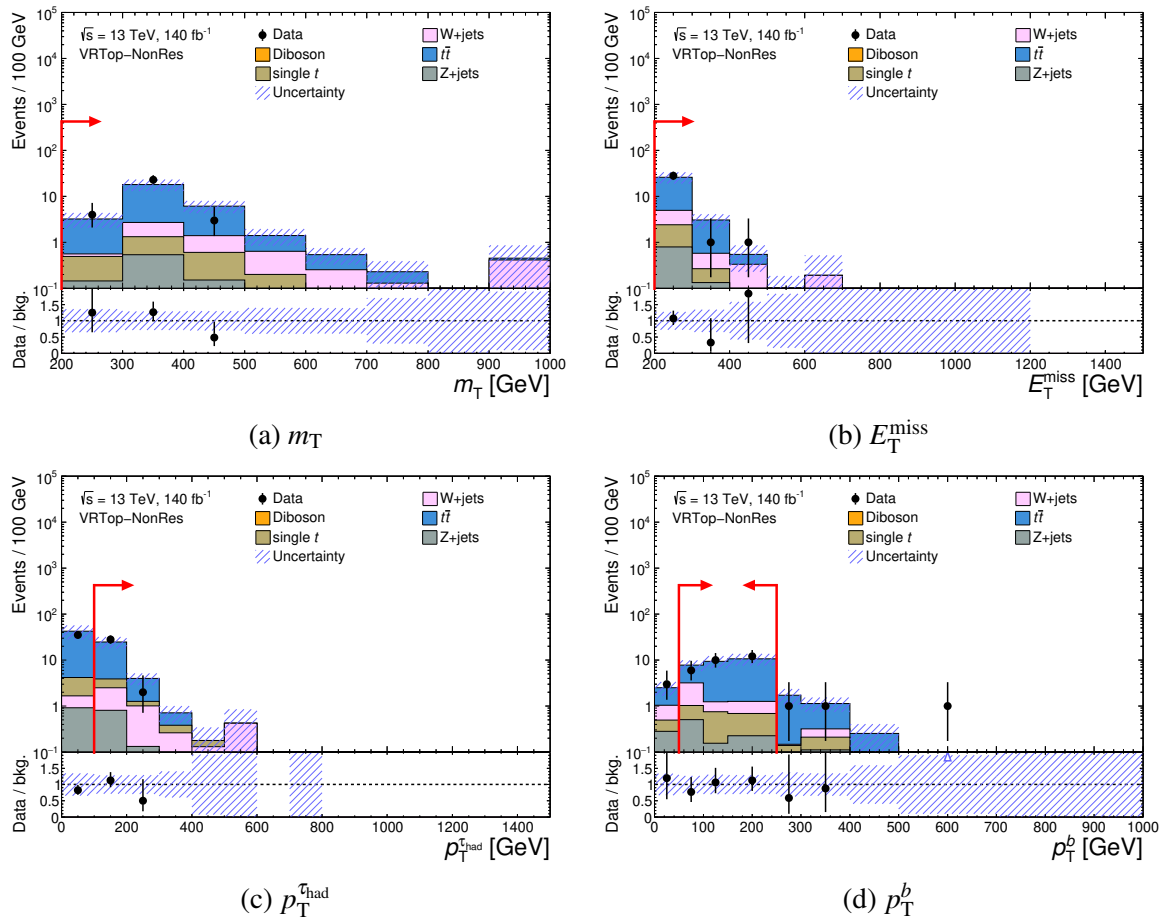


Fig. 8.13 $N - 1$ distributions in VRTop-NonRes after the background-only fit. The hatched bands show the combined statistical and systematic uncertainties. The bottom panels show the ratio of data to background prediction. Red arrows indicate variable thresholds for the VRTop regions. The last bin includes overflow.

2036

Chapter 9

2037

Systematic Uncertainties

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Systematic uncertainties in this analysis includes the experimental and theoretical modelling

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uncertainties. Their sources and evaluation are discussed in this chapter.

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9.1 Experimental uncertainties

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Experimental uncertainties originate from the detector response and object calibrations. They are propagated through the event weights or reconstructed objects. The evaluation and treatment of these uncertainties are summarised below.

2044

Luminosity and pile-up The 2015-2018 integrated luminosity has an uncertainty of 0.83%, as discussed in Section 5.1. It is applied to the signal and backgrounds normalisation. The pile-up uncertainty is evaluated by varying the data-derived μ profile used to calculate the pile-up correction weights for simulated events. Its nominal rescaling factor, $1/1.03$, is changed to $1/1.07$ and $1/0.99$.

2049

Hadronic tau energy scale and efficiencies The TES uncertainty in the low- p_T region is measured using $Z \rightarrow \tau\tau$ events, while the uncertainty in the high- p_T region is evaluated using test-beam data and MC comparisons [123, 124]. Efficiency uncertainties are evaluated for τ_{had} reconstruction and RNN identification using $t\bar{t}$ events, as well as for electron rejection. For the Tight ID working point, the scale-factor uncertainties are 5–11%.

2054

Jets and missing transverse momentum The JES and JER uncertainties are evaluated using the calibration methods described in Section 6.5. The $\vec{p}_T^{\text{miss}}$ uncertainty is evaluated by recalculating the $\vec{p}_T^{\text{miss}}$ for each uncertainty from the hard-object momenta and the soft term. The soft-term uncertainties are derived from measurements in $Z \rightarrow ee$ events [121].

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2058

Jet flavour tagging The flavour-tagging uncertainties are evaluated separately for the b -jet tagging efficiency and the c -jet and light-jet mistagging efficiencies. They include calibration

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sample statistics, background contamination and modelling uncertainties in the efficiency measurements described in Section 6.6.

Electrons and muons Electron efficiency uncertainties are evaluated using $Z \rightarrow ee$ and $J/\psi \rightarrow ee$ events, while energy-scale and resolution uncertainties are obtained by comparing the measured and simulated calorimeter response [105]. The muon efficiency, momentum-scale and resolution uncertainties are determined using $Z \rightarrow \mu\mu$ and $J/\psi \rightarrow \mu\mu$ events [108, 110]. In particular, the resolution uncertainty has a sizeable effect on the high- p_T muons in this analysis.

9.2 Theoretical uncertainties

Considering that the background fit uses the total event yield in each region, the transfer-factor (TF) method is introduced to evaluate the modelling uncertainties in the $W + \text{jets}$ and top-quark backgrounds. Given that the $W + \text{jets}$ (top-quark background) contamination in CRTop (CRW) is below 5% (2%), the modelling uncertainties on these small contributions are neglected. The uncertainties types for the $W + \text{jets}$ and top-quark backgrounds are then introduced.

9.2.1 Transfer-factor method

The background yield in an SR or VR is estimated as

$$N_{bg}^{SR(VR),data} = \lambda \times N_{bg}^{SR(VR),MC} = TF \times N_{bg}^{CR,data}. \quad (9.1)$$

The NF λ is introduced as a free-floating parameter as described in Equation 8.1. TF is defined by

$$TF = \frac{N_{bg}^{SR(VR),MC}}{N_{bg}^{CR,MC}}. \quad (9.2)$$

Suppose a theory uncertainty changes the background yield by $\Delta N_{bg} = N_{bg}^{sysMC} - N_{bg}^{nomMC}$, where $nomMC$ and $sysMC$ denote the nominal and alternative MC predictions, respectively.

2083 Propagating this change to TF gives

$$\begin{aligned}
 2084 \quad TF + \Delta TF &= \frac{N_{bg}^{SR(VR),MC} + \Delta N_{bg}^{SR(VR),MC}}{N_{bg}^{CR,MC} + \Delta N_{bg}^{CR,MC}} \\
 2085 \quad &= TF \times \frac{N_{bg}^{SR(VR),sysMC}}{N_{bg}^{SR(VR),nomMC}} \bigg/ \frac{N_{bg}^{CR,sysMC}}{N_{bg}^{CR,nomMC}}. \quad (9.3)
 \end{aligned}$$

2086 If an uncertainty produces the same fractional change in both regions, it cancels in the double
 2087 ratio, giving $\Delta TF = 0$. The remaining relative uncertainty is applied only to the SR or VR
 2088 yield through the factor $1 \pm \Delta TF / TF$.

2089 9.2.2 $W + \text{jets}$

- 2090 • **QCD scales:** the uncertainty is evaluated from the envelope of seven-point predictions ob-
 2091 tained by varying the matrix-element and parton-shower renormalisation and factorisation
 2092 scales [125].
- 2093 • **PDF:** the uncertainty is evaluated using the Hessian NNPDF3.0 NNLO uncertainty scheme,
 2094 including alternative PDF sets and two alternative α_s values.
- 2095 • **$W + \text{HF}/W + \text{LF}$ ratio:** an uncertainty is assigned to heavy-to-light-flavour ratio when
 2096 extrapolating from the b -vetoed CRW to regions with b jets. A value of 30% is adopted,
 2097 motivated by flavour-dependent normalisation differences observed in an earlier ATLAS
 2098 measurement [126].

2099 9.2.3 Top-quark background

- 2100 • **QCD scales, PDF and PDF α_s :** the uncertainties are evaluated by reweighting the nominal
 2101 samples, as for $W + \text{jets}$.
- 2102 • **Matrix element (NLO matching):** the uncertainty is evaluated by changing the definition
 2103 of the region in which PYTHIA shower emissions are vetoed to avoid overlap with radiation
 2104 generated by POWHEG.
- 2105 • **Parton shower and hadronisation:** the uncertainty is obtained by comparing the nominal
 2106 samples with alternative samples using HERWIG 7.
- 2107 • **ISR (FSR) α_s :** the uncertainty is evaluated using internal weights for initial-state (final-
 2108 state) radiation.
- 2109 • **h_{damp} :** the uncertainty is evaluated by varying h_{damp} , which regulates hard radiation
 2110 recoiling against the $t\bar{t}$ system [127].

- 2111 • **$t\bar{t}$ NNLO:** the uncertainty is derived by reweighting the nominal sample to the NNLO
2112 prediction [128].
- 2113 • **$t\bar{t}$ shower recoil:** the uncertainty is evaluated by assigning the recoil from subsequent
2114 gluon emissions to the parent top quark instead of the b quark in the top quark decay.
- 2115 • **$Wt\text{--}t\bar{t}$ interference:** the uncertainty is evaluated by comparing the nominal Wt sample
2116 with an alternative sample in which the overlapping $t\bar{t}$ contribution is subtracted [129].

2117 9.2.4 Other backgrounds and signal

2118 The $Z \rightarrow \nu\nu$ background with a jet misidentified as a τ_{had} is assigned with a 100% nor-
2119 malisation uncertainty, while diboson and the remaining $Z \rightarrow \tau\tau$ backgrounds are assigned
2120 with a conservative 30% normalisation uncertainty. Signal uncertainties are evaluated using
2121 QCD scale and PDF variations, resulting in a 10–20% uncertainty. The background theory
2122 uncertainties are summarised in Tables 9.1 and 9.2 for the resonant and non-resonant regions,
2123 respectively.

Table 9.1 Theory uncertainties considered in the resonant regions. The last four sources are applied to the overall normalisation rather than the TF.

Source	SR1b	$\Delta TF/TF$ Up (Down) VRW	VRTop
W + jets QCD scales	0.10(−0.17)	0.04(−0.04)	—
W + jets PDF	< 0.001	0.007	—
W + jets PDF α_s	−0.02(−0.001)	0.02(0.001)	—
W + jets EW correction	0.04	−0.02	—
Top NLO matching	−0.06	0.08	0.23
Top parton shower	0.16	0.1	0.11
Top μ_F	0.12(−0.09)	0.03(−0.02)	0.01(−0.01)
Top μ_R	0.15(−0.19)	−0.01(0.01)	0.005(−0.008)
Top ISR	−0.04(−0.01)	0.01(−0.01)	−0.01(0.01)
Top FSR	0.19(−0.24)	−0.04(0.08)	−0.02(0.22)
Top PDF	−0.051	0.043	0.032
Top PDF α_s	0.01(−0.01)	0.01(−0.004)	0.002(0.0002)
Top h_{damp}	0.09	0.20	0.15
$t\bar{t}$ NNLO	−0.05	0.02	0.01
$t\bar{t}$ shower recoil	−0.08	0.04	−0.04
Wt interference	−0.1	−0.18	−0.25
$Z \rightarrow \nu\nu$ with fake τ_{had}	1.0	0	1.0
Diboson modelling	0.3	0.3	0.3
Z + jets modelling	0.3	0.3	0.3
W + jets heavy flavour*	0.3	0.3	0.3

Table 9.2 Theory uncertainties considered in the non-resonant regions. The last four sources are applied to the overall normalisation rather than the TF.

Source	SR1b	$\Delta TF/TF$ Up (Down) VRW	VRTop
W + jets QCD scales	0.14(−0.16)	0.01(−0.03)	—
W + jets PDF	−0.003	0.002	—
W + jets PDF α_s	0.03(−0.03)	0.02(−0.004)	—
W + jets EW correction	0.01	0.01	—
Top NLO matching	−0.15	0.01	−0.03
Top parton shower	−0.01	−0.14	−0.07
Top μ_F	−0.05(0.05)	0.01(−0.01)	0.003(−0.004)
Top μ_R	−0.01(0.02)	0.04(−0.05)	0.01(−0.02)
Top ISR	0.16(−0.13)	−0.0003(0.002)	−0.01(< 0.001)
Top FSR	0.01(−0.25)	0.06(−0.14)	0.003(−0.04)
Top PDF	−0.025	−0.002	0.003
Top PDF α_s	−0.01(0.002)	0.005(−0.001)	0.001(< 0.001)
Top h_{damp}	−0.07	−0.01	0.05
$t\bar{t}$ NNLO	−0.003	−0.004	−0.01
$t\bar{t}$ shower recoil	−0.13	0.09	0.03
Wt interference	0.03	−0.04	−0.009
$Z \rightarrow \nu\nu$ with fake τ_{had}	1.0	0	1.0
Diboson modelling	0.3	0.3	0.3
Z + jets modelling	0.3	0.3	0.3
W + jets heavy flavour*	0.3	0.3	0.3

* The W + jets heavy-flavour uncertainty is applied independently in the resonant and non-resonant regions.

Chapter 10

Statistical Method and Results

This chapter presents the statistical method and results. The statistical model and hypothesis test are introduced first in Section 10.1. The unblinded results are then presented in Section 10.2, followed by the exclusion limits summarised in Section 10.3. A discussion on the future improvement is presented in Section 10.4.

10.1 Statistical method

10.1.1 Likelihood model

The likelihood model is used to test for the presence of a BSM signal. In a maximum-likelihood fit, the model parameters are determined by maximising the likelihood function. This function combines Poisson probabilities for the observed event yields in the fitted regions with constraint terms for the nuisance parameters (NPs):

$$\mathcal{L}(\mu, \vec{\lambda}, \vec{\theta}) = \prod_{\text{SR,CR}} \text{Pois}(N_{\text{obs}} | N_{\text{exp}}) \prod_k C_k(\theta_k), \quad (10.1)$$

where μ is the signal strength, $\vec{\lambda} = \{\lambda_W^{\text{Res}}, \lambda_W^{\text{NonRes}}, \lambda_t^{\text{Res}}, \lambda_t^{\text{NonRes}}\}$ contains the background normalisation factors, and $\vec{\theta}$ denotes the NPs representing the systematic uncertainties introduced in Chapter 9 and the MC statistical uncertainties. $C_k(\theta_k)$ is the constraint term for the corresponding NP and may take a Gaussian, log-normal or Poisson form, depending on the uncertainty. Statistical uncertainties due to the finite size of the simulated background samples are included using the Barlow–Beeston lite prescription [130]. Since only the total event yield in each region is used in the fit, the shape systematics are pruned.

The expected yield in each region is given by

$$N_{\text{exp}} = \mu N_{\text{sig}}(\vec{\theta}) + \sqrt{\mu} N_{\text{int}}(\vec{\theta}) + \lambda_W N_W^{\text{MC}}(\vec{\theta}) + \lambda_t N_t^{\text{MC}}(\vec{\theta}) + N_{\text{others}}^{\text{MC}}(\vec{\theta}). \quad (10.2)$$

Here, N_{sig} and N_{int} are the pure-BSM and interference yields, respectively. The remaining terms are the MC background yields, with the W + jets and top-quark contributions scaled by the normalisation factors introduced in Chapter 8. The appropriate Res or NonRes NFs are used in each region.

$\sqrt{\mu}$ is the parameter of interest (POI) in the analysis since the interference term is scaled by it. $\mu \geq 0$ is therefore required. The background-only model and a signal-plus-background (S+B) hypothesis are obtained for $\mu = 0$ and $\mu > 0$, respectively.

10.1.2 Fit strategy

Two fit configurations are used in the analysis: a background-only fit and an S+B fit. The background normalisations are first determined by a background-only fit with $\mu = 0$ in CRs. The four background NFs and the NPs are allowed to vary to maximise the likelihood. Their fitted values are used to predict the background yields in the SRs and VRs. This fit is also used to assess the constraints on the NPs from the CR data alone. The S+B fit then extends the likelihood fit to the SRs and allows μ to vary to determine the best-fit value $\hat{\mu}$.

10.1.3 Test statistic and NP impacts

The test statistic is defined using the profile-likelihood ratio [131]. Profiling means that at each fixed μ , the background NFs and NPs are fitted again to obtain the maximum likelihood value. It then compares the maximum likelihood at fixed μ with the likelihood at $\hat{\mu}$. Thus,

$$q_\mu = -2 \ln \frac{\mathcal{L}(\mu, \vec{\lambda}_\mu, \vec{\theta}_\mu)}{\mathcal{L}(\hat{\mu}, \vec{\hat{\lambda}}, \vec{\hat{\theta}})}. \quad (10.3)$$

Here, $\vec{\hat{\lambda}}$ and $\vec{\hat{\theta}}$ are the best-fit values of the background NFs and NPs. A large q_μ indicates poor agreement with the tested hypothesis.

Expanding q_μ around $\hat{\mu}$, using $q_{\hat{\mu}} = 0$ and neglecting higher-order terms, gives the local Gaussian approximation:

$$q_\mu \simeq \frac{(\mu - \hat{\mu})^2}{\sigma_\mu^2}, \quad (10.4)$$

where σ_μ is the standard deviation of $\hat{\mu}$. The uncertainty on $\hat{\mu}$, $\Delta\hat{\mu}$, is then determined by scanning μ to find the range satisfying $q_\mu \leq 1$, corresponding to one standard deviation from $\hat{\mu}$.

The impact of each NP on μ is estimated from the post-fit covariance matrix using a local linear approximation:

$$\Delta\mu = \sigma_\mu \rho \sigma_{\text{NP}}, \quad (10.5)$$

where σ_μ and σ_{NP} are the post-fit uncertainties in μ and the NP, respectively, and ρ is their correlation coefficient. A larger $|\Delta\mu|$ indicates a greater impact of the NP on μ .

10.1.4 Hypothesis testing and CL_s method

Hypothesis testing is performed to determine if the observed data are consistent with the SM-only or SM+BSM scenario. The null hypothesis refers to the SM-only scenario, and the alternative hypothesis refers to a scenario with the leptoquark signal. In case no significant deviation is observed from the SM, upper limits are set by the CL_s method with the one-sided statistic

$$\tilde{q}_\mu = \begin{cases} q_\mu, & \hat{\mu} \leq \mu, \\ 0, & \hat{\mu} > \mu. \end{cases} \quad (10.6)$$

The CL_s is defined by the probability of observing a given μ using $\tilde{q}_\mu$ under the S+B and background-only hypotheses:

$$\text{CL}_{s+b} = P(\tilde{q}_\mu \geq \tilde{q}_\mu^{\text{obs}} \mid \mu), \quad \text{CL}_b = P(\tilde{q}_\mu \geq \tilde{q}_\mu^{\text{obs}} \mid 0), \quad (10.7)$$

where $\tilde{q}_\mu^{\text{obs}}$ is evaluated using the observed data. And their ratio gives the CL_s :

$$\text{CL}_s = \frac{\text{CL}_{s+b}}{\text{CL}_b}. \quad (10.8)$$

CL_s is more robust against background fluctuations compared with CL_{s+b} . The upper limit on μ is then found at $\text{CL}_s = 0.05$, which corresponds to 95% confidence level (CL).

10.1.5 Limit calculation

Two methods are applied in this analysis to calculate the upper limit on μ :

Toy experiments The toy calculation uses pseudo-experiments to calculate CL_s :

1. At each tested μ , toy datasets are generated under the S+B and background-only hypotheses, with event yields fluctuated according to their Poisson distributions.
2. Each toy is fitted with μ fixed and free parameters, NFs and NPs, to calculate $\tilde{q}_\mu$.
3. The fractions of toys with $\tilde{q}_\mu \geq \tilde{q}_\mu^{\text{obs}}$ determine the CL_s . The calculation is repeated across μ values, and the $\text{CL}_s = 0.05$ crossing is interpolated to obtain the upper limit.

Asymptotic approximation In general, the toy experiments method gains robustness against background fluctuations at the cost of computational time. On the other hand, the asymptotic calculation uses analytic formulae to approximate the $\tilde{q}_\mu$ distributions. An Asimov dataset is constructed using the expected event yields in the SRs, with NPs and NFs obtained from the background-only fit and a signal scaled by μ . The tail probabilities are then evaluated from these formulae, and the same μ scan is performed. The calculation with this method is faster and avoids finite-toy fluctuations, but the approximation can be less reliable for low event yields. In fact, the results calculated by these two methods differ by at most approximately 5% in the study.

As a result, the low SR yields motivate the use of toys for the left-handed limits, with interference included and 10 000 toys per hypothesis at each scan point. And the right-handed limits in the SR1b are calculated as a supplementary interpretation using the asymptotic approximation without interference.

10.2 Results

10.2.1 Background-only fit

The fitted NFs are given in Table 8.4. The CR and VR distributions are discussed in Sections 8.2–8.4. The background-only fit results in each region are shown in Fig. 10.1.

As a result, no significant deviation from the SM expectation is observed. In SR1b-Res and SR1b-NonRes, the expected background yields are 2.69 ± 0.83 and 1.88 ± 0.71 , compared with observed yields of 3 and 2, respectively. The corresponding predictions in SR0b-Res and SR0b-NonRes are 36.4 ± 7.3 and 38.3 ± 9.9 , compared with observed yields of 33 and 45. Figures 10.2 and 10.3 show the post-fit N-1 distributions. The post-fit yields in the CRs, VRs and SRs are summarised in Tables 10.1, 10.2 and 10.3, respectively.

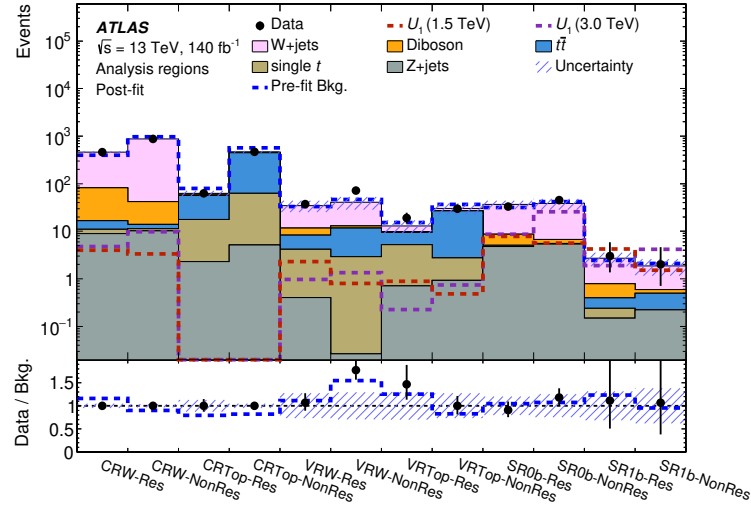


Fig. 10.1 Comparison of the data and predicted background event yields for four CRs, four VRs and four SRs in the upper panel. The background prediction is shown after a background-only fit in the CRs. The signal expectation for $\mu = 1$ is shown for illustrative purposes at the parameter points $(M_U, g_U, \beta_L^{23}) = (1.5 \text{ TeV}, 1.5, 0.6)$ and $(3.0 \text{ TeV}, 2.5, 1.0)$ with a red dashed line and a violet dashed line, respectively. The ratio of the data to the background prediction is shown in the lower panel, separately for post-fit (black points) and pre-fit (dashed blue line) background. The size of the combined statistical and systematic uncertainty in the background prediction is indicated by a blue hatched band.

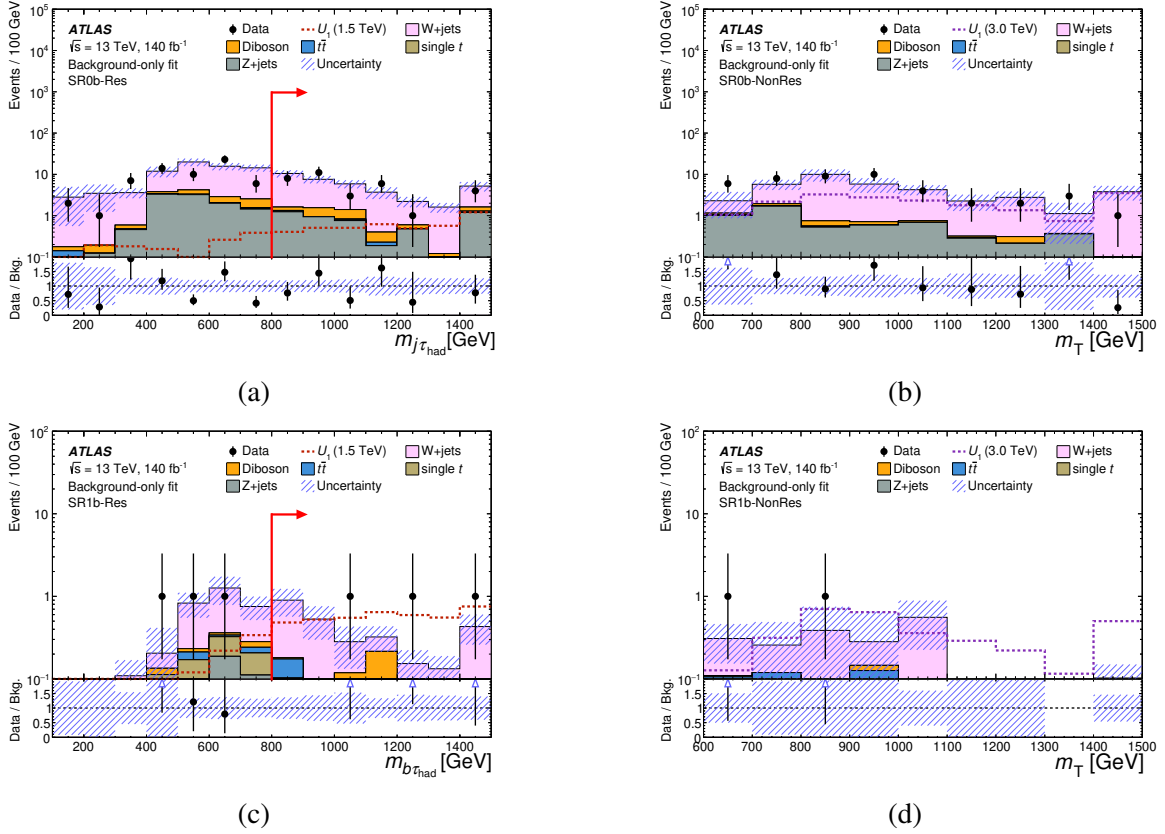


Fig. 10.2 N-1 distributions of (a) $m_{j\tau_{had}}$ in SR0b-Res, (b) m_T in SR0b-NonRes, (c) $m_{b\tau_{had}}$ in SR1b-Res and (d) m_T in SR1b-NonRes. The background-only fit is used for the background prediction. The signal expectations at parameter points $(M_U, g_U, \beta_L^{23}) = (1.5 \text{ TeV}, 1.5, 0.6)$ and $(3.0 \text{ TeV}, 2.5, 1.0)$ are shown as red dashed lines in (a) and (c) and violet dashed lines in (b) and (d), respectively. The hatched bands show the combined statistical and systematic uncertainties. The bottom panels show the ratio of data to background prediction. Blue upward triangles indicate ratio values exceeding the displayed range. Red arrows indicate variable thresholds for the SRs. The last bin includes overflow.

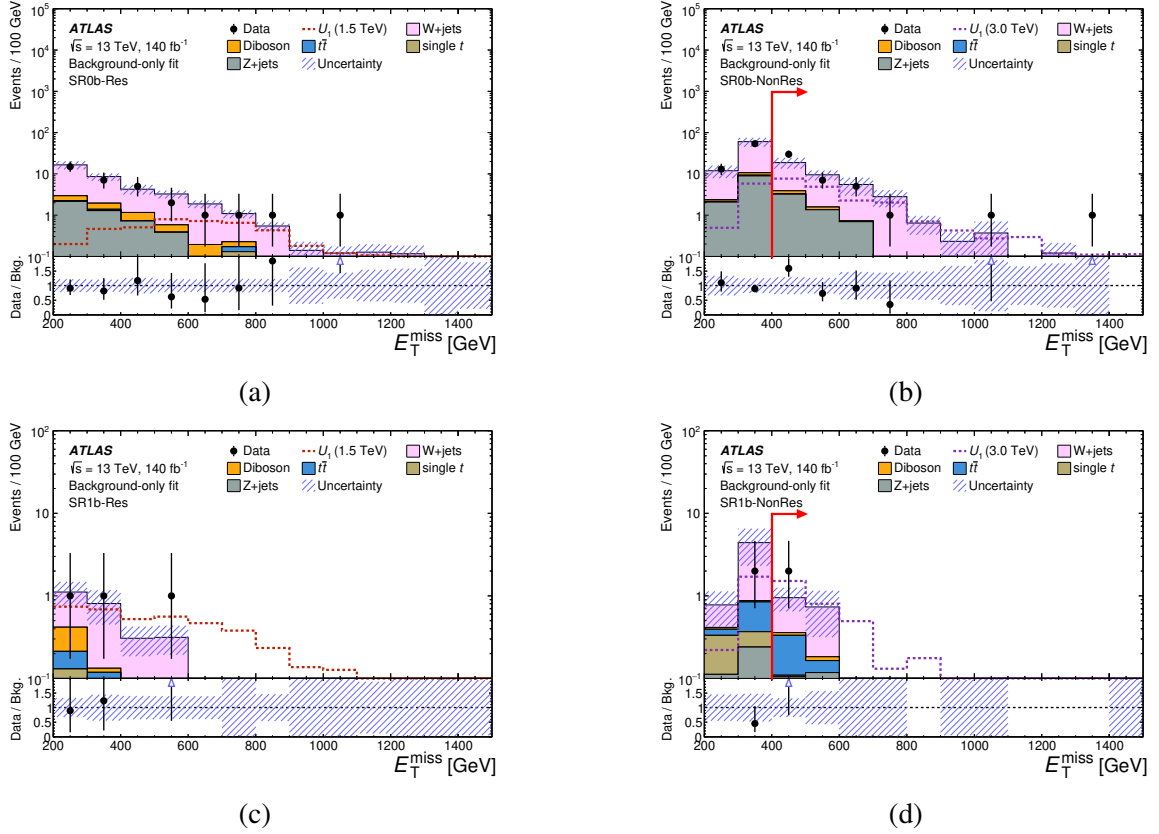


Fig. 10.3 The unblinded N-1 E_T^{miss} distributions in (a) SR0b-Res, (b) SR0b-NonRes, (c) SR1b-Res and (d) SR1b-NonRes before the E_T^{miss} requirement. The background-only fit is used for the background prediction. The signal expectations at parameter points $(M_U, g_U, \beta_L^{23}) = (1.5 \text{ TeV}, 1.5, 0.6)$ and $(3.0 \text{ TeV}, 2.5, 1.0)$ are shown as red dashed lines in (a) and (c) and violet dashed lines in (b) and (d), respectively. The hatched bands show the combined statistical and systematic uncertainties. The bottom panels show the ratio of data to background prediction. Blue upward triangles indicate ratio values exceeding the displayed range. Red arrows indicate variable thresholds for the SRs.

The NP pulls are defined as:

$$\text{pull} = \frac{\hat{\theta} - \theta_0}{\Delta\theta}, \quad (10.9)$$

where θ_0 and $\Delta\theta$ are the nominal NP value and pre-fit uncertainty, respectively. The NP is considered to be constrained by the data if the normalised uncertainty is smaller than one.

The fit results show that the NPs remain close to their pre-fit values and their uncertainties are nearly unchanged. This is expected, as differences between the observed and simulated yields in the CRs can mostly be absorbed by the free NFs.

The correlations among the NPs are also studied. These correlations are included through the post-fit covariance matrix when evaluating the uncertainties in the expected background yields. In the S+B fit, they also affect the uncertainty on $\hat{\mu}$. The leading correlations involve the TES detector NP and the top-quark NFs, while subleading correlations are observed between the muon momentum-resolution and trigger-efficiency NPs and the $W + \text{jets}$ NFs. As shown in Fig. 10.4, pre-fit $\pm 1\sigma$ variations of the TES detector NP change the top-quark background yield by 11.2% and 17.4% in CRTop-Res and CRTop-NonRes, respectively, while the corresponding variations of the two muon NPs change the $W + \text{jets}$ yields in CRW by a few percent. An NP variation that increases a dominant background yield in a CR can be compensated by a decrease in its NF, resulting in these correlations. Their impacts on the μ fit in the S+B fit are discussed in the following section.

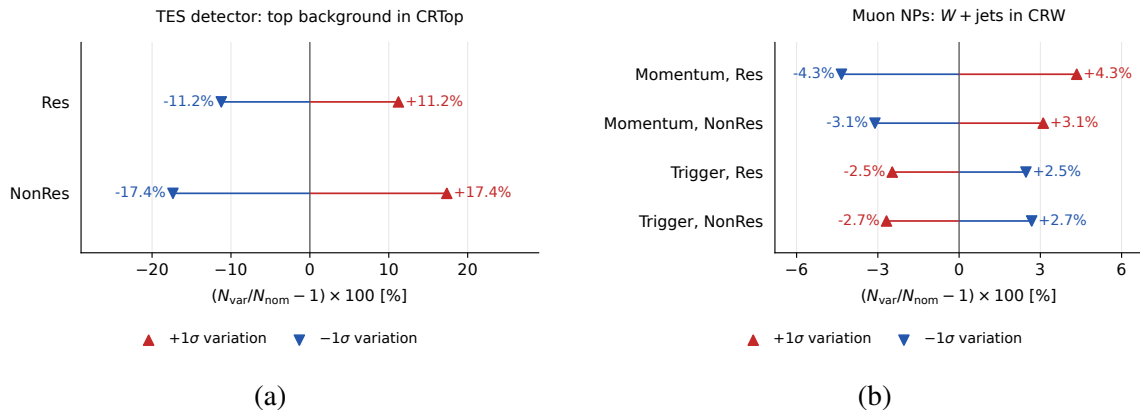


Fig. 10.4 Changes of (a) the expected top-quark background ($t\bar{t}$ and single-top) yields in CRTop for TES detector variations and (b) the expected $W + \text{jets}$ yields in CRW for muon momentum-resolution and trigger-efficiency variations. Red and blue markers show the $+1\sigma$ and -1σ variations, respectively.

Table 10.1 Observed and expected yields in the CRs after a background-only fit.

Process	CRW-Res	CRW-NonRes	CRTop-Res	CRTop-NonRes
$t\bar{t}$	5.6 ± 1.3	2.6 ± 0.6	38.8 ± 5.8	385.5 ± 19.1
Single- t	2.2 ± 0.5	1.1 ± 0.2	15.3 ± 2.4	57.9 ± 3.8
$Z \rightarrow \ell\ell$ +jets	7.0 ± 1.2	9.5 ± 1.2	0.03 ± 0.03	0.42 ± 0.44
$Z \rightarrow \nu\nu$ +jets	1.2 ± 0.3	0.42 ± 0.16	< 0.01	< 0.01
$Z \rightarrow \tau\tau$ +jets	0.72 ± 0.06	0.28 ± 0.08	2.26 ± 0.37	4.73 ± 0.60
W +jets	380 ± 22	830 ± 30	3.0 ± 0.7	15.8 ± 3.1
Diboson	65.3 ± 2.4	28.0 ± 1.9	3.6 ± 0.5	3.7 ± 0.5
Total	462 ± 22	872 ± 30	63.0 ± 7.9	468 ± 22
Data	462	872	63	468

Table 10.2 Observed and expected yields in the VRs after a background-only fit in the CRs.

Process	VRW-Res	VRW-NonRes	VRTop-Res	VRTop-NonRes
$t\bar{t}$	4.1 ± 1.8	8.7 ± 3.3	4.18 ± 1.76	24.0 ± 7.3
Single- t	3.8 ± 1.6	2.9 ± 1.1	4.54 ± 1.93	1.8 ± 0.6
$Z \rightarrow \ell\ell$ +jets	0.37 ± 0.16	< 0.01	< 0.01	< 0.01
$Z \rightarrow \nu\nu$ +jets	0.016 ± 0.003	0.03 ± 0.02	0.27 ± 0.27	0.78 ± 0.78
$Z \rightarrow \tau\tau$ +jets	0.019 ± 0.016	< 0.01	0.44 ± 0.15	0.15 ± 0.06
W +jets	23.1 ± 8.4	27.8 ± 10.6	3.22 ± 1.21	3.3 ± 1.4
Diboson	3.3 ± 1.2	1.18 ± 0.40	0.31 ± 0.11	0.06 ± 0.04
Total	34.7 ± 9.3	40.7 ± 12.0	13.0 ± 3.9	30.1 ± 8.1
Data	37	72	19	30

Table 10.3 Observed and expected yields in the SRs after a background-only fit in the CRs.

Process	SR0b-Res	SR0b-NonRes	SR1b-Res	SR1b-NonRes
$t\bar{t}$	0.24 ± 0.11	0.11 ± 0.10	0.16 ± 0.12	0.27 ± 0.13
Single- t	0.05 ± 0.02	0.02 ± 0.01	0.09 ± 0.11	< 0.01
$Z \rightarrow \ell\ell$ +jets	0.03 ± 0.03	< 0.01	< 0.01	< 0.01
$Z \rightarrow \nu\nu$ +jets	2.8 ± 2.8	2.3 ± 2.3	0.04 ± 0.05	0.05 ± 0.06
$Z \rightarrow \tau\tau$ +jets	1.90 ± 0.66	3.1 ± 1.2	0.11 ± 0.05	0.17 ± 0.06
W +jets	27.7 ± 6.0	31.5 ± 8.7	1.9 ± 0.7	1.3 ± 0.6
Diboson	3.7 ± 1.2	1.18 ± 0.47	0.39 ± 0.18	0.10 ± 0.04
Total	36.4 ± 7.3	38.3 ± 9.9	2.69 ± 0.83	1.88 ± 0.71
Data	33	45	3	2

10.2.2 Unblinded S+B fit

Given that no significant excess is observed in the four SRs, the S+B fits have been performed for various signal parameters to set upper limits.

Two sets of signal parameters are picked to study the NP impacts in different LQ mass regions. Figures 10.5 and 10.6 show the leading NP impacts at $(M_U, g_U, \beta_L^{23}) = (1.5 \text{ TeV}, 1.5, 0.2)$ and $(3.0 \text{ TeV}, 2.0, 1.0)$, respectively, together with the pulls and post-fit uncertainties. The corresponding best-fit values are

$$\hat{\mu} = 0.35_{-0.35}^{+2.62} = 0.35_{-0.29}^{+2.47} (\text{stat})_{-0.34}^{+0.45} (\text{syst}) \quad (10.10)$$

at $(M_U, g_U, \beta_L^{23}) = (1.5 \text{ TeV}, 1.5, 0.2)$, and

$$\hat{\mu} = 0.66_{-0.66}^{+1.28} = 0.66_{-0.64}^{+0.30} (\text{stat})_{-0.51}^{+1.21} (\text{syst}) \quad (10.11)$$

at $(M_U, g_U, \beta_L^{23}) = (3.0 \text{ TeV}, 2.0, 1.0)$.

The data statistical uncertainty dominates for $M_U = 1.5 \text{ TeV}$, while both statistical and systematic uncertainties are important for $M_U = 3.0 \text{ TeV}$. For $M_U = 1.5 \text{ TeV}$, the largest NP impact comes from the $W + \text{jets}$ heavy flavour uncertainty described in Section 9.2.2, followed by the $W + \text{jets}$ QCD-scale uncertainties. For $M_U = 3.0 \text{ TeV}$, the TES detector NP gives the largest impact, followed by the $W + \text{jets}$ QCD-scale uncertainties. This is expected: since the CRs use e/μ in place of τ_{had} , the tau-related uncertainties are not cancelled out and remain important. And the uncertainties from $W + \text{jets}$ also have the large impact since it's the dominant background in the SRs.

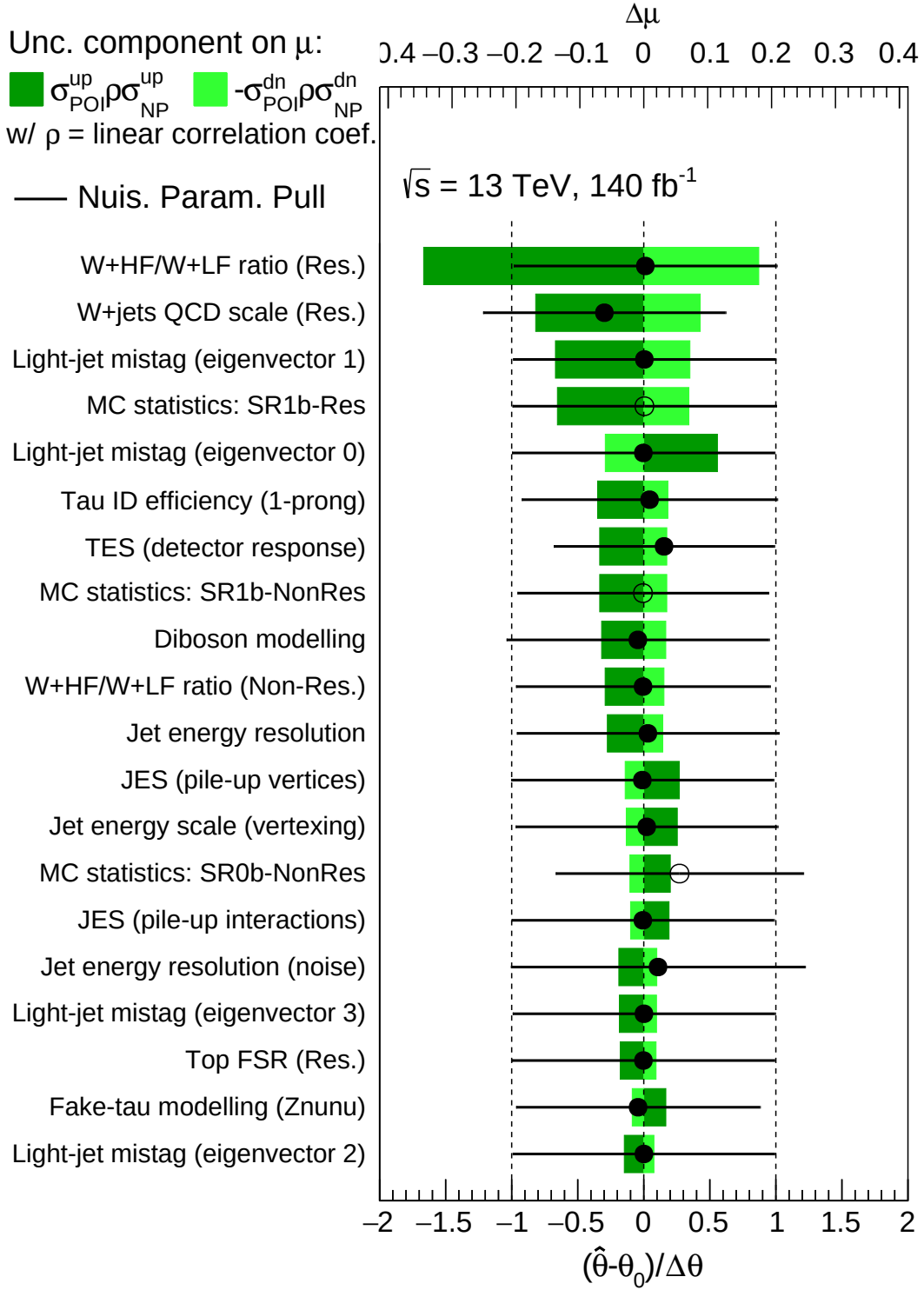


Fig. 10.5 The unblinded ranking plot of NPs for $(M_U, g_U, \beta_L^{23}) = (1.5 \text{ TeV}, 1.5, 0.2)$, using the four CRs and four SRs. NP impacts are obtained from the post-fit covariance matrix and are shown by the green bars. Black markers and horizontal error bars show the NP pulls and their post-fit uncertainties, respectively. The vertical dashed lines at ± 1 indicate the pre-fit $\pm 1\sigma$ range. The quantity labelled $\Delta\mu$ represents the individual NP impacts on the fitted parameter $\sqrt{\mu}$. Most experimental systematic variations are symmetrised.

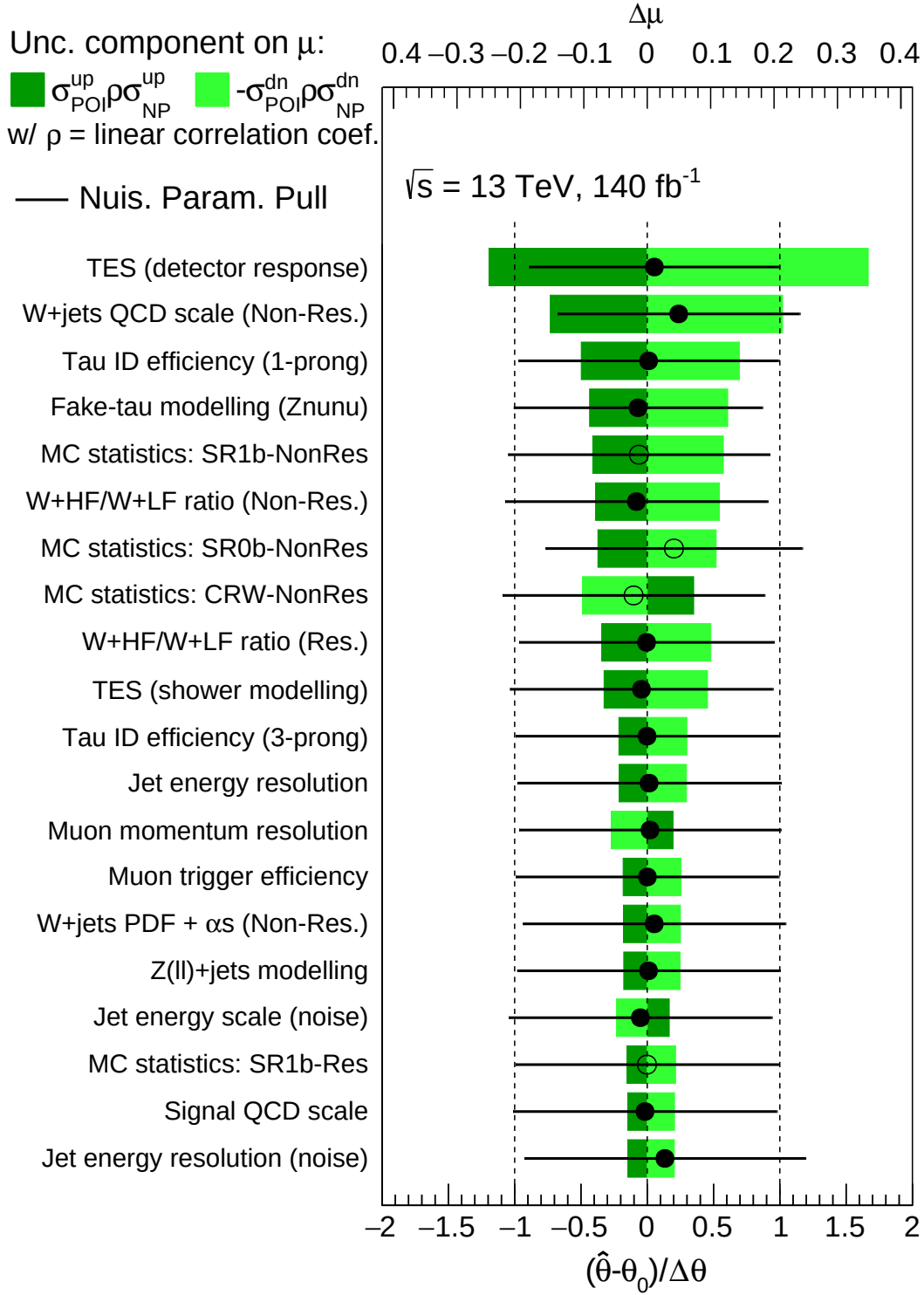


Fig. 10.6 The unblinded ranking plot of NPs for $(M_U, g_U, \beta_L^{23}) = (3.0 \text{ TeV}, 2.0, 1.0)$, using the four CRs and four SRs. NP impacts are obtained from the post-fit covariance matrix and are shown by the green bars. Black markers and horizontal error bars show the NP pulls and their post-fit uncertainties, respectively. The vertical dashed lines at ± 1 indicate the pre-fit $\pm 1\sigma$ range. The quantity labelled $\Delta\mu$ represents the individual NP impacts on the fitted parameter $\sqrt{\mu}$. Most experimental systematic variations are symmetrised.

10.3 Exclusion limits

10.3.1 left-handed scenario

The contributions of the individual SRs are shown in Fig. 10.7 using asymptotic calculations. As shown in Fig. 10.7(a), the resonant SRs provide the stronger sensitivity at low mass, while the non-resonant SRs become more sensitive as on-shell production is suppressed at higher mass. Their combination gives the best expected limit. Meanwhile, as shown in Fig. 10.7(b), SR1b provides most of the sensitivity, while adding SR0b improves the combined limit particularly towards large β_L^{23} and small g_U . This is expected since the larger signal contribution without a b jet would be enhanced by the second-generation coupling. These comparisons explain the role of each SR. The exclusion limits are then obtained with pseudo-experiments.

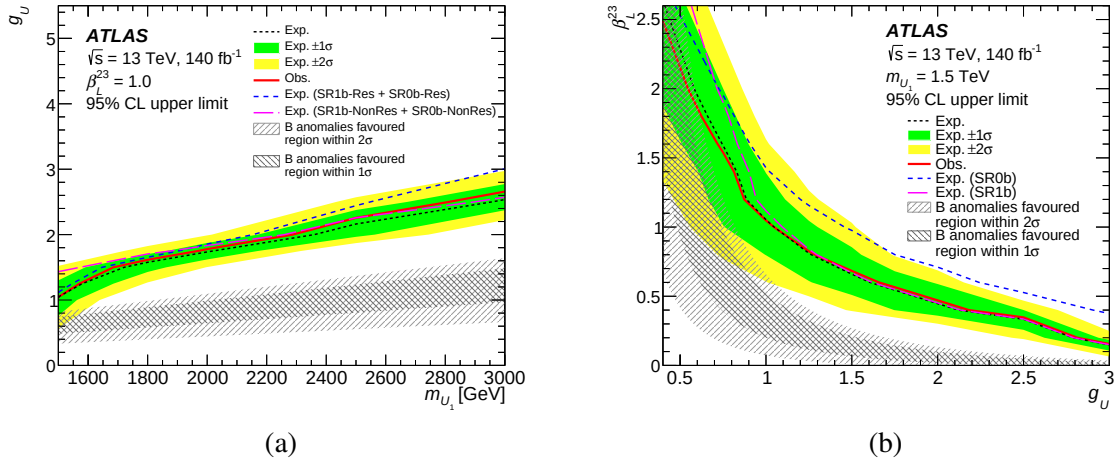


Fig. 10.7 Observed (red full line) and expected (black dashed line) 95% CL upper limits on (a) g_U as a function of M_U at $\beta_L^{23} = 1.0$ and (b) β_L^{23} as a function of g_U at $M_U = 1.5$ TeV. The shaded regions are favoured by B anomalies at the 2σ level, converted from the C_{LL}^c limit in Fig. 3.1 using Eq. (3.19). The individual expected limits from the resonant and non-resonant SRs are also overlaid in (a), and those from SR0b and SR1b are overlaid in (b). In this figure, limits are calculated using the asymptotic formulae [131] rather than pseudo-experiments.

Figures 10.8 and 10.9 present the 95% CL exclusions for the purely left-handed scenario using all four SRs. At $M_U = 1.5$ TeV, the observed (expected) upper limits on g_U are 1.13 (1.11) for $\beta_L^{23} = 1.0$ and 0.52 (0.53) for $\beta_L^{23} = 2.2$. At $M_U = 2.5$ TeV, they are 2.30 (2.19) and 1.22 (1.18), respectively. The hatched regions are obtained by mapping the low-energy constraint on C_{LL}^c shown in Fig. 3.1 through Eq. (3.19). The results test the parameter space at large β_L^{23} and complement the $b\tau$ searches discussed in Section 3.1.4 [55].

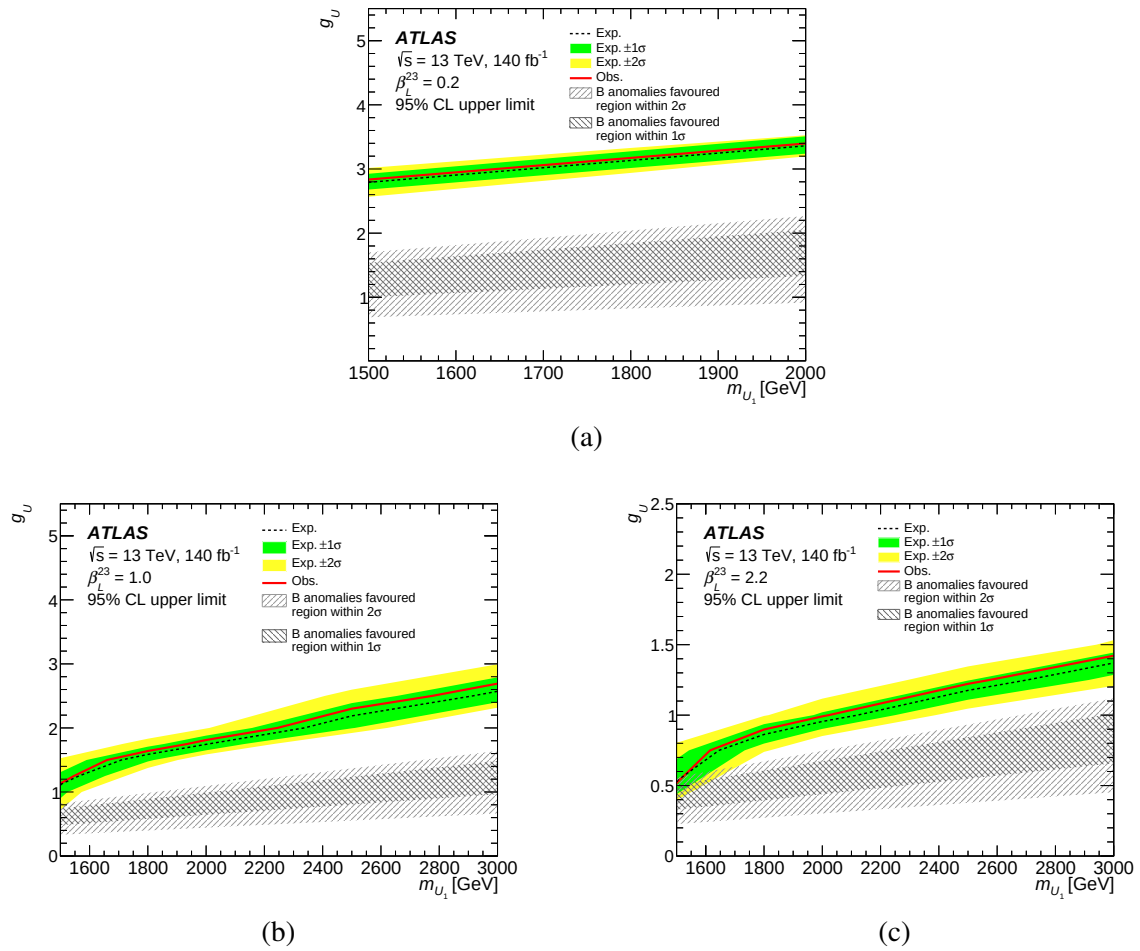


Fig. 10.8 Observed (red full line) and expected (black dashed line) 95% CL upper limits on g_U as a function of M_U , at (a) $\beta_L^{23} = 0.2$, (b) 1.0 and (c) 2.2. The shaded regions are favoured by B anomalies at the 1σ and 2σ levels, converted from the C_{LL}^c limit in Fig. 3.1 using Eq. (3.19).

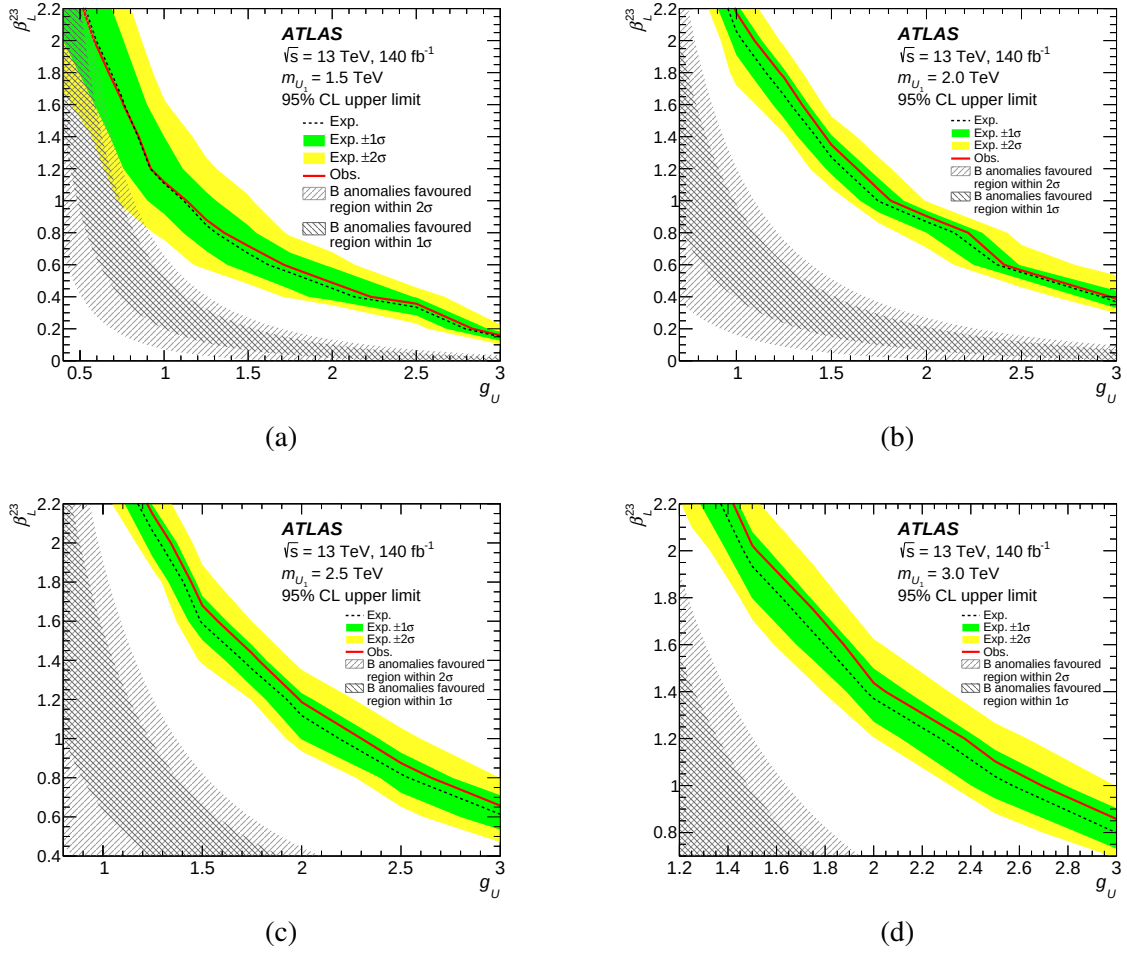


Fig. 10.9 Observed (red full line) and expected (black dashed line) 95% CL upper limits on β_L^{23} as a function of g_U at (a) $M_U = 1.5$ TeV, (b) 2.0 TeV, (c) 2.5 TeV and (d) 3.0 TeV. The shaded regions are favoured by B anomalies at the 1σ and 2σ levels, converted from the C_{LL}^c limit in Fig. 3.1 using Eq. (3.19).

10.3.2 right-handed scenario

The interpretation for the right-handed scenario uses the signals at $|\beta_R^{33}| = 1$ and $\beta_L^{33} = 1$ introduced in Section 3.1.3. The signal yields are obtained with the reweighting procedure described in Section 5.2.

Figure 10.10 compares the observed 95% CL upper limits on the visible cross section for $|\beta_R^{33}| = 0$ and 1 at $g_U = 1$ and $\beta_L^{23} = 0.2$, without considering the interference. It is defined as:

$$\sigma_{\text{vis}}^{95} = \mu^{95} \sigma A \epsilon, \quad (10.12)$$

where σ is the theoretical signal cross section. A and ϵ are the signal acceptance and selection efficiency, respectively. μ^{95} is the observed upper limit on μ .

Compared with the purely left-handed scenario, the non-resonant contribution is enhanced more strongly than the resonant contribution in the right-handed scenario. For non-zero β_R^{33} , an additional contribution to non-resonant $\tau\nu$ production is generated through the $b_R\tau_R$ coupling, while the $c_L\nu\tau$ coupling is left unchanged. For resonant production involve the $b\tau$ coupling, the branching fraction is increased by the additional right-handed coupling, but its enhancement is partly offset by the increase in the total LQ width. Therefore the enhancement is smaller than that of the non-resonant contribution. The visible cross-section limit is therefore lower for $|\beta_R^{33}| = 1$ at $M_U = 4$ TeV.

For the exclusion limit interpretation in the right-handed scenario, the limits are calculated with the asymptotic approximation and without interference, as specified in Section 10.1.5. Figures 10.11 and 10.12 show the 95% CL exclusion limits. At $M_U = 1.5$ TeV, the observed (expected) upper limits on g_U are approximately 0.84 (0.80) and 0.53 (0.51) for $\beta_L^{23} = 1.0$ and 2.2, respectively. At $M_U = 3.0$ TeV, the corresponding limits are 1.98 (1.96) and 1.30 (1.28). A similar dependence on g_U and β_L^{23} to that in the left-handed scenario is observed. Note that the hatched regions are recalculated using Eq. (3.19) with C_{LL}^c values read from the black solid line of $C_{LR}^c = -C_{LL}^c$ in Figure 3.1.

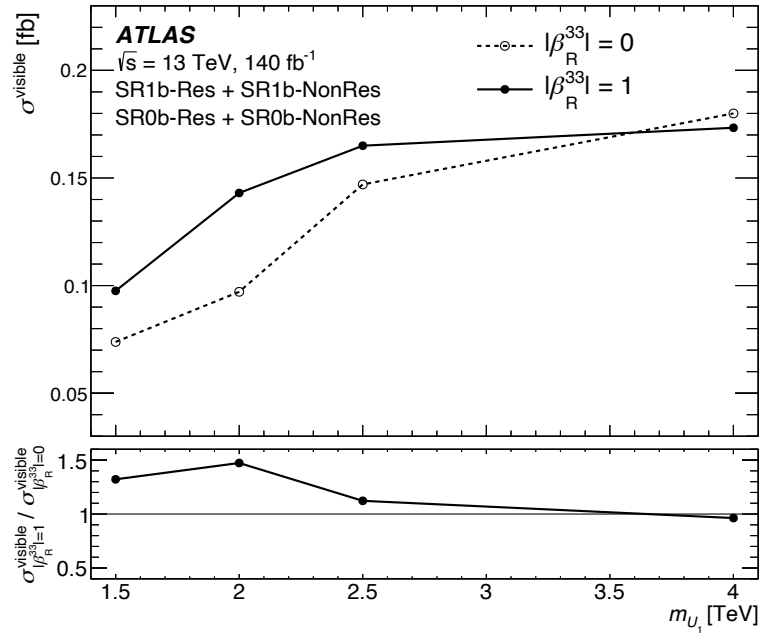


Fig. 10.10 The 95% CL upper limits on the inclusive visible cross section, with $|\beta_R^{33}| = 0$ (blank circle) and $|\beta_R^{33}| = 1$ (full circle), as a function of M_U . The parameters g_U and β_L^{23} are fixed at 1 and 0.2, respectively. The visible cross section is defined as the cross section calculated with MADGRAPH, multiplied by the acceptance and selection efficiency. All signal regions are combined for simplicity. The interference effect with the SM $\tau\nu$ process is not considered.

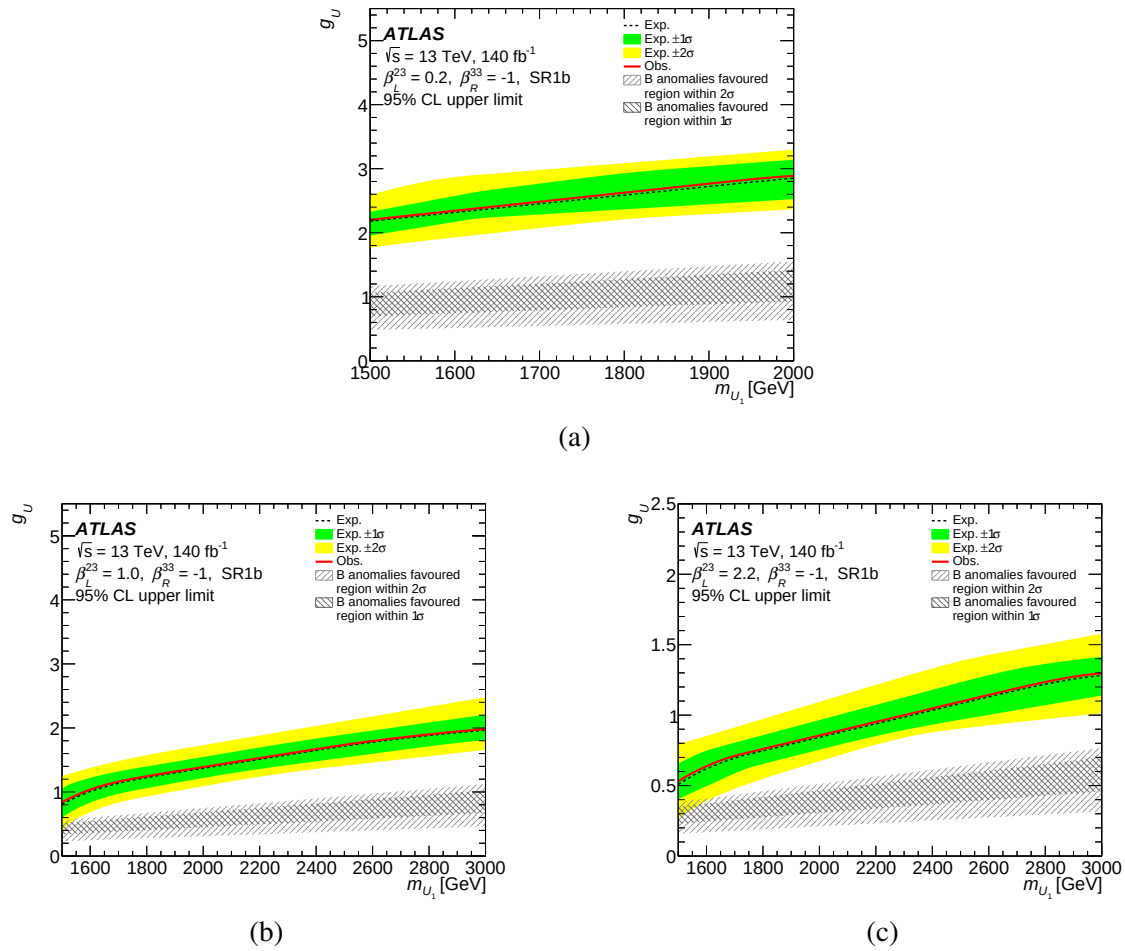


Fig. 10.11 Observed (red full line) and expected (black dotted line) 95% CL upper limits on g_U as a function of M_U for $\beta_R^{33} = -1$, at (a) $\beta_L^{23} = 0.2$, (b) 1.0 and (c) 2.2. The hatched regions are recalculated using Eq. (3.19) with C_{LL}^c values read from the black solid line of $C_{LR}^c = -C_{LL}^c$ in Figure 3.1.

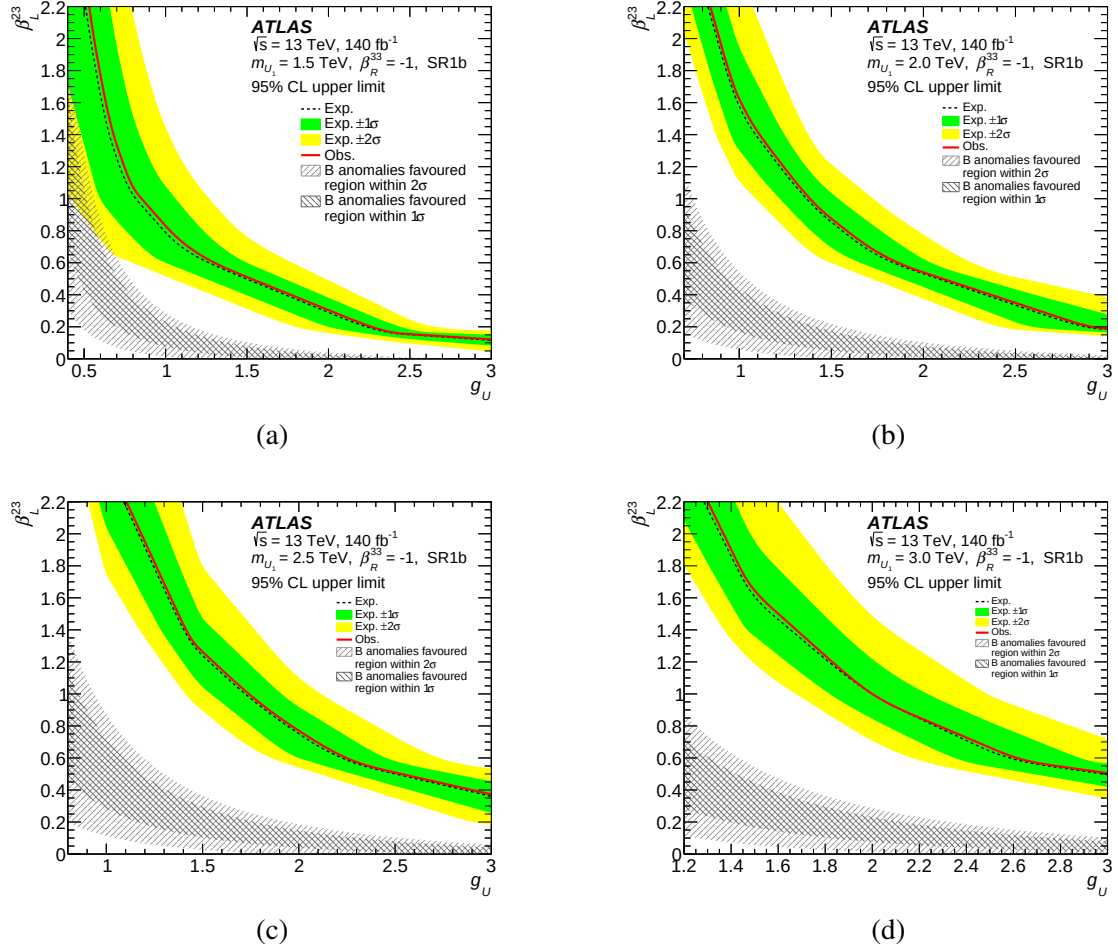


Fig. 10.12 Observed (red full line) and expected (black dotted line) 95% CL upper limits on β_L^{23} as a function of g_U for $\beta_R^{33} = -1$ at (a) $M_U = 1.5$ TeV, (b) 2.0 TeV, (c) 2.5 TeV and (d) 3.0 TeV. The hatched regions are recalculated using Eq. (3.19) with C_{LL}^C values read from the black solid line of $C_{LR}^C = -C_{LL}^C$ in Figure 3.1.

10.4 Future improvements

The coupling limits are expected to improve only modestly with the larger datasets from Run 3 and the HL-LHC. An illustrative projection study compares expected limits at 140 fb^{-1} with those at 450 fb^{-1} and 3000 fb^{-1} using Asimov datasets at $\sqrt{s} = 13 \text{ TeV}$ with the same selections. For $M_U = 3.0 \text{ TeV}$ and $\beta_L^{23} = 1.4$, the expected 95% CL upper limit on g_U improved by approximately 5% and 10% at 450 fb^{-1} and 3000 fb^{-1} , respectively, compared with 140 fb^{-1} . This limited gain is expected due to the approximate g_U^4 dependence of the non-resonant pure-BSM contribution at fixed M_U and β_L^{23} . Further improvements are therefore expected from reoptimising the selections in addition to increasing the integrated luminosity.

The right-handed interpretation should be extended with interference included in the limit calculation. The existing limits neglect this contribution, while the study at $M_U = 1.5 \text{ TeV}$ indicates sizeable interference in SR0b. Additional interference samples covering masses and couplings in the full range of interest are therefore needed.

The sensitivity could be improved through a dedicated optimisation of SR0b, whose selections were inherited from the SR1b. A preliminary study with selected leading systematic uncertainties found improved sensitivity with $E_T^{\text{miss}} > 600 \text{ GeV}$ in SR0b-Res. The higher threshold favours events containing an energetic neutrino from an on-shell $U_1 \rightarrow c\nu_\tau$ decay, while suppressing the softer $W + \text{jets}$ background. The magnitude of the interference compared with the pure-BSM yield is also reduced in such phase space. This optimisation was not applied in this thesis, and its potential gain should be reassessed in a future analysis.

Chapter 11

Conclusion

The search presented in this thesis is the first ATLAS search dedicated to high-mass U_1 leptoquarks in the $\tau\nu(+b)$ final state with a hadronically decaying τ and missing transverse momentum. It uses 140 fb^{-1} of pp collision data recorded during Run 2 at $\sqrt{s} = 13 \text{ TeV}$. The search is motivated by the $R(D^{(*)})$ anomalies, which can be explained by a U_1 contribution to $b \rightarrow c\tau\nu$.

Both resonant production and non-resonant t -channel exchange are probed, extending the sensitivity across different coupling scenarios. For each topology, two SRs are defined by requiring one b jet or applying a b -jet veto, giving four SRs in total to gain better sensitivity. The optimisation is performed for SR1b using a grid search for maximum sensitivity. The fake- τ_{had} background, mainly from $Z \rightarrow \nu\nu + \text{jets}$, is small. The multi-jet background is negligible after the full SR selections. The remaining dominant $W + \text{jets}$ and top-quark backgrounds are estimated using NFs measured in the corresponding CRs, and their extrapolation is tested in dedicated VRs.

As a result, no significant excess above the SM prediction is observed. A profile-likelihood fit including statistical and systematic uncertainties is performed to extract the limit on the signal strength μ . Upper limits are set at 95% CL on the couplings for M_U between 1.5 and 3.0 TeV in the purely left-handed scenario. The limits are obtained using the CL_s method with pseudo-experiments. At $M_U = 1.5 \text{ TeV}$, the observed (expected) upper limits on g_U are 1.13 (1.11) for $\beta_L^{23} = 1.0$ and 0.52 (0.53) for $\beta_L^{23} = 2.2$. At $M_U = 2.5 \text{ TeV}$, the corresponding limits are 2.30 (2.19) and 1.22 (1.18).

A supplementary interpretation in the right-handed scenario in SR1b is also performed using the asymptotic approximation, excluding the interference effect. At $M_U = 1.5 \text{ TeV}$, the observed (expected) upper limits on g_U are 0.84 (0.80) for $\beta_L^{23} = 1.0$ and 0.53 (0.51) for $\beta_L^{23} = 2.2$. At $M_U = 3.0 \text{ TeV}$, the corresponding limits are 1.98 (1.96) and 1.30 (1.28).

These results probe large values of β_L^{23} in a region of parameter space not covered by previous $\tau\tau$ searches. The SR1b regions provide most of the sensitivity, while SR0b improves the sensitivity when the $s\tau$ contribution is enhanced at large β_L^{23} . The $\tau\nu(+b)$ channel therefore complements the existing searches.

Further improvements are expected from larger datasets and a dedicated optimisation of SR0b. Especially the gain from the optimisation is expected to be significant in the low-mass region.

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Appendix A

Multijet background study

The multijet suppression used in Section 7.1.2 is studied with simulated dijet and leptoquark events. The motivation and discriminating variables are introduced there. This appendix presents the study selection, threshold scan and residual-background checks.

A.1 Study selection and distributions

The study selection is kept loose to retain enough simulated multijet events to examine the relevant distributions. It requires the E_T^{miss} trigger, exactly one τ_{had} candidate, no selected electron or muon, and at least one jet, as listed in Table A.1. The final SR thresholds on E_T^{miss} , $p_T^{\tau_{\text{had}}}$, m_T and the visible invariant mass are not applied at this stage. No b -tag multiplicity is required.

Table A.1 Loose selection used to study the multijet suppression. This selection defines the denominator of the rejection fractions shown below.

Requirement	Selection
Trigger	Lowest-threshold unprescaled E_T^{miss} trigger
τ_{had} multiplicity	$N_{\tau_{\text{had}}} = 1$
Electron and muon veto	$N_{\ell} = 0$
Jet multiplicity	$N_j \geq 1$

Figure A.1 shows the two-dimensional distributions after this selection. The multijet sample is concentrated at small $\min \Delta\phi(E_T^{\text{miss}}, p_T^{\text{jet}})$ and small $\frac{E_T^{\text{miss}}}{p_T^{\text{jet}(\min \Delta\phi)}}$. The signal occupies a broader angular range and extends to larger momentum ratios. Approximately 90% of the multijet events have $\frac{E_T^{\text{miss}}}{p_T^{\text{jet}(\min \Delta\phi)}} < 6$, compared with approximately 55% of the signal in the

study sample. The ratio requirement alone would therefore reject a substantial fraction of the signal. It is then used in combination with the angular requirement.

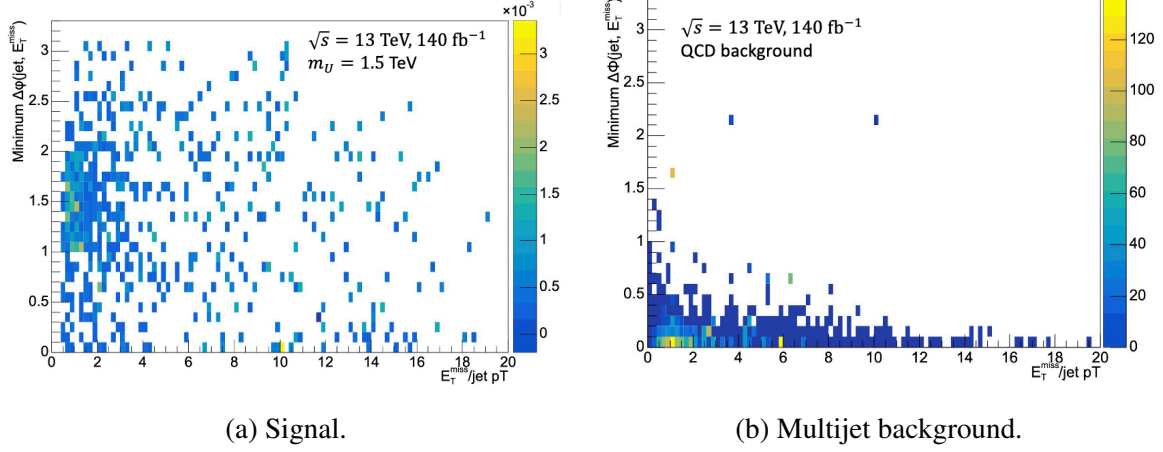


Fig. A.1 Correlation between $\min \Delta\phi(E_T^{\text{miss}}, p_T^{\text{jet}})$ and $\frac{E_T^{\text{miss}}}{p_T^{\text{jet}(\min \Delta\phi)}}$ for (a) simulated signal with $M_U = 1.5$ TeV and (b) simulated multijet background after the loose study selection. The concentration of multijet events at small values of both variables motivates a combined requirement.

A.2 Threshold scan and signal retention

The angular threshold is scanned while the momentum-ratio threshold is fixed at six. For an angular threshold X , the rejected fraction is defined as

$$\text{Rejection ratio} = \frac{N\left(\min \Delta\phi(E_T^{\text{miss}}, p_T^{\text{jet}}) < X \text{ and } \frac{E_T^{\text{miss}}}{p_T^{\text{jet}(\min \Delta\phi)}} < 6\right)}{N(\text{all events after preselection})}. \quad (\text{A.1})$$

The rejected fraction is evaluated separately for signal and multijet events, using each sample's yield after the loose selection as its denominator.

Figure A.2 shows the resulting rejection fractions. At $X = 0.4$, approximately 86% of the multijet events are rejected, while the signal loss is 3.6%. Increasing the angular threshold rejects more multijet events but also drops more signal.

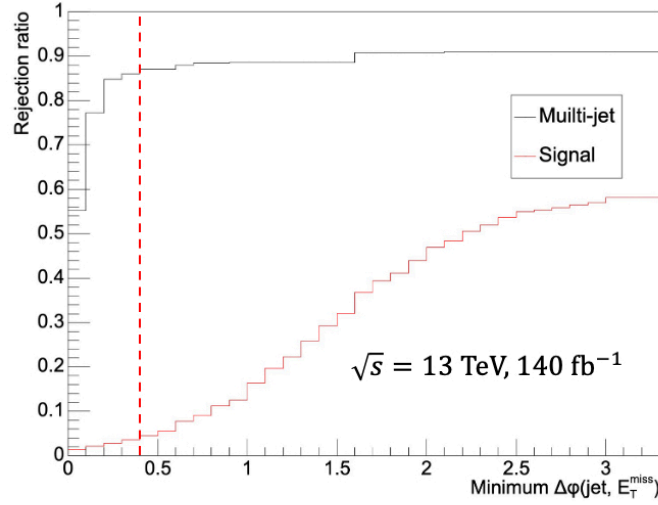


Fig. A.2 Rejected fractions of simulated signal and multijet events as a function of the threshold X on $\min\Delta\phi(E_T^{\text{miss}}, p_T^{\text{jet}})$, with $\frac{E_T^{\text{miss}}}{p_T^{\text{jet}(\min\Delta\phi)}} < 6$ also required. The red and black curves show signal loss and multijet rejection, respectively. The vertical dotted line marks the adopted threshold $X = 0.4$. The loose selection in Table A.1 is applied to both samples.

A.3 Residual multijet contribution

The residual multijet contribution is examined after applying the cleaning cut and the full SR requirements. No simulated multijet events survive those selections in the samples used for the study. This supports neglecting the multijet contribution in the nominal SR prediction. A supplementary selection requiring Medium τ_{had} identification while failing Tight also gives a negligible multijet contribution in the simulated study, supporting the effectiveness of the cleaning beyond the nominal Tight selection.

Appendix B

Zero-b-jet signal-region optimisation

The E_T^{miss} threshold in the SR0b regions defined in Section 7.3 is studied to improve the sensitivity. A higher threshold can suppress both the SM background and the interference, but also drops the signal. The significance is therefore evaluated as the threshold is varied. For simplicity, only a few leading systematic uncertainties are selected in the study.

The E_T^{miss} threshold is scanned from 200 to 1500 GeV in 100 GeV steps. The signal benchmarks use the parameter points at $(M_U, g_U, \beta_L^{23}) = (1.5 \text{ TeV}, 0.5, 1.8)$ for SR0b-Res and $(3.0 \text{ TeV}, 2.0, 1.8)$ for SR0b-NonRes.

The scan result is shown in Figure B.1. For SR0b-NonRes, the significance is maximised at $E_T^{\text{miss}} > 400 \text{ GeV}$. For SR0b-Res, the significance is maximised at $E_T^{\text{miss}} > 600 \text{ GeV}$. The final resonant selection nevertheless retains $E_T^{\text{miss}} > 200 \text{ GeV}$ to keep its kinematic requirements aligned with SR1b-Res.

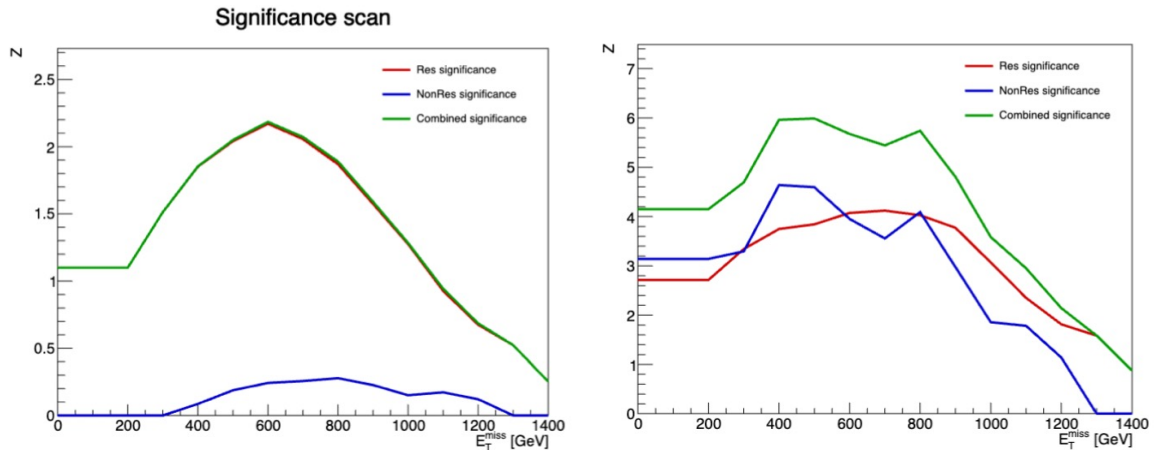


Fig. B.1 Significance as a function of the E_T^{miss} threshold for the SR0b optimisation. The left and right panels use $(M_U, g_U, \beta_L^{23}) = (1.5 \text{ TeV}, 0.5, 1.8)$ and $(3.0 \text{ TeV}, 2.0, 1.8)$, respectively. Only a few leading systematic uncertainties are included.

Appendix C

Fake- τ_{had} estimation

The fake- τ_{had} background from $Z \rightarrow \nu\nu + \text{jets}$ is estimated using the strategy described in Section 8.1. Its modelling is tested in a $Z \rightarrow \ell\ell$ validation region (ZVR).

The ZVR requires two signal electrons or two signal muons and exactly one τ_{had} candidate, with the dilepton mass restricted to the Z-boson mass window. A b -jet veto and looser selections than in the SRs are applied to increase the event yield, and both lepton channels are used. The quantity $|\vec{p}_{\text{T}}^{\ell\ell} + \vec{p}_{\text{T}}^{\text{miss}}|$ is used to emulate the $E_{\text{T}}^{\text{miss}}$ in $Z \rightarrow \nu\nu$ events by adding the dilepton momentum to $\vec{p}_{\text{T}}^{\text{miss}}$. An additional requirement of $p_{\text{T}}^{\ell\ell} > 130$ GeV suppresses the $t\bar{t}$ background and increases the $Z + \text{jets}$ purity. The definition is summarised in Table C.1.

Table C.1 Event selection criteria for the ZVR.

ZVR selections		
Feature	Muon channel	Electron channel
Trigger	Lowest-unprescaled $E_{\text{T}}^{\text{miss}}$ trigger	Single-electron trigger
Lepton multiplicity	$N_{\mu} = 2$	$N_e = 2$
τ_{had} multiplicity	$N_{\tau_{\text{had}}} = 1$	
b -jet multiplicity	$N_b = 0$	
Jet multiplicity	$N_j \geq 1$	
Multijet suppression	$\min \Delta\phi(E_{\text{T}}^{\text{miss}}, p_{\text{T}}^{\text{jet}}) > 0.4$ or $\frac{E_{\text{T}}^{\text{miss}}}{p_{\text{T}}^{\text{jet}(\min \Delta\phi)}} > 6$	
$p_{\text{T}}^{\tau_{\text{had}}}$	> 100 GeV	
$m_{\text{T}}(p_{\text{T}}^{\ell}, E_{\text{T}}^{\text{miss}})$	> 100 GeV	
$m_{\ell\ell}$	$[75, 105]$ GeV	
$ \vec{p}_{\text{T}}^{\ell\ell} + \vec{p}_{\text{T}}^{\text{miss}} $	> 200 GeV	
$p_{\text{T}}^{\ell\ell}$	> 130 GeV	
N_j	≤ 3	

A total of 116 data events are selected in the ZVR, with a simulated $Z + \text{jets}$ purity of approximately 90%. The $p_T^{\tau_{\text{had}}}$ and RNN-score distributions are shown in Figures C.1 and C.2, respectively. The simulation overestimates the data by about 50% at high $p_T^{\tau_{\text{had}}}$. Both the fake- τ_{had} rate and the $Z + \text{jets}$ kinematics can contribute to this disagreement. However, agreement between data and simulation is observed at low RNN scores, in the τ_{had} -veto region. The discrepancy is therefore attributed mainly to the modelling of the fake- τ_{had} rate. This study motivates the conservative normalisation uncertainty specified in Section 9.2.4.

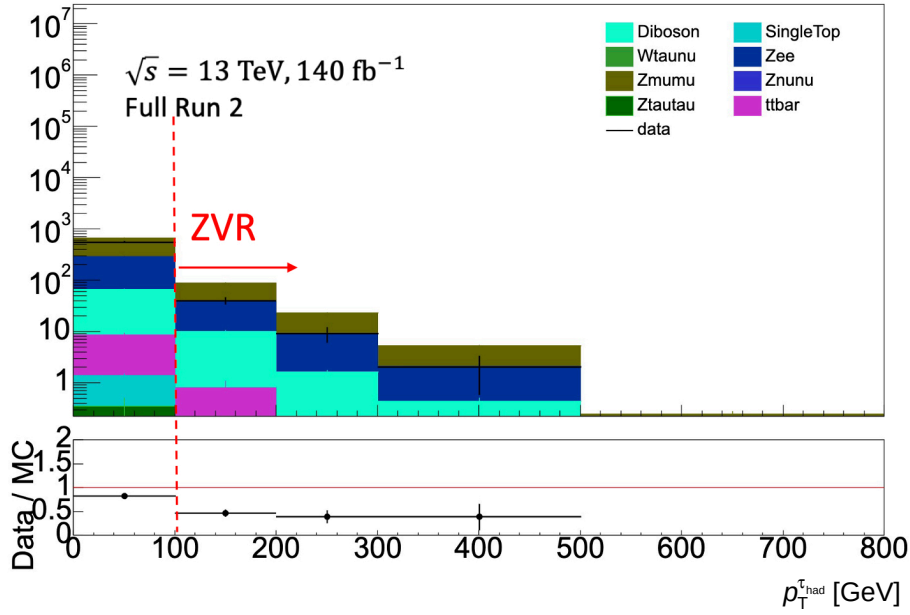


Fig. C.1 $N - 1$ kinematic plot of $p_T^{\tau_{\text{had}}}$ in the ZVR. The mismodelling of fake τ_{had} in the tail region is observed.

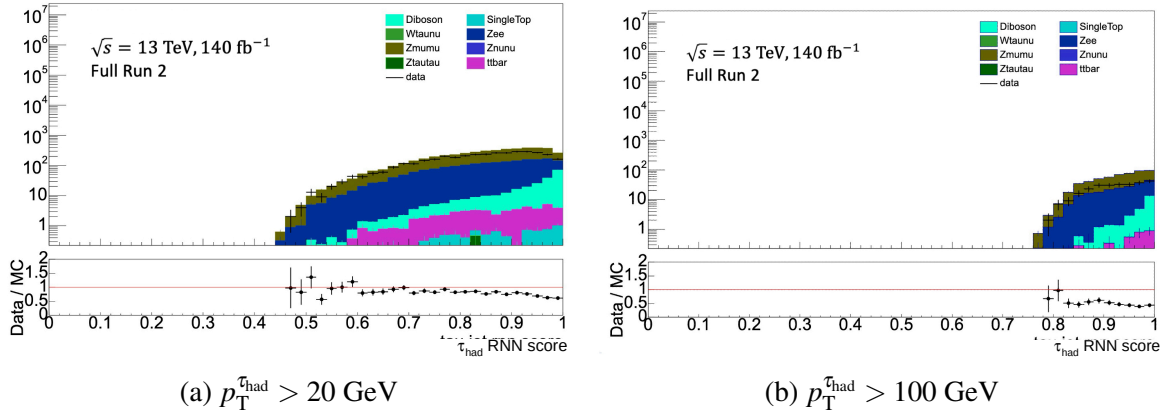


Fig. C.2 $\tau_{\text{had-id}}$ RNN score in the regions of $p_{\text{T}}^{\tau_{\text{had}}} > 20 \text{ GeV}$ (left) and $p_{\text{T}}^{\tau_{\text{had}}} > 100 \text{ GeV}$ (right). Both figures use a loose $\tau_{\text{had-id}}$ selection and a lower m_{T} selection to observe the difference between high RNN score and low RNN score regions.

Appendix D

Investigation of the VRW-NonRes excess

The excess in VRW-NonRes discussed in Section 8.4.1 is examined using the cutflow and kinematic distributions. The pre-fit data and MC simulation yields are consistent through the jet multiplicity requirement, and the discrepancy appears after applying $E_T^{\text{miss}} > 400$ GeV, as shown in Figure D.1. The $N - 1$ distribution with finer E_T^{miss} bins in Figure D.2 shows agreement in most bins, with the excess concentrated in $425 < E_T^{\text{miss}} < 450$ GeV. Lowering the E_T^{miss} threshold to 300 GeV or raising it to 500 GeV reduces the discrepancy to 1.8σ or 1.5σ , respectively, when only statistical uncertainties are considered.

The lepton- p_T distribution within $425 < E_T^{\text{miss}} < 450$ GeV is also shown in Figure D.2. The excess is concentrated in $300 < p_T^\ell < 400$ GeV, well above the 200 GeV threshold, and is therefore not explained by the lepton- p_T threshold. The 12 data events in this interval comprise five electron events and seven muon events. Their lepton and b -jet positions are shown in Figure D.3. The interpretation of these checks is given in Section 8.4.1.

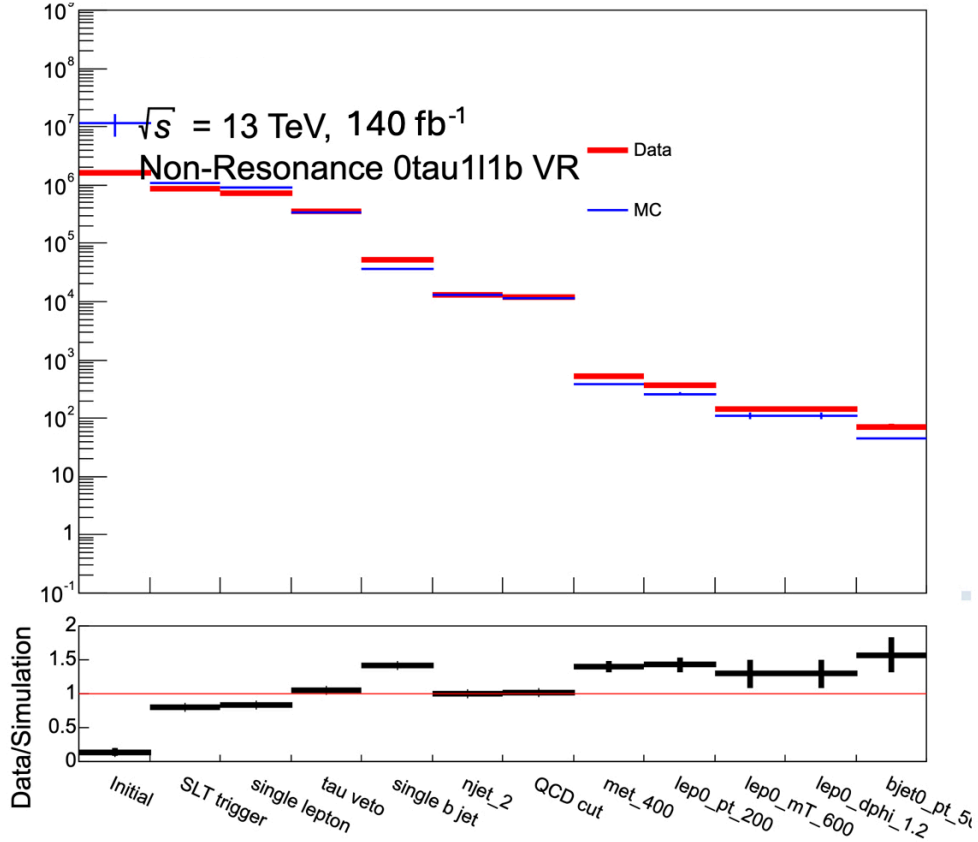


Fig. D.1 Cutflow comparison between the data and MC for VRW-NonRes.

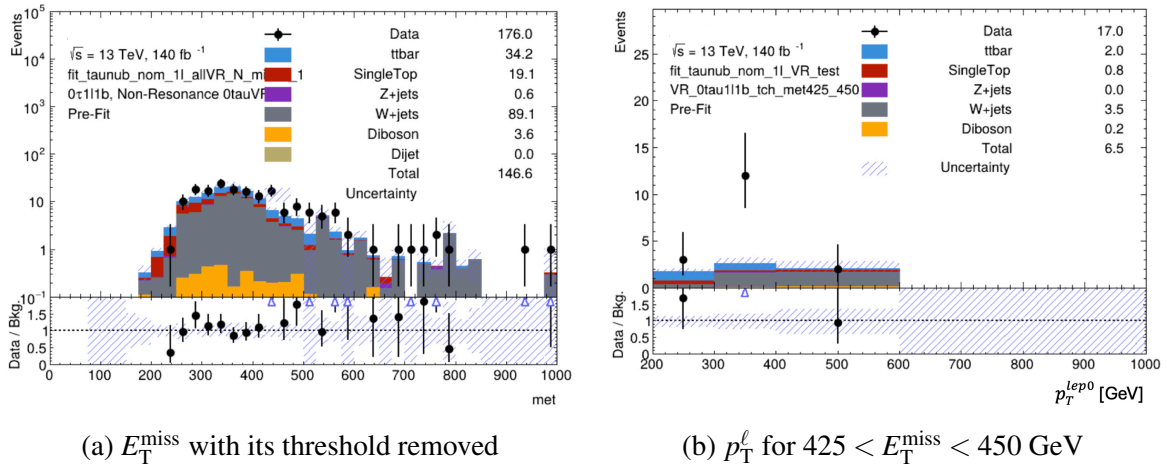
(a) E_T^{miss} with its threshold removed(b) p_T^{ℓ} for $425 < E_T^{\text{miss}} < 450$ GeV

Fig. D.2 Left: $N - 1$ E_T^{miss} distribution in VRW-NonRes. Data and MC are found to be in rather good agreement for most of the bins. Right: the leading-lepton p_T distribution after applying the E_T^{miss} range cut $425 < E_T^{\text{miss}} < 450$ GeV, where a large discrepancy between data and the MC expectation is observed.

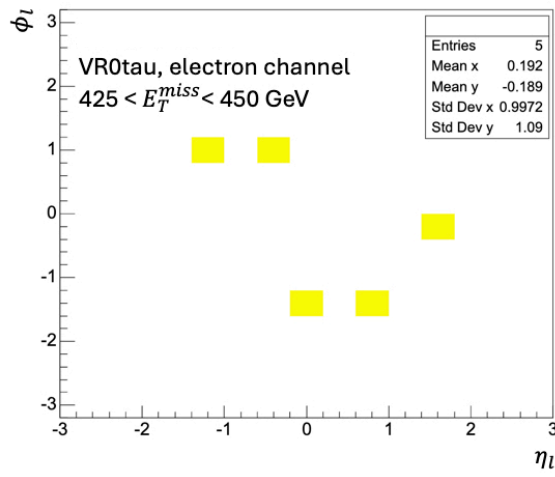
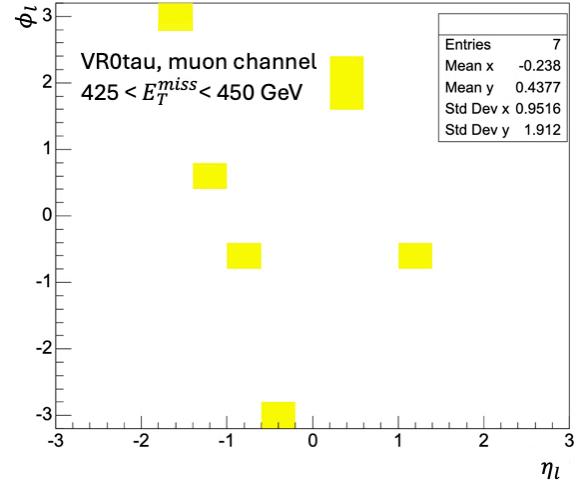
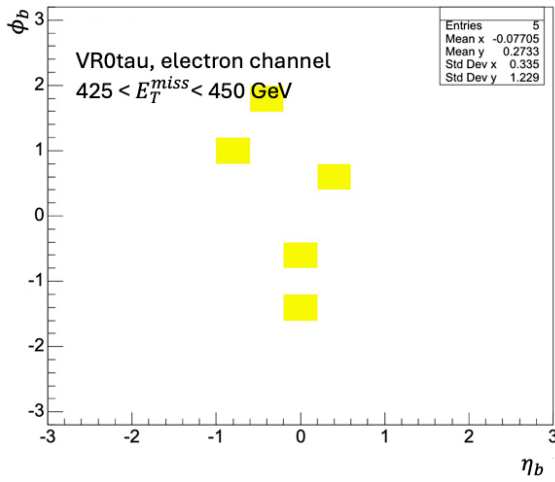
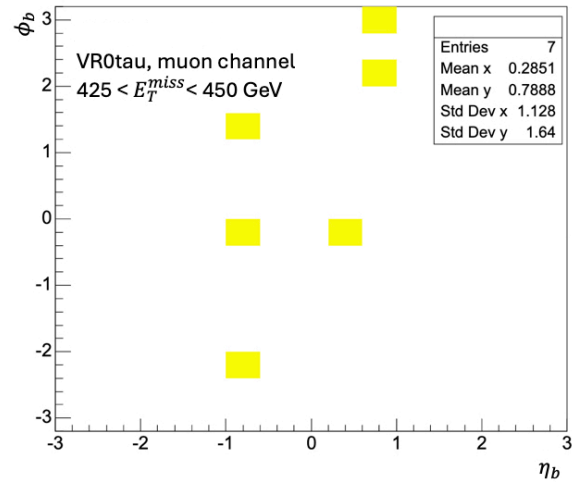
(a) Lepton, e channel.(b) Lepton, μ channel.(c) b -jet, e channel.(d) b -jet, μ channel.

Fig. D.3 η - ϕ map of the lepton and b -jets in VRW-NonRes for the electron channel (left) and the muon channel (right).

D.1 Low-mass VRs

The low-mass VRs provide the additional $1\tau_{\text{had}}0\ell1b$ check introduced in Section 8.2. They are used as a supplementary check of the background prediction following the VRW-NonRes study.

The low-mass VRs are defined by replacing $m_{b\tau_{\text{had}}} > 800$ GeV with $400 < m_{b\tau_{\text{had}}} < 800$ GeV in the resonant region, and $m_T > 600$ GeV with $500 < m_T < 600$ GeV in the non-resonant region. To increase the event yields, the m_T and $p_T^{\tau_{\text{had}}}$ thresholds in the resonant region are both lowered by 100 GeV. In the non-resonant region, the E_T^{miss} threshold is lowered by 200 GeV and the $p_T^{\tau_{\text{had}}}$ threshold by 100 GeV. The resulting selections are summarised in Table D.1. All other requirements follow the corresponding SR selections. Signal contamination is checked before examining the data, using the same benchmarks as in the nominal VRs. It is approximately 2% in the resonant region and 7% in the non-resonant region.

The NFs from a background-only fit to the CRs are then applied to these regions. This cross-check is performed without systematic uncertainties in the fit. The predicted and observed yields are given in Table D.1, and the post-fit comparison is shown in Figure D.4.

Table D.1 Selections and yields in the low-mass VRs. The background-only fit to the CRs is extrapolated to these regions. The systematic uncertainties are not included.

	Resonant	Non-resonant
$p_T^{\tau_{\text{had}}} [\text{GeV}]$	> 100	> 100
$E_T^{\text{miss}} [\text{GeV}]$	> 200	> 200
$m_T [\text{GeV}]$	> 100	$500 < m_T < 600$
$m_{b\tau_{\text{had}}} [\text{GeV}]$	$400 < m_{b\tau_{\text{had}}} < 800$	No requirement
Signal contamination	$\sim 2\%$	$\sim 7\%$
Post-fit background	25.4 ± 1.5	8.0 ± 1.7
Observed data	24	9

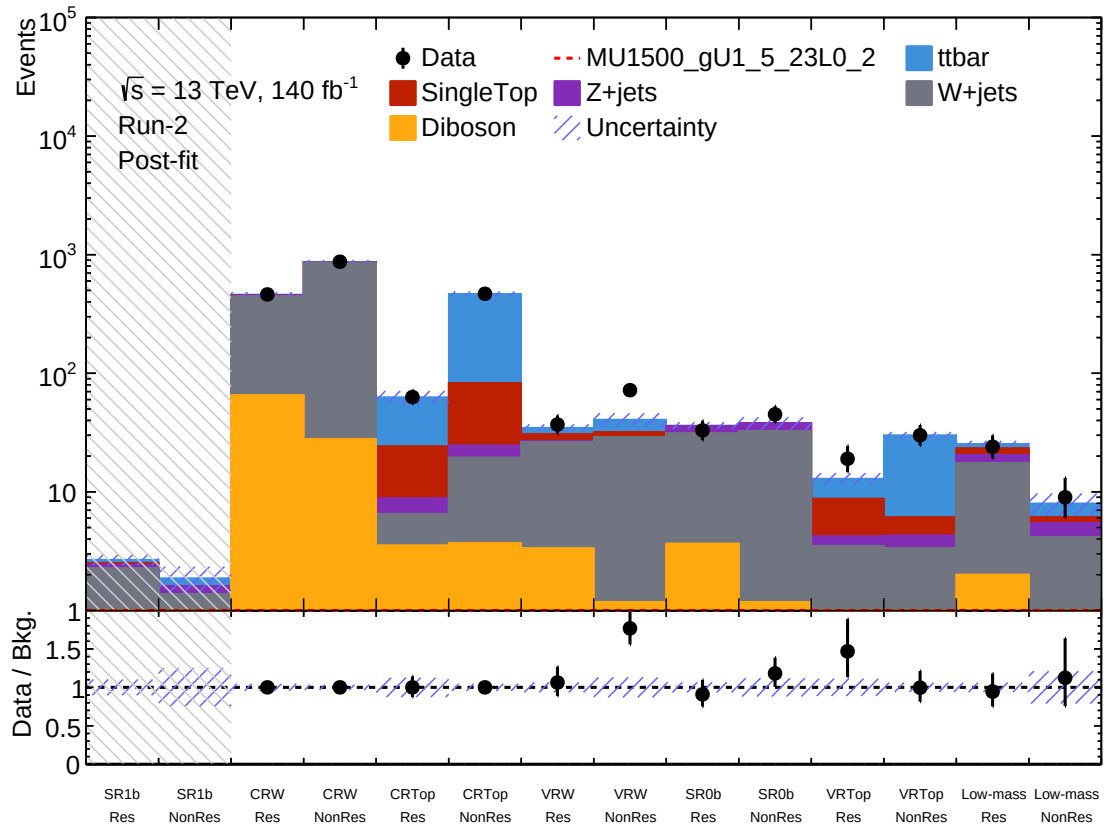


Fig. D.4 Post-fit summary plot for the background-only fit without systematic uncertainties. The last two bins stand for the resonant and non-resonant low-mass VRs, respectively.

2790 **Appendix E**

2791 **Electron and muon consistency in the** 2792 **control regions**

2793 The electron and muon channels in CRW are combined as described in Section 8.2. The
2794 difference between their data/MC ratios is investigated here. Before fitting, the ratios in CRW-
2795 NonRes are 1.00 ± 0.09 and 0.80 ± 0.08 for the electron and muon channels, respectively,
2796 corresponding to a 1.7σ difference. The uncertainties in this study are statistical only.

2797 The scale factors in the two channels are first checked and found to be applied correctly.
2798 The data/MC ratio is then examined after each cut. The difference appears after the $50 <$
2799 $p_T^{j_1} < 250$ GeV requirement in CRW-NonRes. The leading-jet p_T distributions are compared
2800 between channels, separately for data and simulation, in Figure E.1. The electron yield is
2801 slightly higher in data and slightly lower in simulation. An upward fluctuation is observed in
2802 the electron data around 80–100 GeV.

2803 The simulated event weights, including cross-section normalisation, efficiency scale
2804 factors and pile-up weights, are examined as a function of leading-jet p_T in Figure E.2. A
2805 small number of events have weights of approximately 30. Most lie below 50 GeV, but one
2806 such event is found above this threshold in the selected muon sample, with no corresponding
2807 event in the electron sample. It increases the weighted muon prediction and lowers its
2808 data/MC ratio.

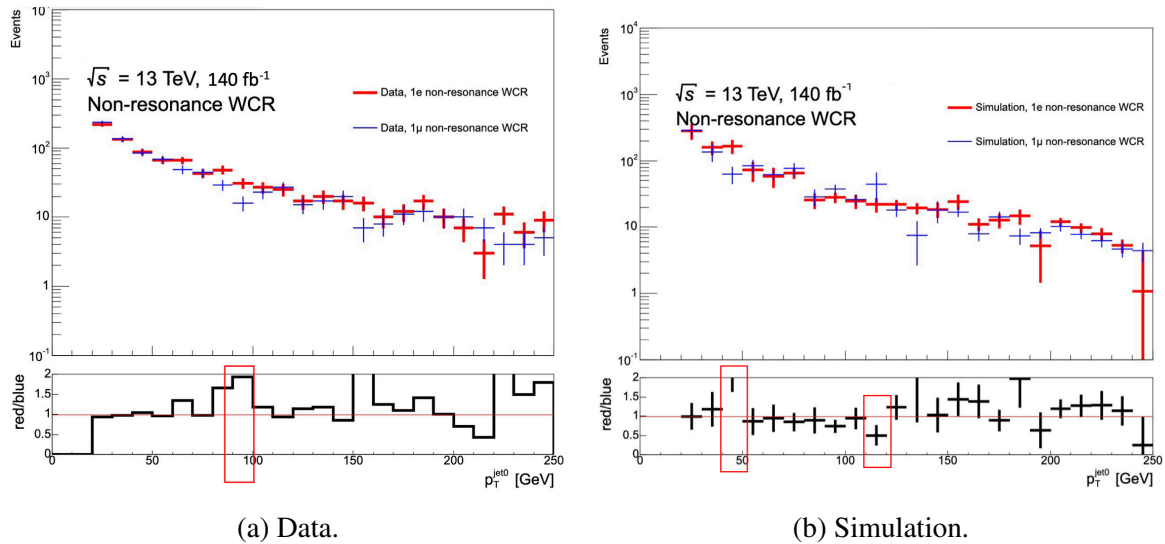


Fig. E.1 Electron and muon channel comparison of leading jet p_T in CRW-NonRes for data (left) and MC (right). The red line shows the leading jet p_T for the electron channel and the blue line shows the one for the muon channel. 2.6σ and 0.7σ discrepancies are observed for data and MC samples, respectively.

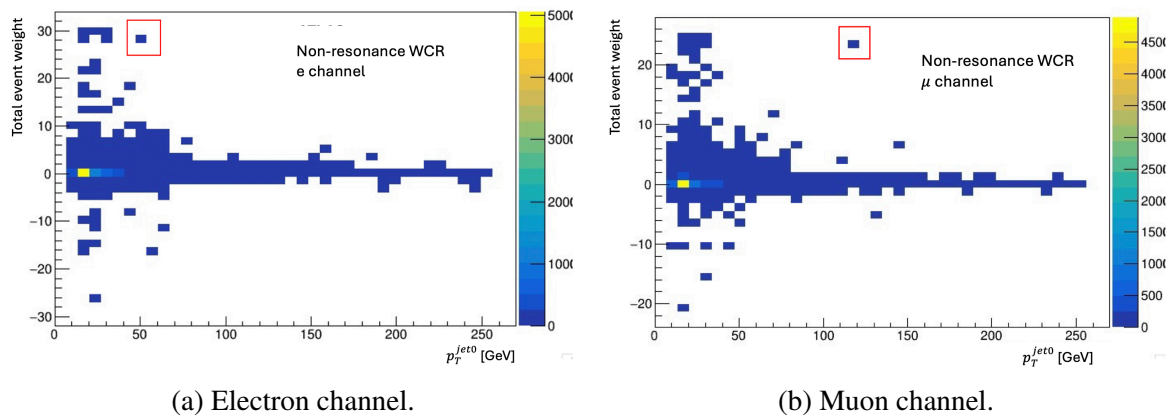


Fig. E.2 Event weight distribution as a function of leading jet p_T in CRW-NonRes for the electron (left) and muon (right) channels. Events that are marked with red rectangles correspond to the deviated bins marked in the simulation e/μ ratio distribution in Figure E.1.

2809 **Appendix F**

2810 **Comparison of one- and two- b -jet** 2811 **samples in CRTop**

2812 The kinematic shapes in the one- and two- b -jet CRTop samples are compared to check their
2813 use in a common CR. The simulated m_T and $m_{b\tau_{\text{had}}}$ distributions are normalised to unit area,
2814 so this comparison tests the shapes independently of their event yields. The electron and
2815 muon channels are examined separately in both topologies, as shown in Figures [F.1](#) and [F.2](#).
2816 No large shape difference is observed. The inclusive $N_b \geq 1$ selection is therefore used to
2817 improve the statistical precision of the top-quark background constraint.

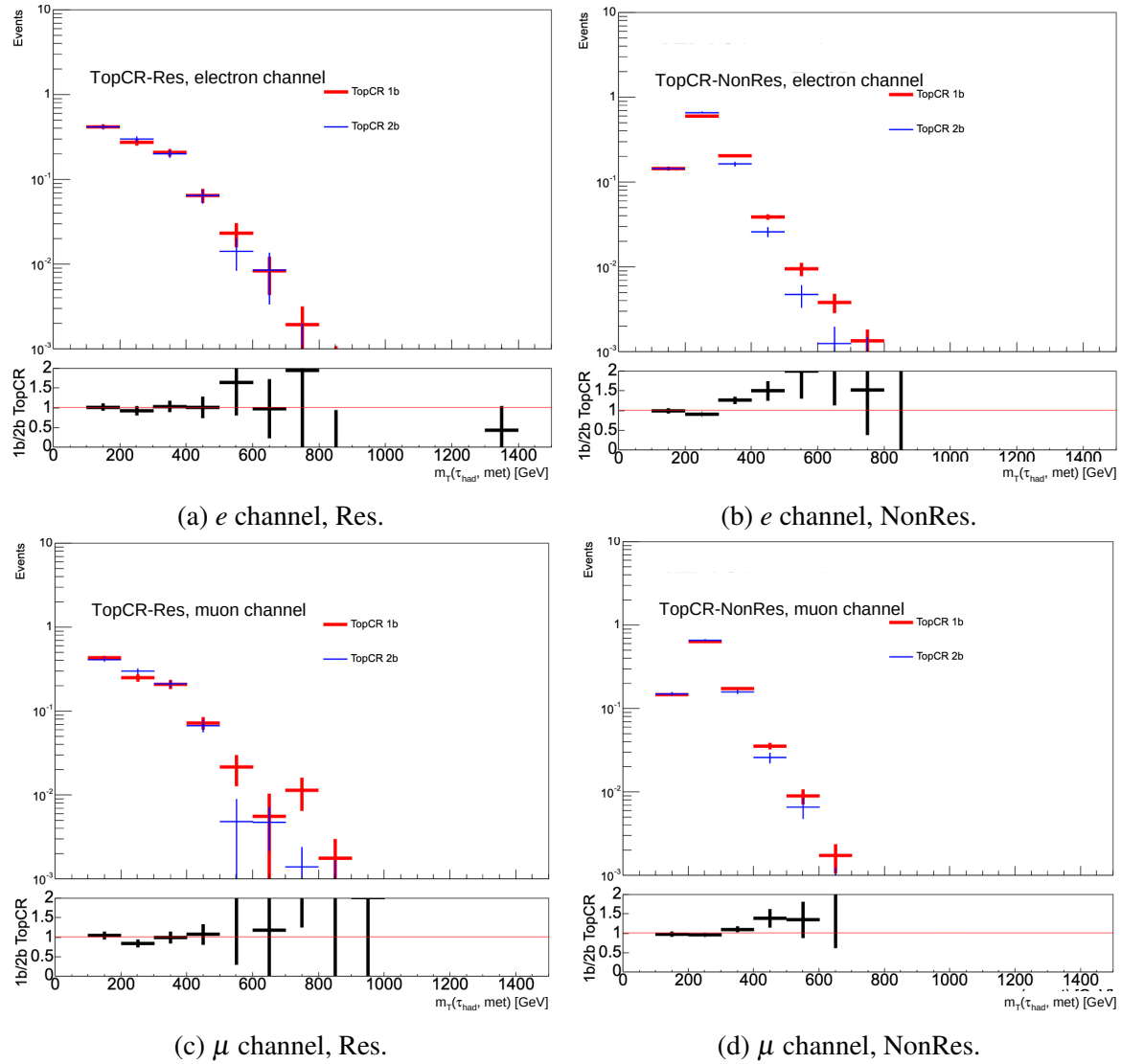


Fig. F.1 Simulated m_T distributions in the one- and two- b -jet CRTop samples. Each distribution is normalised to unit area. The red and blue lines represent the one- and two- b -jet samples, respectively. The electron and muon channels are shown for both topologies.

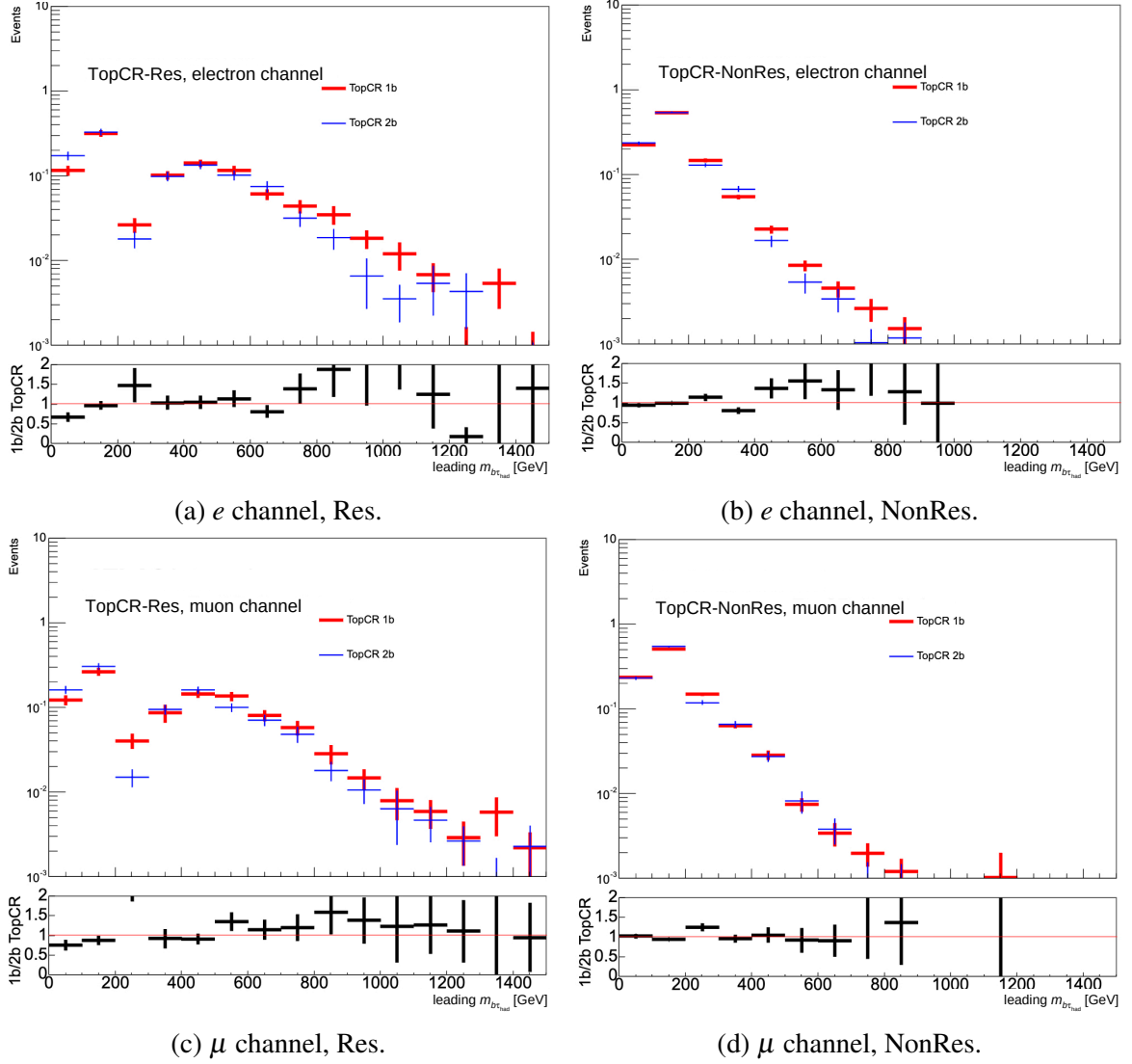


Fig. F.2 Simulated $m_b \tau_{\text{had}}$ distributions in the one- and two- b -jet CReTop samples. Each distribution is normalised to unit area. The red and blue lines represent the one- and two- b -jet samples, respectively. The electron and muon channels are shown for both topologies.

